GEK-41076

INSTRUCTIONS

MK II MS1002

TURBINE CONTROL PANEL

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM MK II MS1002
REV. 2	ISSUED			FIRST MADE FOR		GEK-41076
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. II SH. NO. I

FORM IC 2390 2-72

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AB	-ANTI-END BIAS	MAN	-MANUAL
ACC	-ACCEPTANCE	MAX	-MAXIMUM
ACOM	-ALS, COMBINED	MCC	-MOTOR CONTROL CENTER
ADJ	-ADJUSTMENT	MIN	-MINIMUM
ALR	-ALARM	MT	-MINIMUM TEST
ANN	-ANNUNCIATOR	N	-SPEED
AUX	-AUXILIARY	N.C.	-NORMALLY CLOSED CONTACT
AVG	-AVERAGE	NCE	-NOZZLE CONTROL VOLTAGE
CAP	-CAPACITOR	NHP	-SPEED, HIGH PRESSURE TURBINE
CCOM	-CONTROL COMMON	NLP	-SPEED, LOW PRESSURE TURBINE
CL	-CLEAR	N.O.	-NORMALLY OPEN CONTACT
CNT	-CONTACT	OS	-OVERSPEED
COM	-COMMON	OTA	-OVER TEMPERATURE A
COMP	-COMPARE OR COMPLETE	OTB	-OVER TEMPERATURE B
CONT	-CONTACT	PCR	-PRESSURE CONTROL RELAY
DCOM	-DIGITAL COMMON	PCOM	-P COMMON
DET	-DETECTOR	PK	-PEAK
DSP	-DIGITAL SETPOINT	PR	-PEAK RESERVE
ELEM	-ELEMENTARY	PUL	-PULSE
EXP	-EXPANDER	P/R	-PLUG/RECEPTACLE
FD	-FEEDBACK	PR/A	-PULSE RATE TO ANALOG
FCOM	-FLOATING/SPECIAL COMMON	RD	-RELAY DRIVE
FCV	-FUEL CONTROL VALVE	REF	-REFERENCE
FD	-FLAME DETECTOR	REG	-REGULATOR
FFN	-LIQUID FUEL FLOW	REM	-REMOTE
FLT	-FAULT	R/P	-RECEPTACLE/PLUG
FSNL	-FULL SPEED NO LOAD	SCOM	-SIGNAL COMMON
FU	-FUSE	SEQ	-SEQUENCE
GCE	-PORTION OF VCE CONTROLLING GAS FUEL FLOW	SF	-SPECIAL FILTER
		SH	-SHIELD
GEN	-GENERATOR	SP	-SETPOINT
GND	-GROUND	SRV	-SPEED RATIO VALVE
HP	-HIGH PRESSURE TURBINE	STCK	START CHECK
INC	-INCREASE	STR	STROKE
INH	-INHIBIT	SV	-SERVO VALVE
ISO	-ISOCHRONOUS	SW	-SWITCH
LCE	-PORTION OF VCE CONTROLLING LIQUID FUEL FLOW	D	-TIME DELAY
LEV	-LEVEL	DO	-TIME DELAY DROP OUT

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR	INDEX
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CONT. ON SH. III SH NO. II

CARD PIN NOMENCLATURE

(THE FOLLOWING LIST WILL BE OF ASSISTANCE IN "READING" THE CARD SYMBOLS. THE ABBREVIATIONS USED ARE INTENDED AS CRYPTIC NOTATIONS AND ARE NOT NECESSARILY RIGOROUS.)

AL	-ALARM
ALM	-ALARM
ALR	-ACCELERATION LIMIT RELEASE
AB	-AMBIENT BIAS
ACC	-ACCELERATION
ANN	-ANNUNCIATOR
ACC	-ACCELERATION CONTROL
BOV	-BLOW OFF VALVE
C	-COUNTER OR CARRY
CALIB	-CALIBRATE
CL	-CLEAR
COMP	-COMPARE
DA	-DIFFERENTIAL ALARM
DN	-DOWN
DO	-DROP OUT
EXP	-EXPONENTIAL
ESTOP	-EMERGENCY STOP
FB	-FEEDBACK
FFN	-FEEDBACK, FUEL FLOW
FG	-FULLY ON GAS
FL	-FULLY ON LIQUID
FLT	-FAULT
FS	-FAST START
FS5	-FRAME SIZE 5
G	-GAS
GND	-GROUND
HP	-HIGH PRESSURE TURBINE
HR	-HIGH RANGE
INC	-INCREASE
INH	-INHIBIT
INN	-INPUT, NEGATIVE
INP	-INPUT, POSITIVE
ISO	-ISOCHRONOUS
LIQ	-LIQUID
LP	-LOW PRESSURE TURBINE
LSB	-LEAST SIGNIFICANT BIT
LVDT	-LINEAR VARIABLE DISPLACEMENT TRANSFORMER
M	-METER
MIN	-MINIMUM
MPU	-MAGNETIC PICK-UP
MSB	-MOST SIGNIFICANT BIT
MT	-MINIMUM TEST
N	-SPEED
NC	-SPEED CONTROL/NOZZLE CONTROL
NDO	-NORMAL DROP OUT
NHP	-SPEED, HIGH PRESSURE TURBINE
NLP	-SPEED, LOW PRESSURE TURBINE
NOZ	-NOZZLE
OC	-INVERSE COUNTER INPUT
OFBF	-INVERSE FEEDBACK FAULT
OFLT	-INVERSE FAULT
OR-EXO	-"OR" EXPANDER
OS-	-OVERSPEED
OTA,B	-OVERTEMPERATURE A, B
OUTA,B	-ALARM OUTPUT A, B
P	-PURGE
PCD	-PRESSURE, COMPRESSOR DISCHARGE
PCDN	-PRESSURE, COMPRESSOR DISCHARGE NORMALIZED
PK	-PEAK
PKOP	-PEAK OPERATION
PR/A	-PULSE RATE TO ANALOG
PR	-PEAK RESERVE
PRST	-PRESET
PROT	-PEAK RESERVE OVERTEMPERATURE
PU	-PICK UP
PUL	-PULSE
RA,B	-RANGE A, B, ETC.
RD	-RELAY DRIVER
REF	-REFERENCE
RM	-RUN MODE
RST	-RESET
SB	-SHUTDOWN BIAS
SC	-START-UP CONTROL
SCA,B	-SERVO, CURRENT LOOP A, B, ETC
SER	-SERVO
SFT	-SYSTEM FAULT TRIP
SIG	-SIGNAL
SJ	-SUMMING JUNCTION
SM	-START-UP MODE
SRT	-SERTO-TRIP
SP	-SET POINT
SRV	-SPEED RATIO VALVE
ST	-STEER
TC	-THERMOCOUPLE
TC ADJ	-TIME CONSTANT ADJUST
TP	-TIE POINT
TP	-TEST POINT
TX	-AVERAGE EXHAUST TEMPERATURE
UFD	-UNDER FREQUENCY DROP OUT
US	-UNDERSPEED
VCE	-VARIABLE CONTROL VOLTAGE
VXDR	-VOLTAGE, TRANSDUCER

REV. 1	REV 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CARD PIN NOMENCLATURE
REV. 2	ISSUED			FIRST MADE FOR		GEK-41076
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 1 SH NO. IV

SECTION A

INTRODUCTION

THE MKII TURBINE CONTROL PANEL ALONG WITH SENSORS AND OUTPUT SERVO VALVES CONTROLS THE START-UP, PROTECTION, AND OPERATION OF THE GENERAL ELECTRIC HEAVY DUTY GAS TURBINE. THE PRIMARY FUNCTIONS PROVIDED BY THE SPEEDTRONIC* MKII CONTROL ARE SHOWN IN FIGURE A-1.

CONTROL SYSTEM

THE FUEL CONTROL SYSTEM CONSISTS OF THREE MAJOR LOOPS. THESE ARE START-UP, LP SPEED, AND TEMPERATURE. THE LOOPS ARE ILLUSTRATED IN FIGURE A-2. THESE LOOPS ARE TIED TOGETHER INTO A MINIMUM VALUE GATE SUCH THAT THE LOOP CALLING FOR THE LEAST AMOUNT OF FUEL WILL BE IN COMMAND. THE CONTROL SYSTEM, WHICH IS DESIGNED TO PROVIDE MAXIMUM RUNNING RELIABILITY, IS FURTHER ENHANCED BY "REDUNDANCY BY ASSOCIATION". FOR EXAMPLE: IF THE SYSTEM IS OPERATING ON SPEED CONTROL, THE TEMPERATURE CONTROL WILL BACK IT UP. IN ADDITION TO THE FUEL CONTROL LOOPS, AN OPTIONAL NOZZLE CONTROL LOOP MAY BE PROVIDED WHICH POSITIONS THE ANGLE OF THE NOZZLE TO CONTROL HP SPEED.

SEQUENTIAL

SEQUENTIAL CIRCUITRY IS PROVIDED TO MONITOR THE TURBINE, STARTING MEANS, AND AUXILIARIES TO DETERMINE IF ALL SYSTEMS ARE NORMAL. THE RETURN SIGNALS FROM THESE DEVICES SERVE AS PERMISSIVES TO OPERATE THE TURBINE. ABNORMAL CONDITIONS ARE ANNUNCIATED VIA THE LOCAL PANEL ANNUNCIATOR. THE SEQUENTIAL CIRCUITRY ALSO GENERATES LOGIC SIGNALS TO FACILITATE STARTING AND STOPPING THE TURBINE IN A PRESCRIBED MANNER. THESE LOGIC SIGNALS INCLUDE SPEED SIGNALS, DIGITAL SETPOINT CONTROL, STARTING MEANS CONTROL, TIMERS, AND OTHERS.

PROTECTION

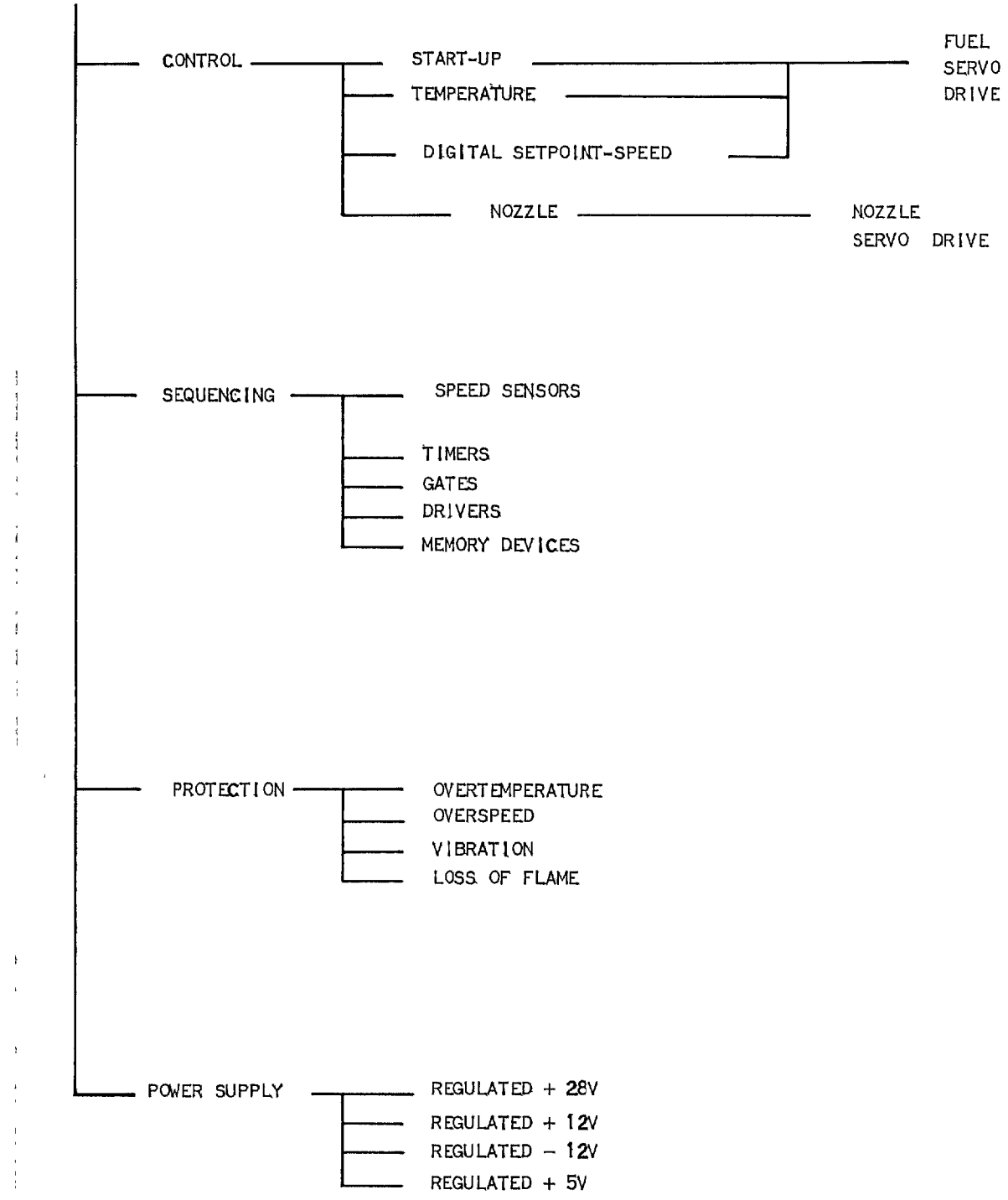
PROTECTION CIRCUITRY IS PROVIDED TO TRIP THE TURBINE WHEN CRITICAL PARAMETERS ARE EXCEEDED OR CRITICAL EQUIPMENT FAILS. ALARM ANNUNCIATION IS PROVIDED TO INDICATE WHEN CRITICAL LEVELS ARE BEING APPROACHED AND TO INDICATE ABNORMAL EQUIPMENT OPERATION.

POWER SUPPLY

THE CONTROL PANEL CONTAINS ITS OWN HIGHLY RELIABLE SOLID STATE POWER SUPPLY. THE POWER SUPPLY CONSISTS OF A DC TO DC CONVERTER AND FINAL REGULATORS FOR P28, P12, P5.3, AND N12 VOLTS. THE CONVERTER WILL OPERATE SATISFACTORILY WITH INPUT VOLTAGE OF 95 TO 140 VOLTS D.C. A SOURCE SELECTOR WHICH WILL PERMIT OPERATING FROM A 95-140 VDC BATTERY, COMMERCIAL POWER, OR INDUCTOR ALTERNATOR IS OPTIONAL.

*TRADEMARK OF GENERAL ELECTRIC COMPANY.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
REV. 2	ISSUED			FIRST MADE FOR		INTRODUCTION	
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 2	SH. NO. 1



SPEEDTRONIC MKII
FIG. A-1

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		INTRODUCTION
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 3 SH. NO. 2

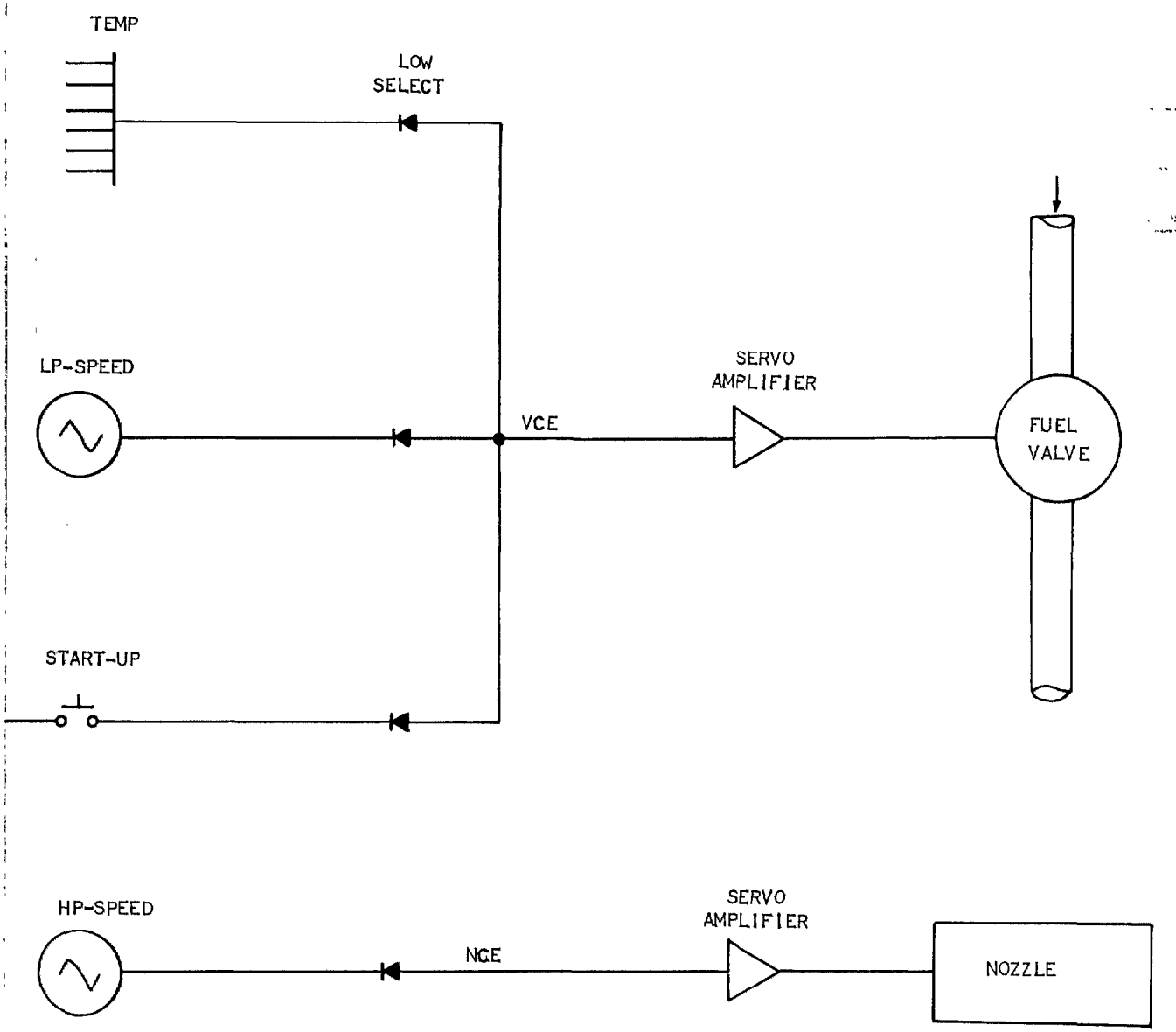


FIG. - A-2

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		INTRODUCTION
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 4 SH NO. 3

PUBLICATIONS AND REPAIR NOTES

THE INTENT OF THIS INSTRUCTION BOOK IS FOR USE IN CONJUNCTION WITH THE (1) TURBINE CONTROL SPECIFICATION, (2) CONTROL PANEL ELEMENTARY DIAGRAM, (3) SCHEMATIC PIPING DIAGRAM (4) OPERATING INSTRUCTIONS, AND (5) TURBINE CONTROL ELEMENTARY READING AID. THIS INSTRUCTION BOOK IS TYPICAL AND DOES NOT PURPORT TO COVER ALL DETAILS OR VARIATIONS THAT MAY ARISE ON A PARTICULAR APPLICATION.

WHEN TROUBLESHOOTING THE TURBINE CONTROL PANEL, THE TEST POINTS PROVIDED ON THE CALIBRATOR (IF AVAILABLE) AND CARD FRONTS SHOULD BE USED WHENEVER POSSIBLE TO AVOID DAMAGE TO THE CARDS THAT OFTEN RESULTS WHEN CONNECTING TEST PROBES TO THE PINS ON BACK OF THE SPEEDTRONIC PAGE. IF IT IS ABSOLUTELY NECESSARY TO TROUBLE-SHOOT ON THE BACK OF THE PAGE, INSULATED TEST PROBES SHOULD BE USED. RECOMMENDED PROCEDURES CONCERNING REMOVING OR REPLACING CARDS WITH POWER APPLIED IS GIVEN IN TECHNICAL BULLETIN 060 TB1-01-001.

FAILED CARDS AND MODULES SHOULD BE RETURNED TO THE GENERAL ELECTRIC COMPANY FOR REPAIRS. NO ATTEMPT SHOULD BE MADE TO REPAIR THE CARD IN THE FIELD. THE PROCEDURE FOR RETURNING CARDS AND MODULES CAN BE OBTAINED FROM YOUR NEAREST GENERAL ELECTRIC REPRESENTATIVE.

SECTION B

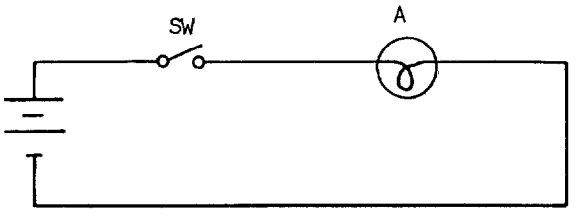
SEQUENTIAL CONTROL

B-1 LOGIC INTERPRETATION

A BRIEF DISCUSSION OF LOGIC SIGNALS IS PRESENTED TO AID IN UNDERSTANDING THE SEQUENTIAL CIRCUITRY. A COMPARISON IS MADE BETWEEN RELAY LOGIC AND SOLID STATE LOGIC TO HELP THOSE FAMILIAR WITH RELAY LOGIC, BUT NOT FAMILIAR WITH SOLID STATE LOGIC, TO READ SOLID STATE LOGIC.

A LOGIC SIGNAL HAS ONLY TWO POSSIBLE STATES. THESE STATES HAVE VARIOUS NAMES IN DIFFERENT LITERATURE. IN THIS INSTRUCTION BOOK THE TWO STATES WILL BE DEFINED AS "ONE" ("1") AND "ZERO" ("0"). A "1" WILL BE USED TO INDICATE THAT SOME EVENT HAS OCCURRED OR IS OCCURRING. A "0" WILL INDICATE THAT SOME EVENT HAS NOT OCCURRED OR IS NOT OCCURRING. IN THE MKII SYSTEM A "1" STATE IS USUALLY REPRESENTED BY +5 VOLTS AND A "0" STATE IS USUALLY REPRESENTED BY ZERO VOLTS.

IF THE STATE OF SOME EVENT IS DESIGNATED BY USING THE VARIABLE A, THEN WHEN A EQUALS "1" THE EVENT IS OCCURRING OR HAS OCCURRED AND WHEN A EQUALS "0" THE EVENT IS NOT OCCURRING OR HAS NOT OCCURRED. FOR EXAMPLE, IN THE CIRCUIT BELOW, THE EVENT IS DEFINED AS THE ILLUMINATION OF THE LIGHT, AND VARIABLE A IS USED TO REPRESENT THE STATE OF THE EVENT. WHEN THE SWITCH IS CLOSED, A IS EQUAL TO "1" (LAMP ILLUMINATED) AND WHEN THE SWITCH IS OPEN, A IS EQUAL TO "0" (LAMP IS OUT).



THE MATHEMATICS OF LOGIC SIGNALS IS REFERRED TO AS BOOLEAN ALGEBRA. A VERY BRIEF DISCUSSION OF BOOLEAN ALGEBRA IS GIVEN SUBSEQUENTLY. FOR EACH BOOLEAN OPERATION, BOTH RELAY LOGIC AND DIRECTO-MATIC *II LOGIC TO PERFORM THE OPERATION IS GIVEN.

THERE ARE THREE BASIC OPERATIONS ASSOCIATED WITH BOOLEAN ALGEBRA. THESE ARE:

(1) LOGIC ADDITION ("OR" FUNCTION): LOGIC ADDITION IS DESIGNATED BY THE SYMBOL "+". THE ADDITION OPERATION IS DEFINED AS:

$$1 + 1 = 1$$

$$1 + 0 = 1$$

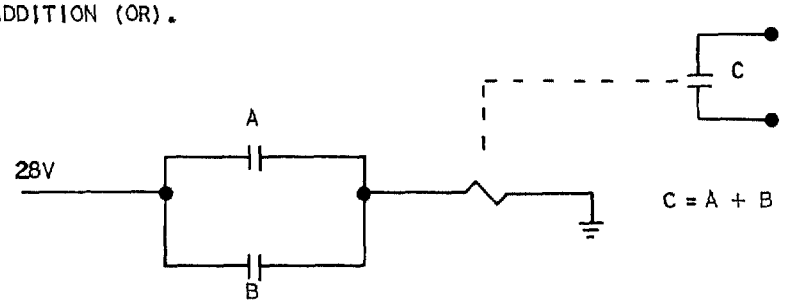
$$0 + 1 = 1$$

$$0 + 0 = 0$$

*TRADEMARK OF GENERAL ELECTRIC COMPANY.

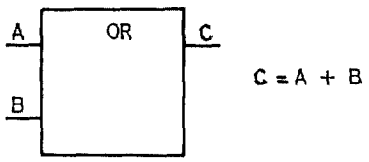
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SEQUENTIAL
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(A) RELAY LOGIC FOR ADDITION (OR).



RELAY C WILL BE PICKED UP IF EITHER CONTACT CONTACT B IS CLOSED.

(B) DIRECTO-MATIC II SOLID STATE LOGIC FOR ADDITION (OR).



OUTPUT C WILL BE "1" (5+ VOLTS) IF EITHER INPUT A OR B IS A "1" (+5 VOLTS).

(2) LOGIC MULTIPLICATION ("AND" FUNCTION): LOGIC MULTIPLICATION IS DESIGNATED BY THE SYMBOL "●". THE MULTIPLICATION OPERATION IS DEFINED AS

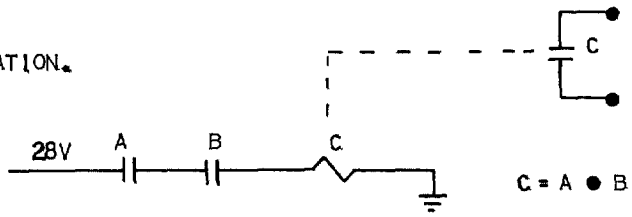
$1 \bullet 1 = 1$

$1 \bullet 0 = 0$

$0 \bullet 1 = 0$

$0 \bullet 0 = 0$

(A) RELAY IMPLEMENTATION.

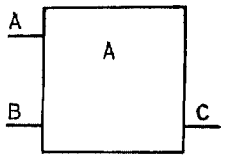


IF A AND B ARE MADE, RELAY C WILL BE PICKED UP.

NOTE: THE SYMBOL " ● " IS OFTEN OMITTED, I.E. A● B = AB.

REV. 1	REV 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SEQUENTIAL
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(B) DIRECTO-MATIC II LOGIC IMPLEMENTATION.



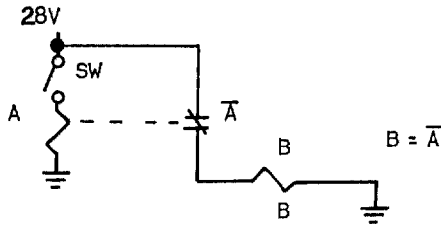
C = A ● B

IF A = "1" AND B = "1" THEN C WILL BE "1". FOR ANY OTHER COMBINATION OF A AND B, C IS "0".

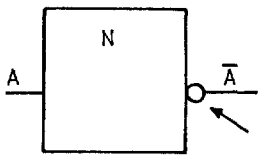
(3) LOGIC NEGATION ("NOT" FUNCTION): LOGIC NEGATION (ALSO CALLED INVERSE OR COMPLEMENT) IS DESIGNATED BY PLACING A BAR ABOVE THE VARIABLE. (FOR EXAMPLE: IF A IS THE VARIABLE, THEN $\overline{A}$ IS THE INVERSE OF A.) THE INVERSE OF A IS DEFINED AS FOLLOWS:

WHEN A = "1", $\overline{A}$ = "0"
WHEN A = "0", $\overline{A}$ = "1"

(A) RELAY IMPLEMENTATION



(B) DIRECTO-MATIC II IMPLEMENTATION.



LOGIC OPERATIONS CAN BE COMBINED TO GIVE LOGIC EQUATIONS SUCH AS:

A = (C ● B) + D

NOTE: THE PRIORITY OF THE OPERATIONS ARE ●, +, NOT. THEREFORE, IF NO PARENTHESIS ARE INCLUDED, ALL THE " ● " OPERATIONS ARE COMPLETED FIRST; THEN THE "+" AND THE NEGATION IS LAST. EXAMPLE: A = C + B ● D IS UNDERSTOOD TO BE A = C + (B ● D).

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		SEQUENTIAL
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LOGIC EQUATIONS CONTAINING TERMS SUCH AS $A = \overline{B \bullet D}$ CAN BE SIMPLIFIED USING THE FOLLOWING TECHNIQUE. TO SIMPLIFY AN EXPRESSION SUCH AS $\overline{B \bullet D}$, SUBSTITUTE "+" FOR "$\bullet$" AND COMPLEMENT EACH INDIVIDUAL VARIABLE. FOR EXAMPLE, THE EXPRESSION $\overline{B \bullet D}$ SIMPLIFIES TO $\overline{B} + \overline{D}$. LIKEWISE $\overline{B + D}$ SIMPLIFIES TO $\overline{B} \bullet \overline{D}$. THIS PROCEDURE CAN BE EXTENDED TO MORE COMPLEX EXPRESSIONS. THE FOLLOWING EXAMPLE ILLUSTRATES THE TECHNIQUE USED.

EXAMPLE:

$$E = \overline{\overline{A \bullet B} \bullet (C + D)}$$

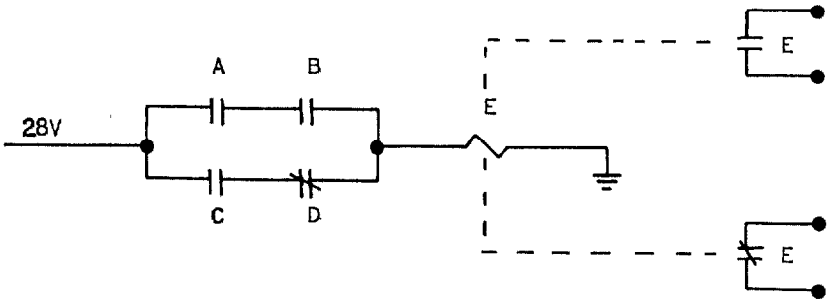
(1) $E = \overline{\overline{A \bullet B}} + \overline{(C + D)}$

(2) $E = A \bullet B + \overline{(C + D)}$

(3) $E = A \bullet B + (\overline{C} \bullet \overline{D})$

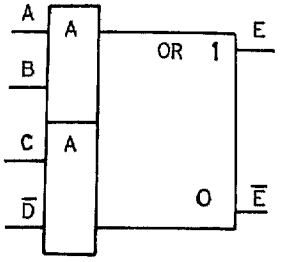
THE IMPLEMENTATION OF LOGIC EQUATIONS USING BOTH RELAY AND DIRECTO-MATIC II LOGIC IS ILLUSTRATED BELOW.

(A) RELAY:



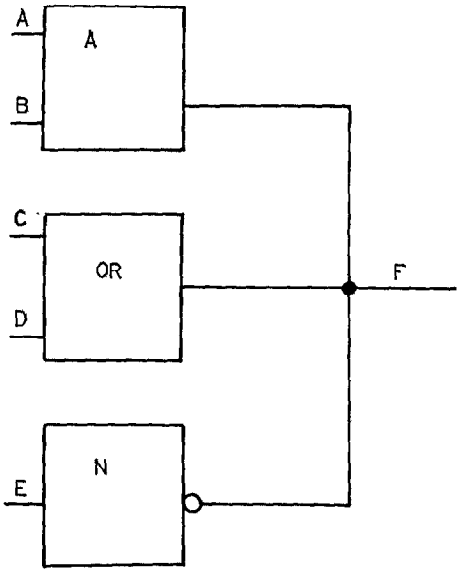
$E = (A \bullet B) + (C \bullet \overline{D})$

(B) DIRECTO-MATIC II



$E = A \bullet B + C \bullet \overline{D}$

THE 1 AND 0 LOCATED INSIDE THE SYMBOL ADJACENT TO OUTPUTS E AND $\overline{E}$ RESPECTIVELY ARE OFTEN USED TO DESIGNATE THE NORMAL AND INVERSE OUTPUTS. THE CIRCUIT IS SAID TO BE IN THE TRUE STATE WHEN THE OUTPUT ADJACENT TO 1 IS A LOGIC ONE. IT IS SAID TO BE IN THE FALSE STATE WHEN THE OUTPUT ADJACENT TO 0 IS A LOGIC ONE. IT SHOULD BE NOTED HERE THAT IN THE CASE WHERE THE OUTPUTS OF MORE THAN ONE LOGIC ELEMENT ARE TIED TOGETHER, A "0" WILL DOMINATE OVER A "1". THE FOLLOWING CIRCUIT ILLUSTRATES THIS.



F IS EQUAL TO "0" IF ANY OF THE FOLLOWING CONDITIONS ARE SATISFIED.

(1) A + B = 0

(2) C ● D = 0

(3) E = 1

PRECAUTION: A "1" CAN BE FORCED TO A "0" BY JUMPERING THE SIGNAL TO COMMON; HOWEVER, TRYING TO FORCE A "0" TO A "1" BY JUMPERING THE SIGNAL TO +5.3 VOLTS COULD RESULT IN DAMAGE TO THE CIRCUIT.

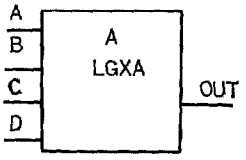
B-2 DIRECTO-MATIC II LOGIC CARDS AND SYMBOLS*

*THIS SECTION IS MERELY TO FAMILIARIZE THE READER WITH DIRECTO-MATIC II LOGIC CARDS. FOR SPECIFIC INFORMATION CONSULT THE DIRECTO-MATIC II CARD APPLICATION DATA BOOK.

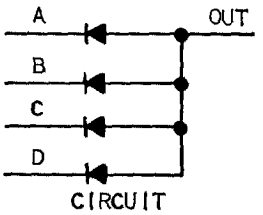


THE PRESENCE OF THE "●" AT THE INPUT TERMINAL OF AN ELEMENT DENOTES THE ABILITY TO ACCEPT EITHER NORMAL LOGIC OUTPUTS OR THE OUTPUT OF A LGXA.

IC3600LGXA - "AND" GATE EXPANDER



9 PER CARD

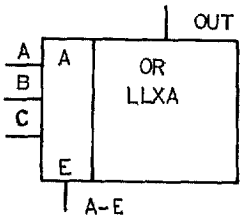


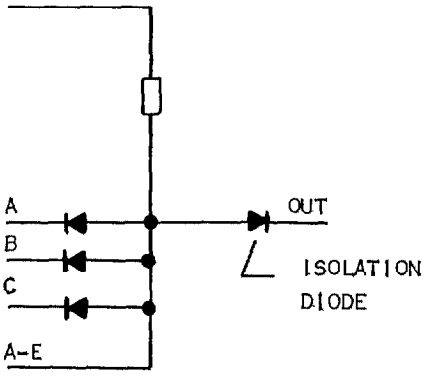
CIRCUIT

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		SEQUENTIAL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 10 SH. NO 9

THE LGXA PROVIDES A CLUSTER OF DIODES FOR EXPANDING THE "AND" INPUTS OF MANY DIRECTO-MATIC II LOGIC CARDS. AN ILLUSTRATION OF HOW THE LGXA IS USED IS GIVEN IN THE LCDA DISCUSSION. WHEN THE OUT TERMINAL OF THE LGXA IS CONNECTED TO THE A-E INPUT OF ANOTHER LOGIC ELEMENT, A "0" ON ANY INPUT WILL PULL OUT TO "0". A "1" ON ALL INPUTS WILL ALLOW "OUT" TO GO TO A "1".

IC3600LLXA - "OR" LOGIC EXPANDER



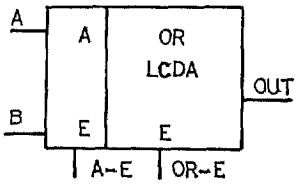


9 CIRCUITS PER CARD

THE LLXA PROVIDES CIRCUITRY FOR INPUT EXPANSION OF LOGIC ELEMENTS HAVING AN "OR-E" INPUT. IF ALL INPUTS ARE "1", THE ISOLATION DIODE WILL BE FORWARD BIASED AND PULL "OR-E" INPUT OF CONNECTED LOGIC ELEMENT TO A "1" WHICH MAKES THE NORMAL OUTPUT OF THAT ELEMENT A "1". IF ANY INPUT IS A "0", THE ISOLATION DIODE WILL BE REVERSED BIASED AND ALLOW THE OUTPUT OF THE CONNECTED ELEMENT TO BE CONTROLLED BY ITS NORMAL INPUTS. AN "A-E" INPUT IS PROVIDED FOR EXPANDING THE "AND" INPUTS USING A LGXA. THE USE OF THE LLXA WILL BE ILLUSTRATED IN THE LCDA DISCUSSION.

NOTE: FOR MOST DIRECTO-MATIC II CARDS, AN UNUSED INPUT LEFT OPEN IS RECOGNIZED AS A "1".

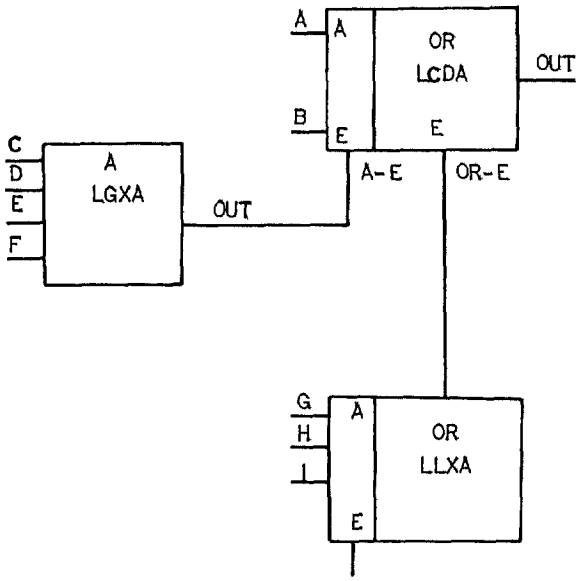
IC3600LCDA - CLOCK DRIVER



9 CIRCUITS PER CARD

THE LCDA CLOCK DRIVER CONTAINS A POWER AMPLIFIER WHICH WILL DRIVE 32 "L" LOGIC LOADS. IN ADDITION TO THE TWO NORMAL "AND" INPUTS, "A-E" AND "OR-E" INPUTS ARE PROVIDED. IF NO CONNECTION IS MADE TO "A-E" OR "OR-E," THE CIRCUIT PERFORMS AS A NORMAL TWO INPUT "AND" GATE WITH POWER AMPLIFIER. AN ILLUSTRATION OF HOW THE "A-E" AND "OR-E" INPUTS COULD BE USED IS ILLUSTRATED AND FOLLOWED BY ITS TRUTH TABLE.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM: SEQUENTIAL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 11 SH. NO. 10



$(A \bullet B \bullet C \bullet D \bullet E \bullet F) + (G \bullet H \bullet I) = "1"$

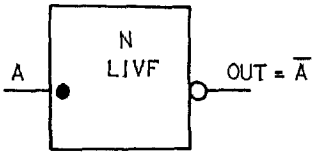
TRUTH TABLE*

A	B	C	D	E	F	G	H	I	OUT
0	0	0	1	1	1	0	1	1	0
ø	ø	ø	1	1	1	1	1	1	1
0	0	1	1	1	1	0	1	1	0
0	1	1	1	1	1	0	1	1	0
1	0	0	1	1	1	0	1	1	0
1	0	1	1	1	1	0	1	1	0
1	1	0	1	1	1	0	1	1	0
1	1	1	1	1	1	0	1	1	1
1	1	1	1	1	1	1	1	1	1

*IN ORDER TO SIMPLIFY TABLE, ASSUME INPUTS H,I,D,E AND F ARE ALWAYS "1". I.E. ONLY INPUTS A,B,C AND G CAN CHANGE STATES.
ø EQUALS "DON'T CARE" IF A "1" OR "0".

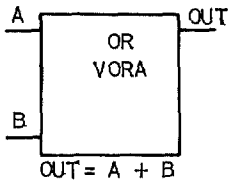
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SEQUENTIAL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH. 12 SH. NO. 11

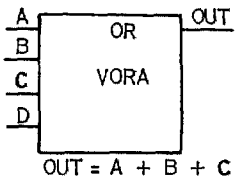
IC3600 LIVF - INVERTER

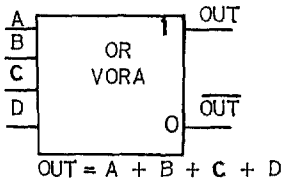


THE LIVF PROVIDES THE FUNCTION OF LOGIC INVERSION. A "1" IN WILL GIVE A "0" OUT AND A "0" IN WILL GIVE A "1" OUT.

IC3600 VORA-LOGIC OR

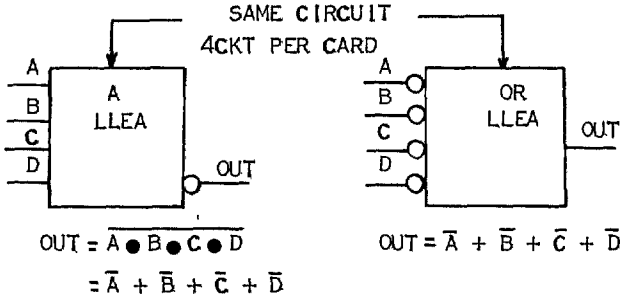


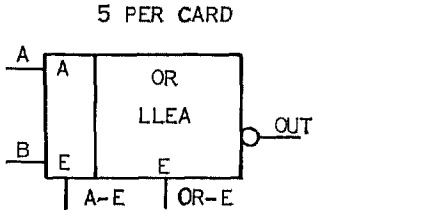




THIS CARD PROVIDES A TRUE "OR" FUNCTION. A "1" ON ANY INPUT WILL GIVE "1" ON "OUT". TWO CIRCUITS PER CARD PROVIDE AN INVERSE OUTPUT, "$\overline{OUT}$". FOR THESE CIRCUITS, A "1" ON ANY INPUT WILL GIVE A "1" ON "OUT" AND A "0" ON "$\overline{OUT}$". IF ALL INPUTS ARE "0", "OUT" IS "0" AND "$\overline{OUT}$" IS "1".

IC3600LLEA - LOGIC ELEMENT

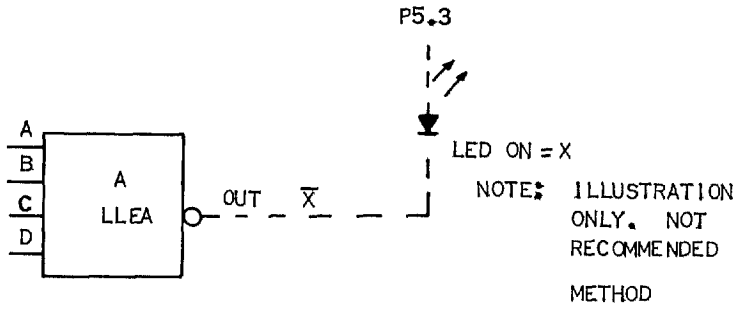




REV. 1	REV. 4	REV 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		SEQUENTIAL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 13 SH. NO. 12

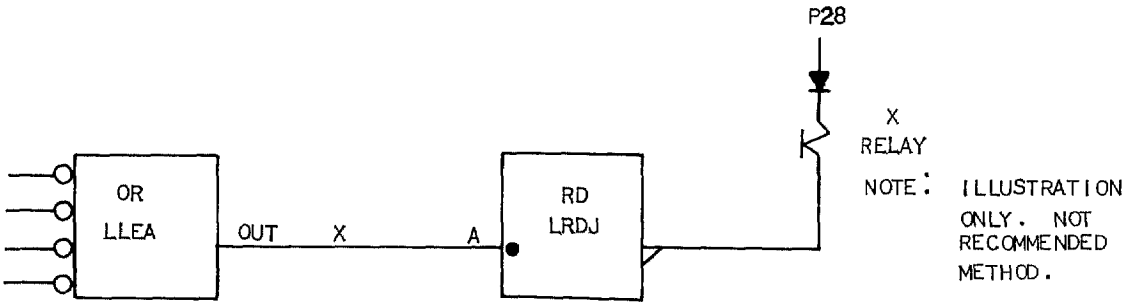
THE LLEA PERFORMS THE NAND (NOT AND) FUNCTION. FIVE CIRCUITS PER CARD HAVE "A-E" AND "OR-E" EXPANDER INPUTS FOR CONNECTION TO LGXA AND LLXA RESPECTIVELY. THE REMAINING 4 CIRCUITS PER CARD ARE TYPICAL NAND GATES. EITHER OF THE TWO LEFT MOST SYMBOLS ABOVE MAY BE USED FOR THESE CIRCUITS. THE SYMBOL USED IS DETERMINED BY THE LOGIC LEVEL DESIRED AT "OUT" TO REPRESENT THE ACTIVE CONDITION OF THE INPUTS. THE DESIRED LOGIC LEVEL IS DEFINED BY THE OUTPUT SIGNAL NOMENCLATURE. THE FOLLOWING EXAMPLES WILL ILLUSTRATE THIS.

EXAMPLE 1:



THE ACTIVE CONDITION IS WHEN THE LED IS ON. IF A "0" IS DESIRED TO REPRESENT THE ACTIVE CONDITION, THE ABOVE SYMBOL IS USED AND THE OUTPUT IS LABELED "$\overline{X}$." THE DESIRED LOGIC LEVEL IS DICTATED BY CIRCUIT THAT IS BEING DRIVEN BY THE LLEA.

EXAMPLE 2:



RELAY PICKED UP WHEN INPUT TO LRDJ = "1"

THE ACTIVE CONDITION IS RELAY X ENERGIZED. IF A "1" IS DESIRED AT "OUT" TO REPRESENT ACTIVE CONDITION AND PICK UP RELAY, THIS SYMBOL IS USED AND OUTPUT OF LLEA LAVELED "X".

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		SEQUENTIAL
REV. 3	MADE BY			I.C. NO	SALEM, VA. U.S.A.	CONT. ON SH. 14 SH. NO. 13

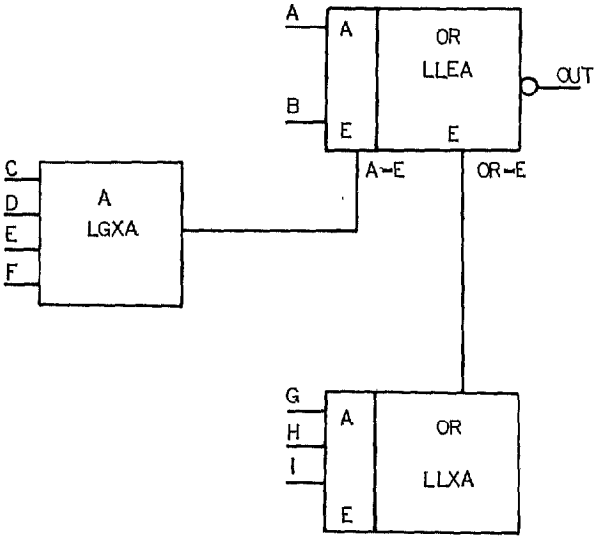
LLEA TRUTH TABLE

FOR CIRCUITS WITHOUT EXPANDER INPUTS

A	B	C	D	OUT
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		SEQUENTIAL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 15 SH. NO. 14

THE FOLLOWING CIRCUIT ILLUSTRATES THE USE OF THE EXPANDER INPUTS. THE USE OF BOTH "A-E" AND "OR-E" ARE ILLUSTRATED; HOWEVER, EITHER MAY BE USED SEPARATELY.



TRUTH TABLE FOR EXPANDER CIRCUIT

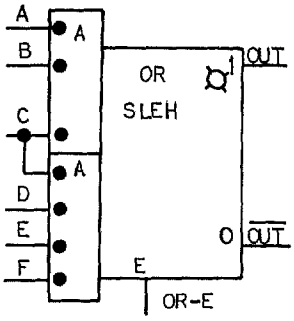
(ASSUME D, E, F, H, I ARE "1" AT ALL TIMES)

A	B	C	G	OUT
0	0	0	0	1
Ø	Ø	Ø	1	0
0	0	1	0	1
0	Ø	0	0	1
0	1	1	0	1
Ø	0	0	0	1
Ø	0	1	0	1
1	1	0	0	1
1	1	1	Ø	0

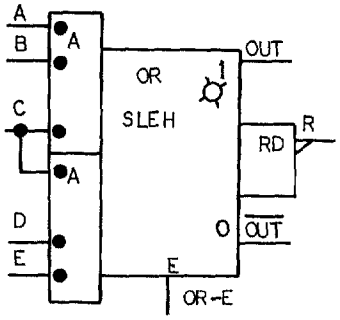
Ø EQUALS "DON'T CARE" IF "1" OR "0".

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SEQUENTIAL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C NO.	SALEM, VA. U.S.A.	CONT. ON SH 16 SH. NO. 15

IC3600SLEH - LOGIC ELEMENT



2 CIRCUITS PER CARD



3 CIRCUITS PER CARD

$OUT = A \bullet B \bullet C + D \bullet E \bullet F \bullet C$

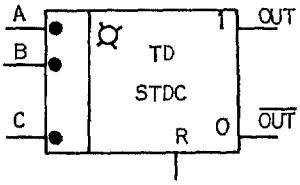
$\overline{OUT} = (\bar{A} + \bar{B} + \bar{C}) \bullet (\bar{D} + \bar{E} + \bar{F} + \bar{C})$

$OUT = A \bullet B \bullet C + D \bullet E \bullet C$

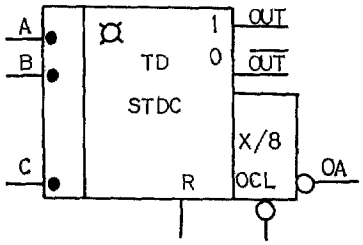
$\overline{OUT} = (\bar{A} + \bar{B} + \bar{C}) \bullet (\bar{D} + \bar{E} + \bar{C})$

THE BASIC SLEH CARD CONSISTS OF AN "OR" LOGIC ELEMENT GATED BY TWO "AND" GATES. IF EITHER "AND" GATE HAS ALL "1" INPUTS, "OUT" WILL BE A "1". AN "OR-E" EXPANDER INPUT IS PROVIDED FOR USE WITH THE LLXA. THREE CIRCUITS PER CARD PROVIDE A RELAY DRIVER OUTPUT. THE RELAY DRIVER PROVIDES A CLOSURE TO COMMON WHEN "OUT" IS A "1".

IC3600STDC - TIME DELAY



3 CIRCUITS PER CARD



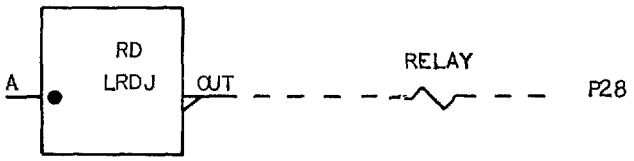
1 CIRCUIT PER CARD

THIS CARD PROVIDES A TIME DELAY FUNCTION FOR RELATIVELY LONG PERIODS OF ONE SECOND TO FIVE MINUTES. THE TIMING IS INITIATED BY ALL INPUTS BEING A "1". IF ALL INPUTS REMAIN A "1", "OUT" WILL BECOME A "1" AND "$\overline{OUT}$" A "0" AFTER THE TIME DELAY SET BY ADJUSTMENT POTENTIOMETER. "OUT" WILL GO FROM A "1" TO A "0" IMMEDIATELY, IF ANY INPUT GOES TO A "0". THE LED WILL BE ON FROM THE INSTANT TIMING IS INITIATED UNTIL "OUT" BECOMES A "1".

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SEQUENTIAL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 17 SH. NO. 16

THE TIME DELAY PROVIDED IS DIVIDED INTO TWO RANGES TO GIVE MORE ADJUSTMENT RESOLUTION. THE SHORT RANGE (PIN R OPEN) IS ADJUSTABLE FROM 1 TO 25 SECONDS. THE LONG RANGE (PIN R TO N12) IS ADJUSTABLE FROM APPROXIMATELY 20 SECONDS TO 5 MINUTES. THERE ARE FOUR CIRCUITS PER CARD. ONE CIRCUIT PROVIDES AN EIGHT COUNT COUNTER FOR ADDITIONAL TIME DELAY. BY CONNECTING $\overline{OUT}$ OF THIS CIRCUIT TO ONE OF THE INPUTS, AND QA TO ANOTHER, THE TIMER WILL CYCLE EIGHT TIMES BEFORE STOPPING WITH QA EQUAL "0". THIS WILL PROVIDE UP TO 40 MINUTES.

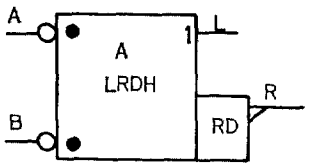
IC3600 LRDJ - RELAY DRIVER



16 CIRCUITS PER CARD

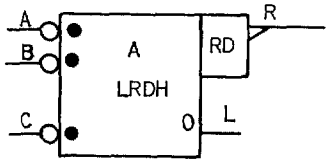
THIS CIRCUIT (ABOVE) PROVIDES A CLOSURE TO COMMON IN RESPONSE TO A ONE INPUT, AN OPEN CIRCUIT IN RESPONSE TO A ZERO INPUT. THE CIRCUIT IS INTENDED TO DRIVE A LOAD SUCH AS A RELAY CONNECTED BETWEEN "OUT" AND POSITIVE SUPPLY VOLTAGE NOT GREATER THAN P28.

IC3600 LRDH - RELAY DRIVER



$L = \bar{A} \bullet \bar{B}$

5 PER CARD



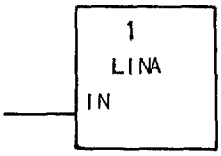
$L = \overline{\bar{A} \bullet \bar{B} \bullet \bar{C}}$

$L = A + B + C = \text{"1"}$

2 PER CARD

THESE CIRCUITS (ABOVE) PROVIDE A CLOSURE TO COMMON FOR A LOAD BETWEEN P28 AND THE "R" OUTPUT IN RESPONSE TO PROPER INPUT LOGIC. IN ADDITION, THEY ALSO PROVIDE A LOGIC OUTPUT SIGNAL AT "L" IN RESPONSE TO THE INPUT LOGIC. CIRCUITS 1 THRU 5 REQUIRE BOTH INPUTS TO BE A LOGIC "0" TO PROVIDE CLOSURE TO COMMON AND TO GENERATE A LOGIC "1" AT THE OUTPUT. CIRCUITS 6 AND 7 ARE SIMILAR TO 1 THRU 5 EXCEPT THAT A 3 INPUT "AND" GATE IS PROVIDED INSTEAD OF A 2-INPUT GATE AND THE LOGIC OUTPUT IS A "0" WHEN CLOSURE TO COMMON IS PROVIDED.

IC3600 LINA - LIGHT INDICATOR

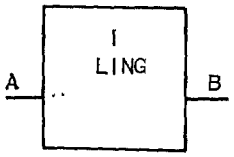


16 CIRCUITS PER CARD

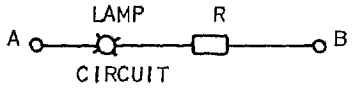
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SEQUENTIAL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 18 SH. NO. 17

THE LINA INCLUDES A LIGHT AMPLIFIER AND INDICATING LIGHT. THE LIGHT WILL BE ILLUMINATED WHEN THE INPUT IS A "1". INPUTS FOR UNUSED CIRCUITS MUST BE TIED TO COMMON TO PREVENT LIGHT STAYING ON.

IC3600 LING - INDICATOR

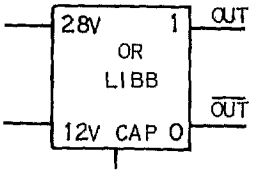


16 CIRCUITS PER CARD



THE LING PROVIDES INDICATING LAMPS TO BE DRIVEN BY DISPLAY DRIVERS, RELAY DRIVERS, OR CONTACT CLOSING. EACH CIRCUIT IS COMPLETELY INDEPENDENT. THIS CARD CANNOT BE OPERATED BY A NORMAL LOGIC SIGNAL.

IC3600 LIBB - INPUT BUFFER

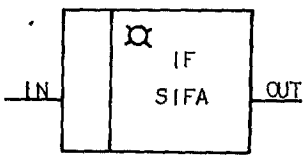


9 CIRCUITS PER CARD

THE LIBB CONVERTS RELAY CONTACT INPUTS TO LOGIC LEVELS. INPUTS ARE PROVIDED FOR CONTACT CLOSURE TO EITHER 28 VOLTS OR 12 VOLTS. WHEN CONTACT TO SUPPLY IS CLOSED, "OUT" IS A "1" AND "$\overline{\text{OUT}}$" IS A "0". THE "28V" INPUT SHOULD NOT BE USED OFF PAGE. WHEN 28 VOLTS IS CONNECTED TO AN OFF PAGE CONTACT, THE SIGNAL MUST BE CONDITIONED BY AN INPUT FILTER CIRCUIT (SIFA).

NOTE: LIBA CARD WORKS THE SAME AS LIBB BUT HAS DIFFERENT RATING AND TIMING. DO NOT INTERCHANGE CARDS.

IC3606 SIFA - NON-ISOLATED INPUT FILTER

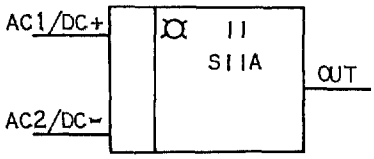


9 CIRCUITS PER CARD

THE SIFA IS DESIGNED TO RECEIVE A 28 VOLT SIGNAL, FILTER THE SIGNAL, AND PROVIDE DRIVE TO THE 12V INPUT OF THE LIBB CARD. AN LED IS "ON" WHEN INPUT IS AT 28 VOLTS.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		SEQUENTIAL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 19 SH NO. 18

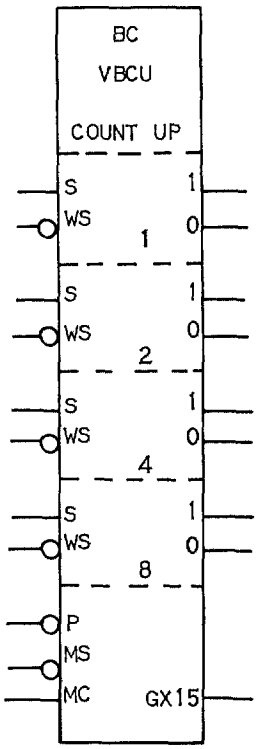
IC3606 SIIA - ISOLATED INPUT FILTER



9 CIRCUITS PER CARD

THE SIIA PROVIDES ISOLATION AND FILTERING OF HIGH LEVEL (125 VOLTS DC OR 115 VOLTS AC) SIGNALS. THE OUTPUT IS DESIGNED TO DRIVE THE 12 VOLT INPUT OF THE LIBB CARD.

IC3600 VBCU-BINARY COUNTER



2 CIRCUITS PER CARD

THE VBCU CARD CONTAINS TWO 4 BIT COUNTERS. THE COUNTER WILL COUNT UP ONE COUNT EACH TIME THE SIGNAL ON INPUT P CHANGES FROM A "0" TO A "1". EACH BIT HAS A "NORMAL" AND "INVERSE" OUTPUT. A LOGIC "1" ON MC WILL CLEAR ALL BITS. A LOGIC "0" ON MS WILL SET ALL BITS WHOSE S INPUT IS A "1". INPUT WS IS PROVIDED FOR WIRED OR EXPANSION OF SET OPERATION. S MUST BE A "0", OR MS MUST BE A "1" FOR THIS FUNCTION. OUTPUT GX15 WILL BE A "1", IF ALL FOUR BITS ARE SET. THE TWO 4 BIT COUNTERS MAY BE OPERATED AS AN 8 BIT COUNTER BY CONNECTING THE "0" OUTPUT OF THE LOWER ORDER BIT 8 COUNT TO THE P INPUT OF THE SECOND COUNTER.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		SEQUENTIAL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 20 SH. NO. 19

B-3 ANNUNCIATOR

THE ANNUNCIATOR ON THE DIRECTO-MATIC II MS1002 TURBINE CONTROL PANEL PROVIDES THE CIRCUITRY TO ANNUNCIATE 30 ALARM CONDITIONS. THE ALARMS ARE LATCHING AND WILL REMAIN, EVEN IF THE ALARM CONDITION IS MOMENTARY, UNTIL CLEARED BY A PUSHBUTTON ON THE PANEL. AN ALARM MAY BE ANNUNCIATED BY THE SOUNDING OF A BELL AND THE FLASHING OF THE ALARM'S ASSOCIATED LIGHTS.

THE ALARM INDICATORS ARE REFERRED TO AS DROPS. SPECIAL FEATURES WHICH WILL BE DISCUSSED SUBSEQUENTLY ARE PROVIDED TO HANDLE MULTIPLE ALARMS. A BLOCK DIAGRAM IS GIVEN IN FIG. B-2. IT CONSISTS OF ONE ANNUNCIATOR MASTER CARD (VANB) AND 8 ANNUNCIATOR CARDS (VANA). THE STANDARD SYSTEM CONSISTS OF 30 DROPS.

THERE ARE THREE OPERATOR PUSHBUTTONS ASSOCIATED WITH THE ANNUNCIATOR. (1) THE "LAMP TEST" PUSHBUTTON PROVIDES A MEANS TO CHECK FOR BURNED OUT BULBS AND FAULTY LAMP DRIVERS. (2) THE "ALARM RESET" PUSHBUTTON WILL SILENCE THE ALARM BELL AND CAUSE THE FLASHING LIGHT TO BECOME STEADY ON. IT ALSO ENABLES AN ADDITIONAL ALARM TO COME ON FLASHING. (3) THE "LAMP RESET" PUSHBUTTON WILL CLEAR THE LIGHT IF THE ALARM CONDITION HAS BEEN REMOVED.

THE FOLLOWING FLOW CHART IN FIG. B-1 IS INCLUDED TO ILLUSTRATE SOME TYPICAL SEQUENCES THAT MAY OCCUR IN THE CASE OF MULTIPLE ALARM CONDITIONS. ALARMS MUST OCCUR AT LEAST 36 MILLISECONDS APART FOR THE ANNUNCIATOR TO DETERMINE WHICH OCCURRED FIRST. IF THERE IS LESS THAN 36 MILLISECONDS BETWEEN ALARMS, BOTH WILL COME ON FLASHING.

A B C D E F G H J K L

01

03

FLOW CHART

05

FIRST ALARM CONDITION OCCURS

07

CORRESPONDING INDICATOR LAMP COMES ON FLASHING AND BELL SOUNDS

09

DEPRESSING LAMP RESET HAS NO EFFECT

11

13

ALARM RESET IS DEPRESSED	SECOND ALARM CONDITION OCCURS
BELL IS SILENCED AND FLASHING LIGHT BECOMES STEADY	CORRESPONDING LIGHT COMES ON STEADY
SECOND ALARM CONDITION OCCURS	ALARM RESET IS DEPRESSED
CORRESPONDING LAMP COMES ON FLASHING AND BELL SOUNDS	FLASHING LIGHT BECOMES STEADY ON
FIRST FAULT CONDITION DISAPPEARS LAMP REMAINS ON	LAMP RESET IS DEPRESSED NO EFFECT
LAMP RESET IS DEPRESSED	BOTH ALARM CONDITIONS ARE REMOVED
LAMP FOR FIRST FAULT GOES OUT BUT NO EFFECT ON SECOND FAULT	BOTH LAMPS REMAIN STEADY ON
ALARM RESET IS DEPRESSED	LAMP RESET IS DEPRESSED
BELL IS SILENCED AND FLASHING LIGHT BECOMES STEADY ON	BOTH LAMPS GO OUT
SECOND ALARM CONDITION IS REMOVED	
LAMP REMAINS ON	
LAMP RESET IS DEPRESSED	
LAMP GOES OUT	

15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47

49

FIG. B-1

51

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SEQUENTIAL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 22 SH. NO. 21

THE VANB CARD PROVIDES THE CONTROL SIGNALS TO THE VANA CARD. (REFER TO FIG. B-2). IT ALSO CONTAINS THE DRIVER FOR RELAY 30X WHOSE CONTACTS ARE USED TO DRIVE THE ALARM BELL. VANB OUTPUT FL PROVIDES THE FLASHING SIGNAL TO ALL VANA CARDS. IT IS PRESENT AT ALL TIMES.

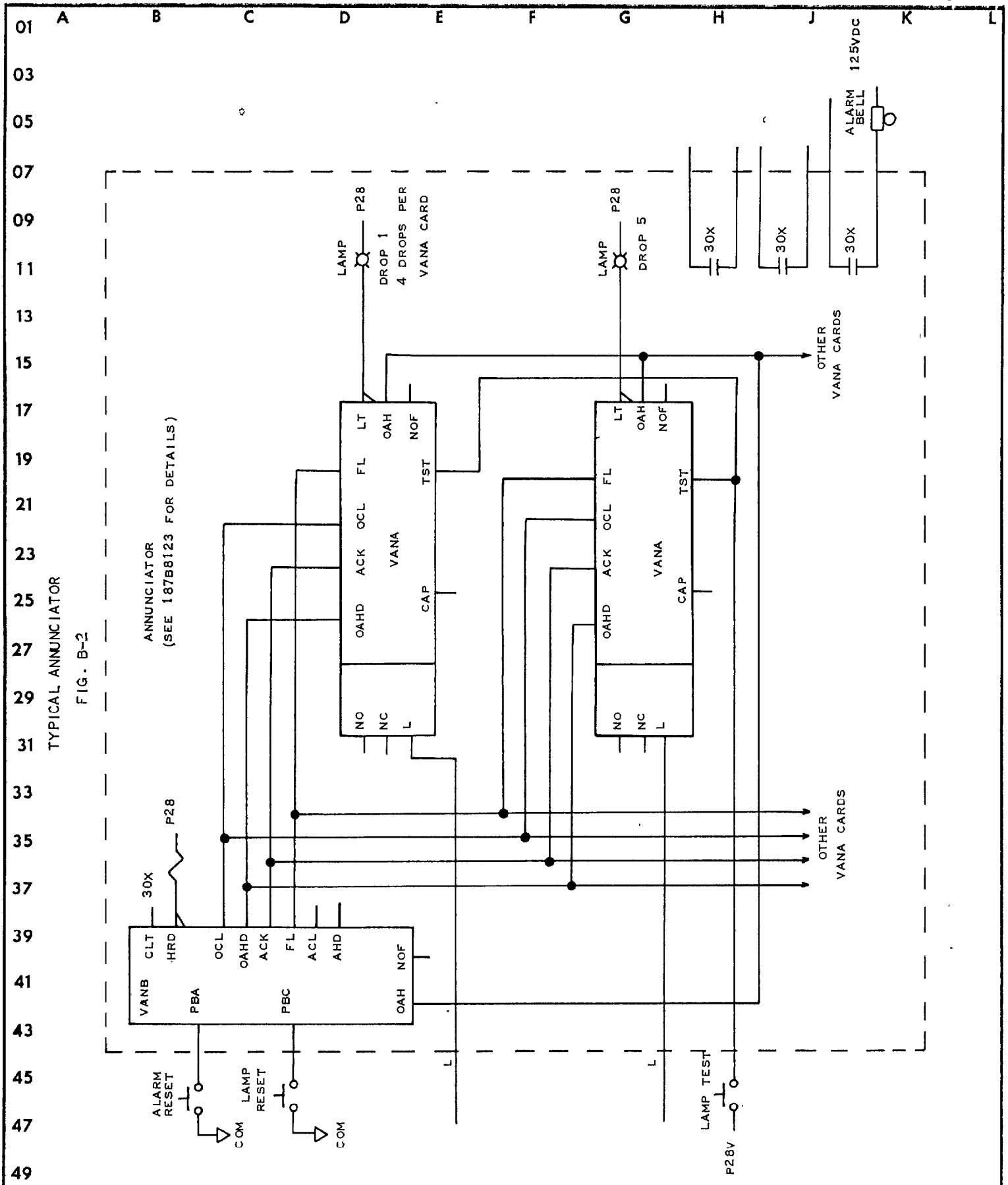
THE SIGNAL FLOW FOR A TYPICAL ALARM CONDITION WILL BE GIVEN TO AID IN EXPLAINING THE ANNUNCIATOR.

ASSUMING THAT NO ALARM CONDITION IS PRESENT, OUTPUT $\overline{AHD}$ OF VANB WILL BE A "1". THIS IS A FLASHING PERMISSIVE TO ALL VANA CARDS. NOW ASSUME THAT A FAULT OCCURS (A "1" ON L). ON THE VANA CARD, THE FLASHING SIGNAL APPEARS ON LT AND THE CORRESPONDING INDICATOR LIGHT FLASHES, AND OUTPUT $\overline{AH}$ BECOMES A "0". THIS "0" GOES TO VANB INPUT $\overline{AH}$ AND PERFORMS TWO FUNCTIONS. IT PLACES A "0" ON HRD AND PICKS UP RELAY 30X WHICH CAUSES THE BELL TO SOUND, AND OUTPUT $\overline{AHD}$ BECOMES A "0" WHICH REMOVES THE FLASHING PERMISSIVE SIGNAL.

THE FIRST FAULT REMAINS FLASHING BUT ANY MORE FAULTS OCCURRING WILL COME ON STEADY. SUPPOSE THAT THE "ALARM RESET" PUSHBUTTON IS NOW DEPRESSED, THE VANB OUTPUT ACK WILL BECOME A "1" WHICH ALSO PERFORMS TWO FUNCTIONS IN THE VANA CARD. IT REMOVES THE FLASHING SIGNAL FROM LT AND REPLACES IT WITH A STEADY "0" WHICH MAKES THE LIGHT REMAIN STEADY ON. IT ALSO CAUSES OUTPUT AH TO BECOME A "1". THIS "1" ON $\overline{AH}$ GOES TO VANB AND SILENCES THE BELL. IT ALSO ALLOWS VANB OUTPUT $\overline{AHD}$ TO BECOME A "1". THIS "1" $\overline{AHD}$ IS A PERMISSIVE TO ALL OTHER VANA CARDS TO FLASH IF AN ADDITIONAL ALARM OCCURS.

NOW, ASSUME THAT NO OTHER ALARM CONDITION OCCURS, AND THE PRESENT FAULT CONDITION IS CLEARED. DEPRESSING PUSHBUTTON "LAMP RESET" PLACES A "0" ON THE $\overline{CL}$ BUS AND THE LT SIGNAL ON VANA BECOMES A "1" WHICH TURNS THE LAMP OFF.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SEQUENTIAL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 23 SH. NO. 22



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		SEQUENTIAL
REV. 3	MADE BY			I.C NO	SALEM, VA. U.S.A.	CONT. ON SH 24 SH NO. 23

B-4 SPEED SENSORS

THE SPEED SENSOR CARDS RECEIVE ANALOG NHP AND NLP SIGNALS FROM THE PR/A CONVERTER ON THE SSZB CARDS, AND FREQUENCY INPUTS FROM MAGNETIC PICK-UPS. THESE SENSOR CARDS PROVIDE LOGIC SIGNALS TO INDICATE THAT THE TURBINE SHAFT, OR OTHER ROTATING DEVICE HAS REACHED A CERTAIN SPEED.

SVSD CARD (SEE FIG. B-3, B-4)

THIS CARD CONTAINS SIX SPEED SENSORS WITH GIVEN (FIXED OR ADJUSTABLE) PICKUP AND DROPOUT CHARACTERISTICS. THESE ARE:

14 HM - H.P. TURBINE MINIMUM SPEED
14 HA - H.P. TURBINE ACCELERATE SPEED
14 HS - H.P. TURBINE OPERATING SPEED
14 LS - L.P. TURBINE OPERATING SPEED
14 SR - STARTING TURBINE ZERO SPEED
14 ST - STARTING TURBINE TOP SPEED

THE FIRST THREE SENSORS OPERATE FROM A COMMON INPUT SIGNAL NHP. THE 14 LS SENSOR OPERATES FROM THE INPUT SIGNAL NLP. 14 SR AND 14 ST OPERATE DIRECTLY FROM A COMMON ELECTROMAGNETIC PICKUP.

THE 14 HM PICKUP AND DROPOUT POINTS ARE PROGRAMMABLE BY A LOGIC SIGNAL ON "4" INPUT. A "1" ON "4" PROGRAMS PICKUP TO 20% OF RATED SPEED WITH DROPOUT AT 18%. A "0" ON "4" REPROGRAMS THE PICKUP POINT TO 2% AND DROPOUT TO 1.8%. RESISTORS CAN BE SWITCHED BETWEEN FS AND M TO REPROGRAM THE PICKUP AND DROPOUT. THE DROPOUT OF 14 HM IS FIXED AT 90% OF PICKUP; NO SEPARATE ADJUSTMENT IS PROVIDED. THE 14 HM SENSOR OUTPUTS ARE ON PINS HM AND $\overline{HM}$

THE 14 HA SENSOR LOOKS AT NHP AND PROVIDES LOGIC OUTPUTS ON HA AND $\overline{HA}$. THE 14 HA PICKUP IS SET AT 40% AND DROPOUT IS AT 30%. THE 14 HS SENSOR LOOKS AT NHP AND PROVIDES LOGIC OUTPUTS ON HS AND $\overline{HS}$. THE 14 HS PICKUP IS SET AT 70% AND DROPOUT IS AT 60%. THE PICKUP POINTS FOR 14 HA AND 14 HS ARE ADJUSTABLE ONLY BY ADDING EXTERNAL RESISTANCES BETWEEN THE NHP INPUT AND THE A AND R INPUTS RESPECTIVELY. THE DROPOUT POINTS ARE DEPENDENT UPON THE PICKUP POINTS AND CANNOT BE ADJUSTED SEPARATELY.

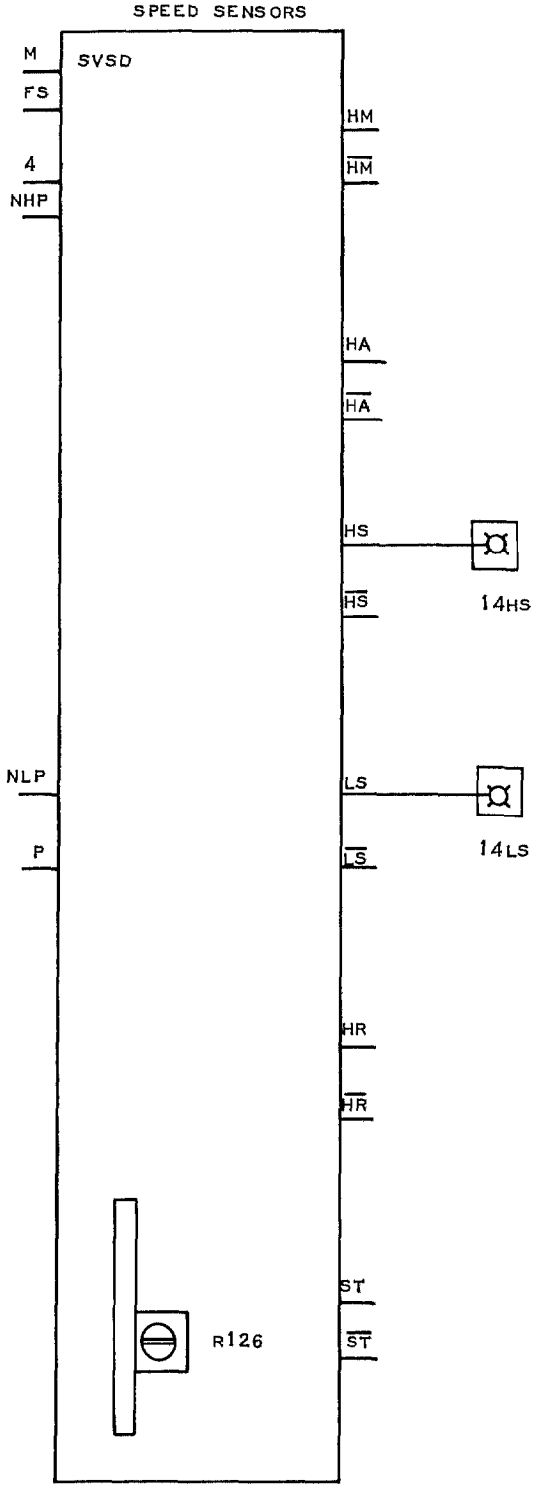
THE 14 LS SENSOR LOOKS AT NLP AND PROVIDES LOGIC OUTPUTS ON LS AND $\overline{LS}$. FOR THE MS1002 THE NLP AND P INPUTS ARE JUMPERED TOGETHER SO THAT 14 LS PICKS UP AT 45%. BY REMOVING THE JUMPER 14 LS WILL PICK UP AT 95%. DROPOUT OCCURS AT 80% OF THE PICKUP VALUE.

THE 14 HR/SR SENSOR RECEIVES A PULSE INPUT PROPORTIONAL TO RPM AND GENERATES AN ANALOG SIGNAL. THIS ANALOG SIGNAL IS COMPARED TO A FIXED REFERENCE AND GIVES A "1" ON HR IF THE ANALOG SIGNAL IS LESS THAN THE REFERENCE. UNLIKE OTHER SENSORS 14 HR/SR IS PICKED UP AT ZERO SPEED AND DROPS OUT AS SPEED IS INCREASED.

THE 14 ST SENSOR ALSO RECEIVES A PULSE INPUT PROPORTIONAL TO RPM AND GENERATES AN ANALOG SIGNAL. THIS SIGNAL IS THEN COMPARED TO AN ADJUSTABLE REFERENCE. TERMINAL ST GOES TO A "1" IF THE SIGNAL LEVEL EXCEEDS THE REFERENCE. THE REFERENCE IS ADJUSTABLE FROM 2000 TO 6000 HZ. THE DROPOUT IS FIXED AT 95% OF THE PICKUP VALUE.

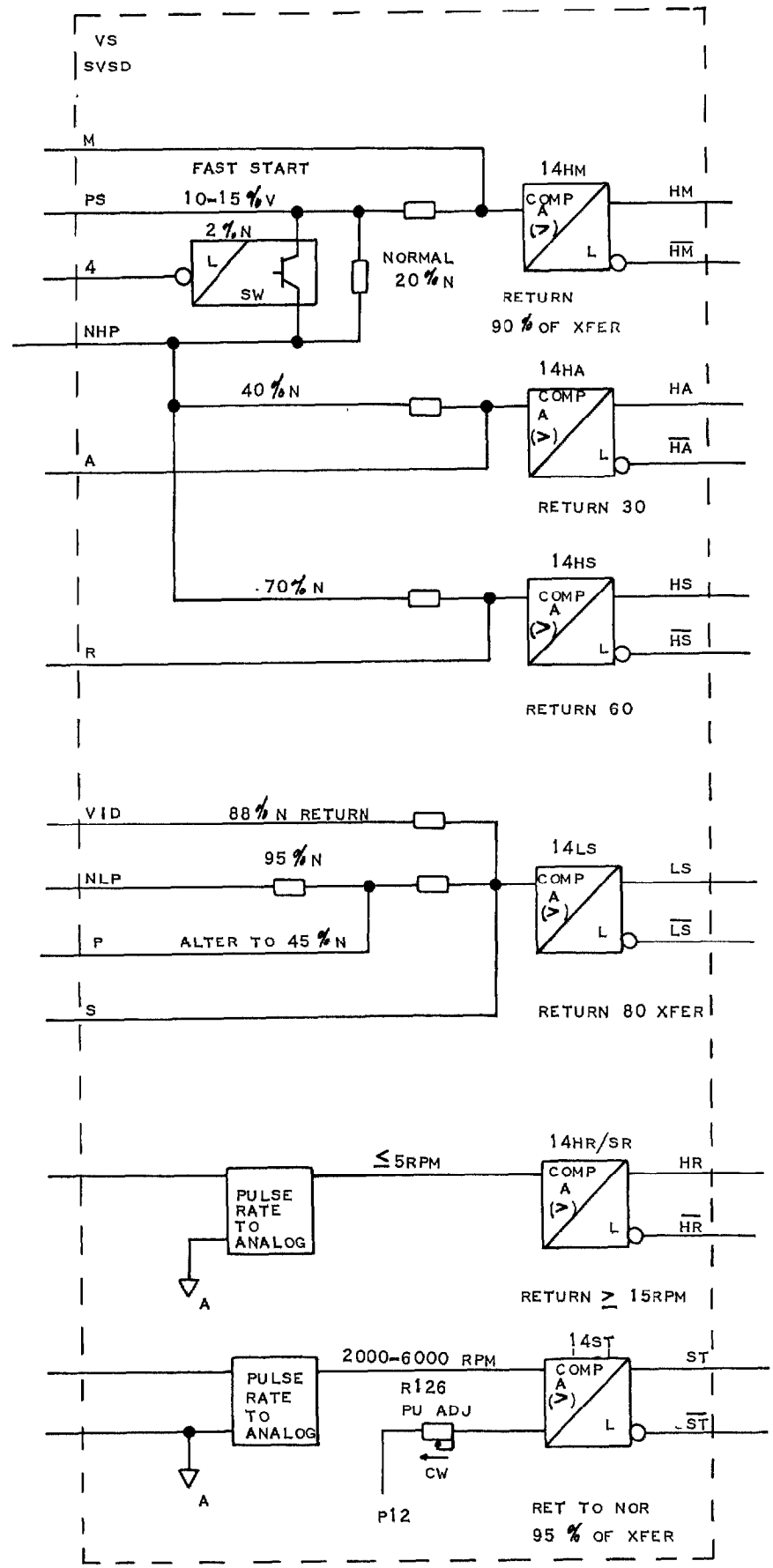
ADJUSTMENT

14 ST PU-R126 2000-6000 RPM



SPEED SENSORS
FIG. B-3

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM SEQUENTIAL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 26 SH. NO. 25



SVSD CARD
FIG. B-4

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM: SEQUENTIAL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 28 SH. NO. 26

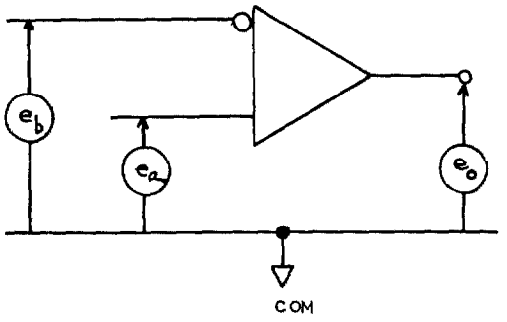
SECTION C

CONTROL

C-1 OPERATIONAL AMPLIFIERS

THE DIRECTO-MATIC II SPEEDTRONIC CONTROL USES THE OPERATIONAL AMPLIFIER (NORMALLY REFERRED TO AS OP AMP) EXTENSIVELY; THEREFORE, TO UNDERSTAND THE CONTROL PANEL, ONE MUST HAVE A BASIC UNDERSTANDING OF THE OP AMP. A BRIEF DISCUSSION OF THE OP AMP ENHANCED WITH SEVERAL EXAMPLES IS GIVEN SUBSEQUENTLY. THE INTENT OF THIS TEXT IS NOT TO PRESENT A RIGOROUS ANALYTICAL DESCRIPTION OF THE OP AMP, BUT AN INTRODUCTION TO THE OP AMP IN SIMPLIFIED TERMS. IT IS HOPED THAT FROM THIS GROUNDWORK THE USER CAN FOLLOW THE TURBINE CONTROL ELEMENTARY AND GAIN AN UNDERSTANDING OF THE SPEEDTRONIC CONTROL.

THE OPERATIONAL AMPLIFIER IS GENERALLY CONSIDERED A HIGH GAIN, HIGH INPUT INPEDANCE, LOW OUTPUT IMPEDANCE AMPLIFIER CAPABLE OF AMPLIFYING D-C AS WELL AS A-C SIGNALS. THE STANDARD SYMBOL FOR THE OP AMP IS SHOWN BELOW.



THE INPUT WITH THE CIRCLE IS THE INVERTING INPUT AND THE OTHER INPUT IS NON-INVERTING. THE OUTPUT IS

$$e_o = K(e_a - e_b)$$

WHERE K IS THE GAIN OF THE AMPLIFIER. AS MENTIONED PREVIOUSLY, THE OP AMP HAS A VERY HIGH GAIN (TYPICALLY 10^5 IN DIRECTO-MATIC II'), THEREFORE, A VERY SMALL DIFFERENCE BETWEEN e_a AND e_b WILL GIVE A LARGE e_o. HOWEVER e_o IS LIMITED BY THE SUPPLY VOLTAGES OF THE OP AMP. WHEN e_o REACHES APPROXIMATELY THE SUPPLY VOLTAGE, IT IS SAID TO BE IN SATURATION. TYPICAL SUPPLY VOLTAGES IN DIRECTO-MATIC II ARE +12 VOLTS. USING A +12 VOLT SUPPLY AND A GAIN OF 10^5 THE VALUE OF $e_a - e_b$ FOR SATURATION IS

$$12 = 10^5(e_a - e_b)$$

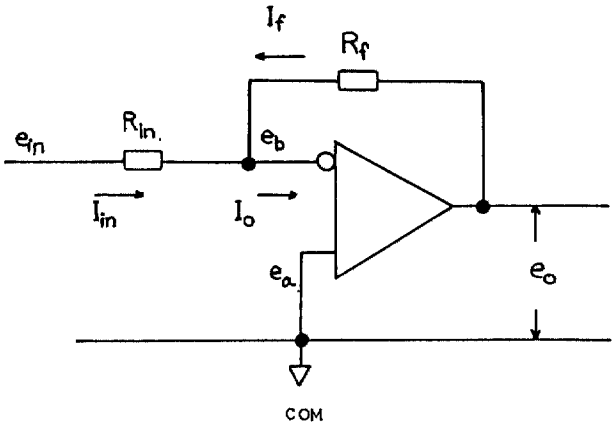
$$(e_a - e_b) = \frac{12}{10^5} = 12 \times 10^{-5}$$

$$(e_a - e_b) = 120 \text{ MICROVOLTS}$$

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 29 SH. NO. 28

BECAUSE OF THIS HIGH GAIN THE OP AMP IS PRACTICALLY USELESS WHEN USED ALONE; HOWEVER, WITH NEGATIVE FEEDBACK ITS USES ARE LIMITED ONLY BY THE DESIGNERS IMAGINATION.

THE MOST COMMON CIRCUIT OF THE OP AMP IS SHOWN BELOW:



IN THIS CASE THE NON-INVERTING INPUT e_a IS TIED TO COMMON. THE USEFUL OPERATING REGION OF THE OP AMP IS WHEN IT IS NOT IN SATURATION; THEREFORE, e_b IS APPROXIMATELY EQUAL TO e_a IN THIS CASE e_a IS TIED TO COMMON; THEREFORE, e_b IS APPROXIMATELY ZERO VOLTS. OPERATING IN THIS MODE e_b IS OFTEN REFERRED TO AS VIRTUAL COMMON. TO DETERMINE e_o AS A FUNCTION OF e_{in}, THE CURRENTS AT e_b ARE SUMMED TOGETHER. THIS JUNCTION IS OFTEN REFERRED TO AS THE SUMMING JUNCTION.

$$I_{in} + I_f - I_o = 0$$

THE OP AMP HAS A VERY HIGH INPUT IMPEDANCE AND I_o CAN BE CONSIDERED ZERO. THEN

$$I_{in} + I_f = 0$$

THE OP AMP HAS A VERY LOW OUTPUT IMPEDANCE AND WILL SUPPLY THE NECESSARY CURRENT TO MAKE THE ABOVE EQUATION TRUE.

I_{in} AND I_f CAN BE FOUND AS FOLLOWS:

$$I_{in} = \frac{e_{in}}{R_{in}} \qquad I_f = \frac{e_o}{R_f}$$

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C NO	SALEM, VA. U.S.A.	CONT. ON SH. 30 SH. NO. 29

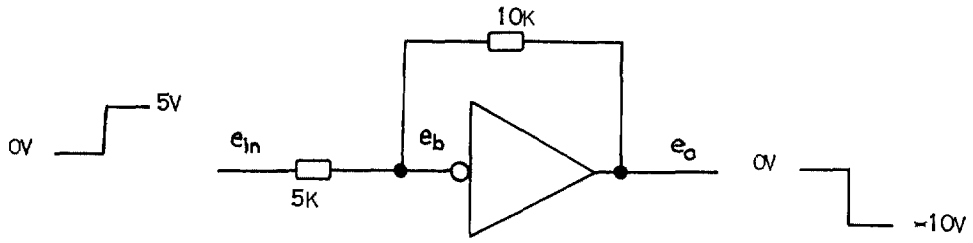
SUBSTITUTING GIVES:

$$\frac{e_{in}}{R_{in}} + \frac{e_o}{R_f} = 0$$

$$\frac{e_o}{R_f} = -\frac{e_{in}}{R_{in}}$$

$$e_o = -e_{in}\frac{R_f}{R_{in}}$$

THIS EQUATION APPLIES AS LONG AS THE OP AMP IS NOT IN SATURATION. IF IT GOES INTO SATURATION, e_b WILL NO LONGER BE AT VIRTUAL COMMON AND THE EQUATION LOSES VALIDITY. TO FURTHER ILLUSTRATE THE INVERTER, THE CIRCUIT IS REPEATED BELOW WITH VALUES ASSIGNED. THE CONNECTION OF e_a TO COMMON ON THE SYMBOL IS OFTEN OMITTED WITH THE UNDERSTANDING THAT IT EXISTS.

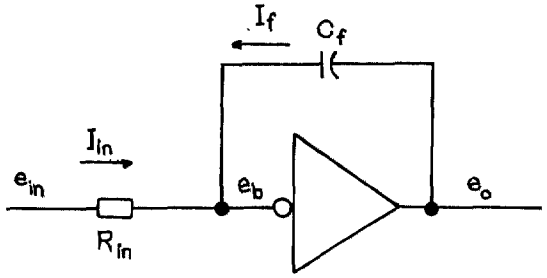


$$e_o = -e_{in}\frac{10K}{5K} = -2e_{in}$$

e_{in} IS A +5V SIGNAL, THEN

$e_o = -2(5)V = -10$ VOLTS

THE SECOND OP AMP CIRCUIT DISCUSSED IS THE INTEGRATOR. THE FOLLOWING IS AN INTEGRATOR CIRCUIT:



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
REV. 2	ISSUED			FIRST MADE FOR		CONTROL	
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 31	SH. NO. 30

AGAIN, SUMMING THE CURRENTS AT e_b GIVES:

$$I_{in} + I_f = 0 \quad \text{WHERE}$$

$$I_{in} = \frac{e_{in}}{R_{in}} \quad \text{AND}$$

$$I_f = C_f \frac{d(e_o)}{dt}$$

$$\frac{e_{in}}{R_{in}} + C_f \frac{d(e_o)}{dt} = 0$$

$$C_f \frac{d(e_o)}{dt} = -\frac{e_{in}}{R_{in}}$$

$$d(e_o) = -\frac{e_{in}}{R_{in}C_f} dt$$

$$\int d(e_o) = -\int \frac{e_{in}}{R_{in}C_f} dt$$

$$e_o = -\frac{1}{R_{in}C_f} \int e_{in} dt$$

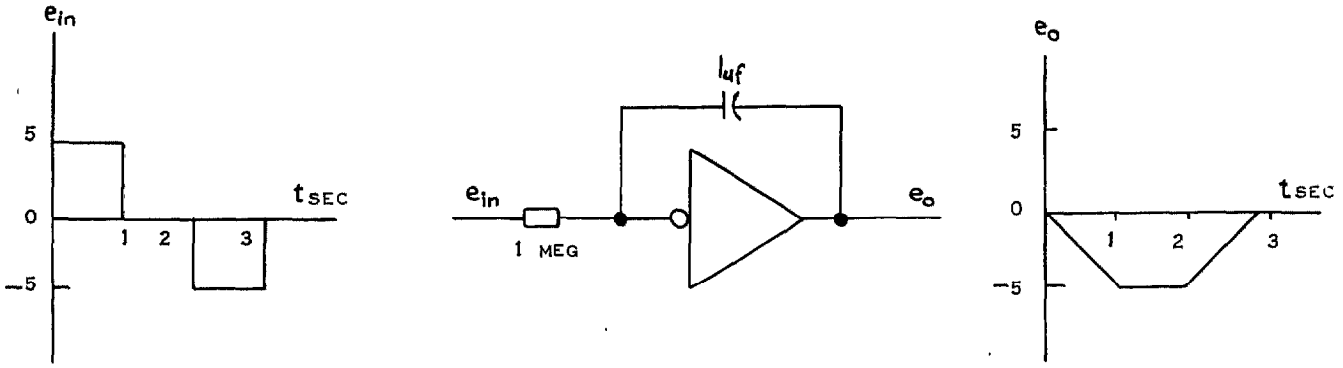
IF e_{in} IS A CONSTANT OR STEP INPUT, THEN

$$e_o = -\frac{e_{in}}{R_{in}C_f} t \quad \text{OR}$$

$$e_o = -\frac{I_{in}}{C_f} t \qquad t = \underline{\text{SECONDS}}$$

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM: CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 32 SH. NO. 31

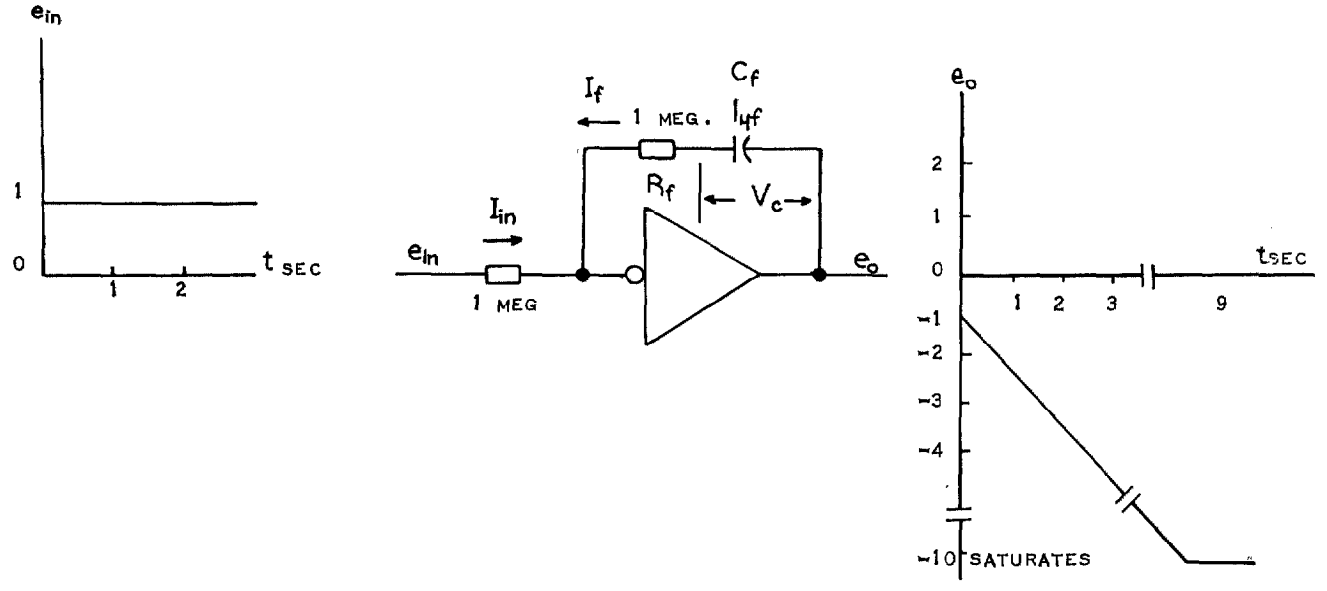
THE INTEGRATOR CIRCUIT IS REDRAWN BELOW WITH VALUES ASSIGNED.



$$e_o = -\frac{5}{(10^6)(10^6)} \qquad t = -5t$$

AS CAN BE SEEN FROM THE CURVES ADJACENT TO THE CIRCUIT, IN RESPONSE TO A STEP INPUT THE OUTPUT INCREASES ON A RAMP. IF THE INPUT RETURNS TO ZERO, THE CAPACITOR REMAINS CHARGED AND e_c REMAINS AT THE LEVEL PRIOR TO INPUT GOING TO ZERO. IF A STEP INPUT OF OPPOSITE POLARITY IS APPLIED, THE CAPACITOR WILL BEGIN DISCHARGING.

FOR THE NEXT CIRCUIT, THE INPUT-OUTPUT RELATIONSHIP WILL NOT BE DEVELOPED MATHEMATICALLY; INSTEAD AN INTUITIVE APPROACH WILL BE PRESENTED. THE CIRCUIT IS OFTEN REFERRED TO AS A PROPORTIONAL PLUS INTEGRAL.



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
REV. 2	ISSUED			FIRST MADE FOR		CONTROL	
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 33	SH. NO 32

TO BEGIN ASSUME THAT e_{in}, AND e_o ARE ZERO ($V_C = 0$). NOW ASSUME A STEP INPUT OF 1 VOLT IS APPLIED AT e_{in}. AS BEFORE

$$I_{in} = \frac{e_{in}}{R_{in}} = \frac{1V}{1M} = I_{ua}$$

$$I_f = -I_{in} = -I_{ua}$$

NOW e_o IS EQUAL TO THE VOLTAGE ACROSS R_f AND C_f. THE VOLTAGE ACROSS R_f IS

$$V_t = I_f R_f = -I_{in} R_f$$

$$V_t = -1 \text{ VOLT}$$

THE VOLTAGE ACROSS A CAPACITOR SUBJECTED TO A CONSTANT CURRENT IS

$$V_c = \frac{I}{C} t$$

THEREFORE, IF e_{in} IS A STEP INPUT

$$e_o = -I_{in} R_f - \frac{I_{in}}{C} t$$

AT THE INSTANT THAT THE STEP IS APPLIED ($t = 0$)

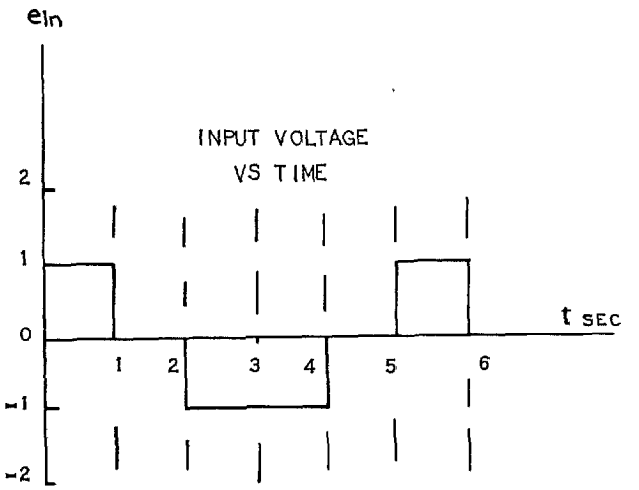
$$e_o = -I_{in} R_f$$

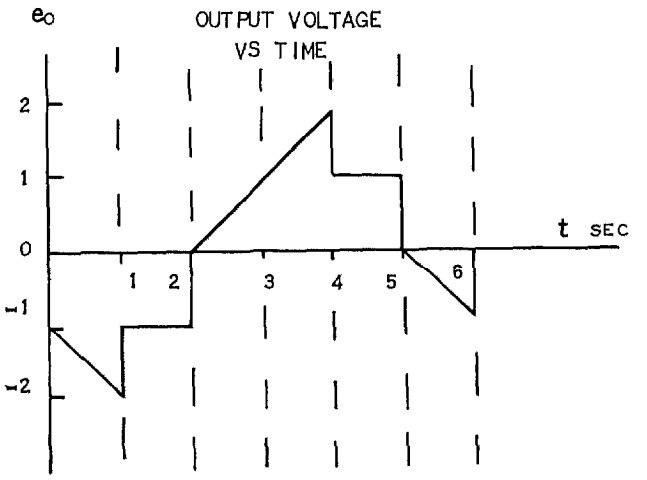
$$e_o = -(I_{ua})(I_{meg})$$

$$e_o = -1 \text{ VOLT}$$

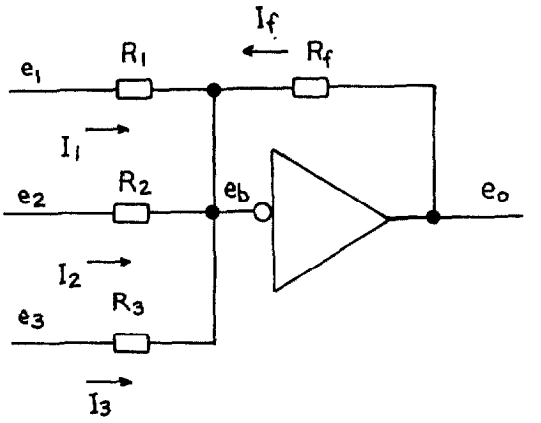
AS TIME INCREASES, THE CAPACITOR CHARGES UP AS SHOWN IN THE GRAPH ADJACENT TO CIRCUIT. FOLLOWING ARE CURVES SHOWING OUTPUT RESPONSE TO A VARYING INPUT VOLTAGE.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 34 SH. NO. 33





THE NEXT OP AMP CIRCUIT DISCUSSED IS THE SUMMER. THE CIRCUIT IS SHOWN BELOW:



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 35 SH. NO 34

SUMMING THE CURRENTS AT e_b YIELDS:

$$I_1 + I_2 + I_3 + I_f = 0$$

$$\frac{e_1}{R_1} + \frac{e_2}{R_2} + \frac{e_3}{R_3} + \frac{e_o}{R_f} = 0$$

$$\frac{e_o}{R_f} = -\left(\frac{e_1}{R_1} + \frac{e_2}{R_2} + \frac{e_3}{R_3}\right)$$

$$e_o = -\left(\frac{R_f e_1}{R_1} + \frac{R_f e_2}{R_2} + \frac{R_f e_3}{R_3}\right)$$

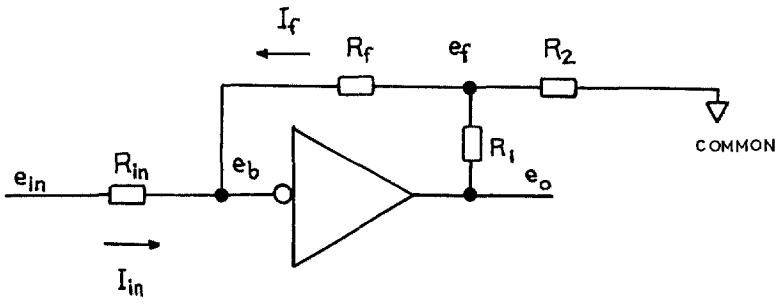
AS IMPLIED BY ITS NAME, THE CIRCUIT SUMS THE INPUTS WITH EACH INPUT WEIGHTED BY THE RATIO OF ITS INPUT RESISTOR TO THE FEEDBACK RESISTOR, R_f .

THE MULTIPLE INPUT CONFIGURATION CAN ALSO BE APPLIED TO THE INTEGRATOR. REFERRING BACK TO THE INTEGRATOR EQUATION OF A STEP INPUT

$$e_o = -\frac{I_{in}}{C_f}t$$

FOR MULTIPLE INPUTS, THE ALGEBRAIC SUM OF THE INPUT CURRENTS IS SUBSTITUTED FOR I_{in} .

OCCASIONALLY THE FEEDBACK VOLTAGE IS NOT DERIVED DIRECTLY FROM e_o .FOLLOWING IS AN EXAMPLE OF SUCH A CIRCUIT:



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C NO.	SALEM, VA. U.S.A.	CONT. ON SH. 36 SH. NO 35

AGAIN, SUMMING THE CURRENTS AT e_b GIVES

$$I_{in} + I_f = 0 \text{ (SUMMING JUNCTION EQUATION) WHERE}$$

$$I_{in} = \frac{e_{in}}{R_{in}}$$

$$I_f = \frac{e_f}{R_{in}} \quad (I_f \text{ EQUATION) BUT}$$

$$e_f = e_o \left(\frac{\frac{R_f R_2}{R_f + R_2}}{R_1 + \frac{R_f R_2}{R_f + R_2}} \right) \text{ LET}$$

$$R_{eq} = \frac{R_f R_2}{R_f + R_2} \quad \text{THEN}$$

$$e_f = e_o \left(\frac{R_{eq}}{R_1 + R_{eq}} \right)$$

SUBSTITUTING THIS IN ABOVE I_f EQUATION GIVES

$$I_f = \frac{e_o}{R_f} \left(\frac{R_{eq}}{R_1 + R_{eq}} \right)$$

THE SUMMING JUNCTION EQUATION NOW BECOMES

$$\frac{e_o}{R_f} \left(\frac{R_{eq}}{R_1 + R_{eq}} \right) = \frac{e_{in}}{R_{in}}$$

$$e_o = -e_{in} \frac{R_f}{R_{in}} \left(\frac{R_1 + R_{eq}}{R_{eq}} \right)$$

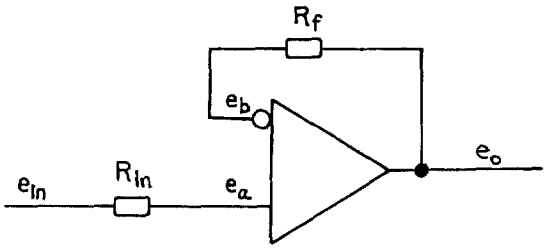
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 37 SH NO. 36

UP TO THIS POINT, WE HAVE ONLY DISCUSSED THE INVERTING MODE OF THE OP AMP. BEFORE GOING INTO THE NON-INVERTING MODE, TWO IMPORTANT FEATURES OF THE OP AMP DISCUSSED PREVIOUSLY WILL BE REPEATED.

1. THE OP AMP HAS VERY HIGH INPUT IMPEDANCE; THEREFORE, IT IS SAFE TO ASSUME THAT IT DRAWS ZERO INPUT CURRENT.

2. A VERY SMALL DIFFERENCE BETWEEN e_a AND e_b CAUSES THE OP AMP TO GO INTO SATURATION, THEREFORE, IN THE ACTIVE REGION, IT IS ASSUMED THAT e_a EQUALS e_b.

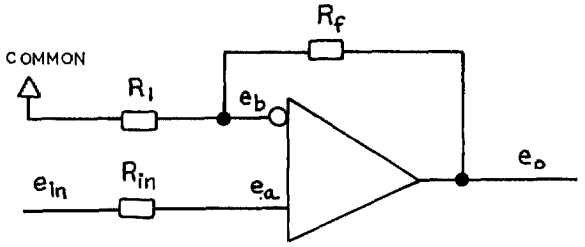
THE FIRST NON-INVERTING OP AMP CIRCUIT TO BE DISCUSSED IS THE VOLTAGE FOLLOWER. THE CIRCUIT IS SHOWN BELOW.



SINCE THE OP AMP INPUT DRAWS NO CURRENT, e_a IS EQUAL TO e_{in}. PER FEATURE NUMBER 2 ABOVE e_b EQUALS e_a AND NO CURRENT IS DRAWN BY THE OP AMP; THEREFORE, e_o EQUALS e_b WHICH IN TURN IS EQUAL TO e_{in}. THE INPUT-OUTPUT EQUATION IS

$$e_o = e_{in}$$

TO OBTAIN A GAIN GREATER THAN ONE, THE FOLLOWING CIRCUIT IS USED:



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH 38 SH. NO. 37

AS IN PREVIOUS CIRCUIT e_b EQUALS e_a WHICH EQUALS e_{in} BECAUSE NO CURRENT FLOWS INTO THE OP AMP. e_b IS ALSO EQUAL TO

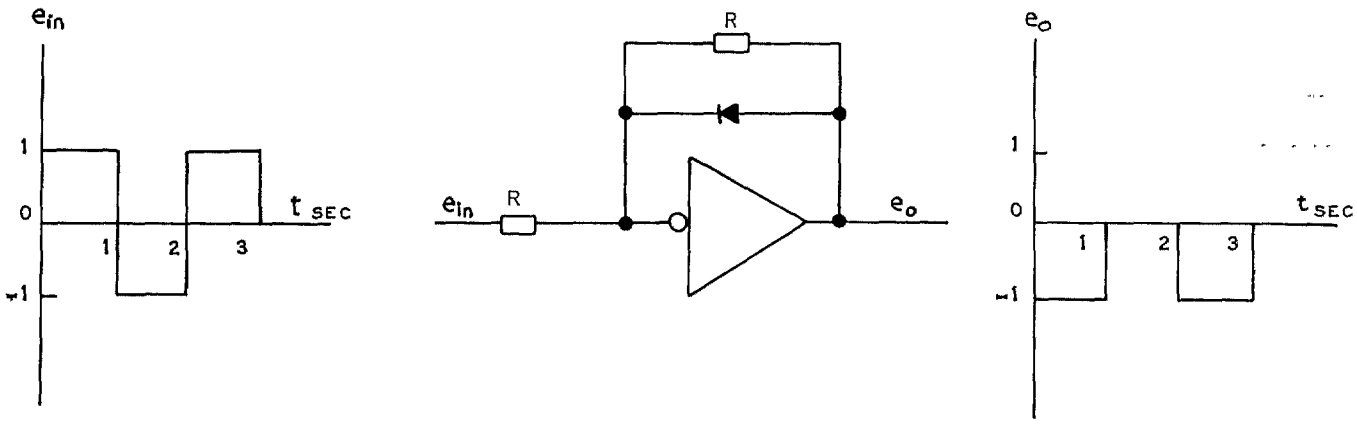
$$e_b = e_o \frac{R_1}{R_1 + R_f} \quad \text{OR}$$

$$e_o = \left(\frac{R_1 + R_f}{R_1}\right) e_b$$

$$e_o = \left(\frac{R_1 + R_f}{R_1}\right) e_{in} \qquad e_b = e_{in}$$

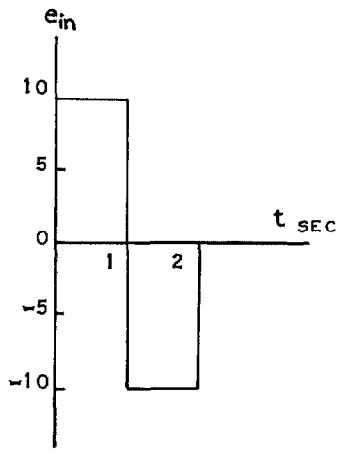
$$e_o = \left(1 + \frac{R_f}{R_1}\right) e_{in}$$

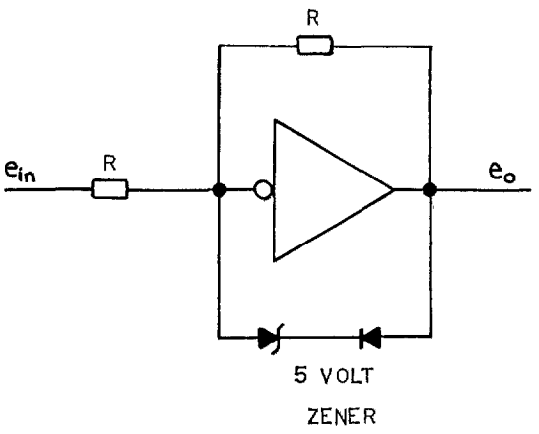
AS SEEN FROM THIS EQUATION THE GAIN OF THIS CIRCUIT IS ALWAYS GREATER THAN ONE. IT IS OFTEN DESIRED TO CLAMP THE OP AMP OUTPUT AT SOME VALUE. THE FOLLOWING CIRCUITS ILLUSTRATE THE USE OF DIODES AND ZENER DIODES TO CLAMP THE OUTPUT.

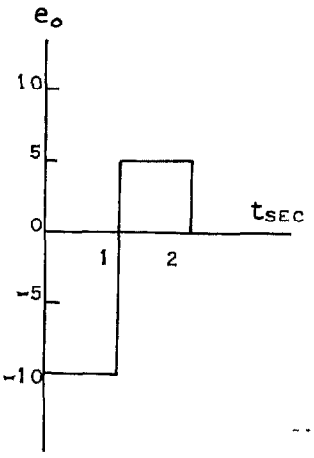


ABOVE CIRCUIT LIMITS POSITIVE SWING TO APPROXIMATELY ZERO.

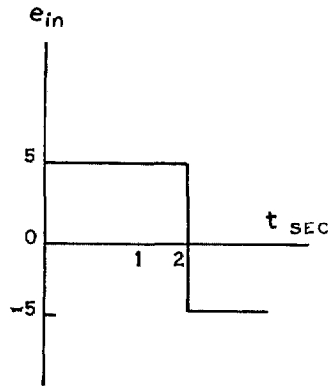
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 39 SH. NO. 38

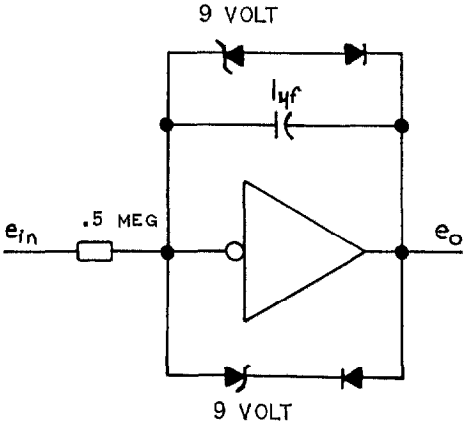


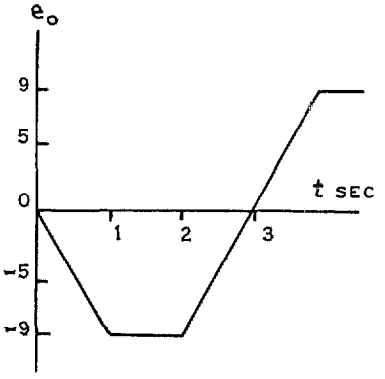




ABOVE CIRCUIT LIMITS POSITIVE SWING TO APPROXIMATELY 5 VOLTS.







ABOVE CIRCUIT LIMITS POSITIVE AND NEGATIVE SWINGS TO 9 VOLTS: THIS CIRCUIT IS OFTEN USED TO PREVENT OP AMP FROM SATURATING AND SUMMING FUNCTIONS FROM RISING ABOVE COMMON.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 40 SH. NO 39

C-2 START-UP

THE TURBINE CONTROL PANEL CONTAINS START-UP, L.P. SPEED, AND TEMPERATURE CONTROL LOOPS WHICH MODULATE VCE. VCE SERVES AS A REFEREMCE TO THE FUEL CONTROL LOOPS WHICH REGULATE FUEL FLOW TO THE TURBINE. A NOZZLE CONTROL LOOP THAT MODULATES NCE TO CONTROL H.P. SPEED IS OPTIONAL. A TYPICAL CONTROL SYSTEM IS SHOWN IN FIG. C-1.

THE START-UP CONTROL IS AN OPEN LOOP SYSTEM THAT OPERATES IN CONJUNCTION WITH THE SEQUENTIAL CONTROL CIRCUITRY TO PROVIDE PRESET FUEL LEVELS TO THE TURBINE DURING START-UP. IN ADDITION TO THE ABOVE, IT ALSO SETS THE MAXIMUM AND MINIMUM LIMITS ON VCE AFTER THE START-UP PERIOD IS OVER.

A TYPICAL VCE START-UP CURVE IS SHOWN IN FIG. C-2. A TYPICAL CONTROL SYSTEM IS SHOWN IN FIG. C-3.

SSKC CARD (SEE FIG. C-4)

A "1" ON INPUT "OFF" OR $\overline{\text{FIRE}}$ WILL CLAMP OUTPUT VCE TO ZERO VOLTS. IF OFF IS A "0" THEN A "0" OCCURRING ON $\overline{\text{FIRE}}$ WILL REMOVE THE CLAMP ON VCE. TO FACILITATE DESCRIBING THE FUNCTION OF THE CARD, ASSUME THAT OFF IS A "0", ALR IS A "0", AND $\overline{\text{FIRE}}$, $\overline{\text{WU}}$ AND $\overline{\text{ACC}}$ ARE "1". UNDER THESE CONDITIONS, VCE IS CLAMPED TO ZERO VOLTS. NOW ASSUME THAT A "0" OCCURS ON INPUT $\overline{\text{FIRE}}$. VCE WILL JUMP TO FIRE LEVEL AS SET BY R102. A "0" OCCURRING NOW ON $\overline{\text{WU}}$ WILL CAUSE VCE TO DROP FROM FIRE TO WARM-UP LEVEL WHICH IS SET BY R101. VCE WILL REMAIN AT THIS LEVEL UNTIL ACC BECOMES "0" OR VCE IS PULLED DOWN BY SOME OTHER CONTROL LOOP. UPON RECEIVING A "0" ON $\overline{\text{ACC}}$, VCE WILL BE RELEASED TO INCREASE ON AN EXPONENTIAL CURVE UNTIL A LIMIT SET BY R100 IS REACHED. THIS LIMIT ON VCE WILL REMAIN UNTIL INPUT ALR BECOMES A "1". A "1" ON ALR WILL REMOVE THE LIMIT AND ALLOW VCE TO INCREASE TO A MAXIMUM LIMIT SET BY R106.

A MINIMUM VCE CIRCUIT MONITORS VCE AND PROVIDES A NEGATIVE BIAS AT OUTPUT MIN WHEN VCE DECREASES BELOW A MIN LEVEL SET BY R104.

TYPICALLY, THE BIAS OUT OF MIN IS CONNECTED TO THE SPEED AND ACCELERATION CONTROLS AND PREVENTS THEM FROM DRIVING VCE BELOW THE MIN LEVEL. A TEST CIRCUIT IS PROVIDED THAT SUPPLIES A POSITIVE BIAS OUT OF MT WHEN MIN PUSHBUTTON IS DEPRESSED THAT CAN BE USED AS AN INPUT TO THE SPEED CARD TO SIMULATE SPEED FEEDBACK WHICH TELLS THE SPEED CARD TO DECREASE VCE. THE MIN CIRCUIT SHOULD THEN PREVENT VCE FROM GOING BELOW THE PRESET LEVEL.

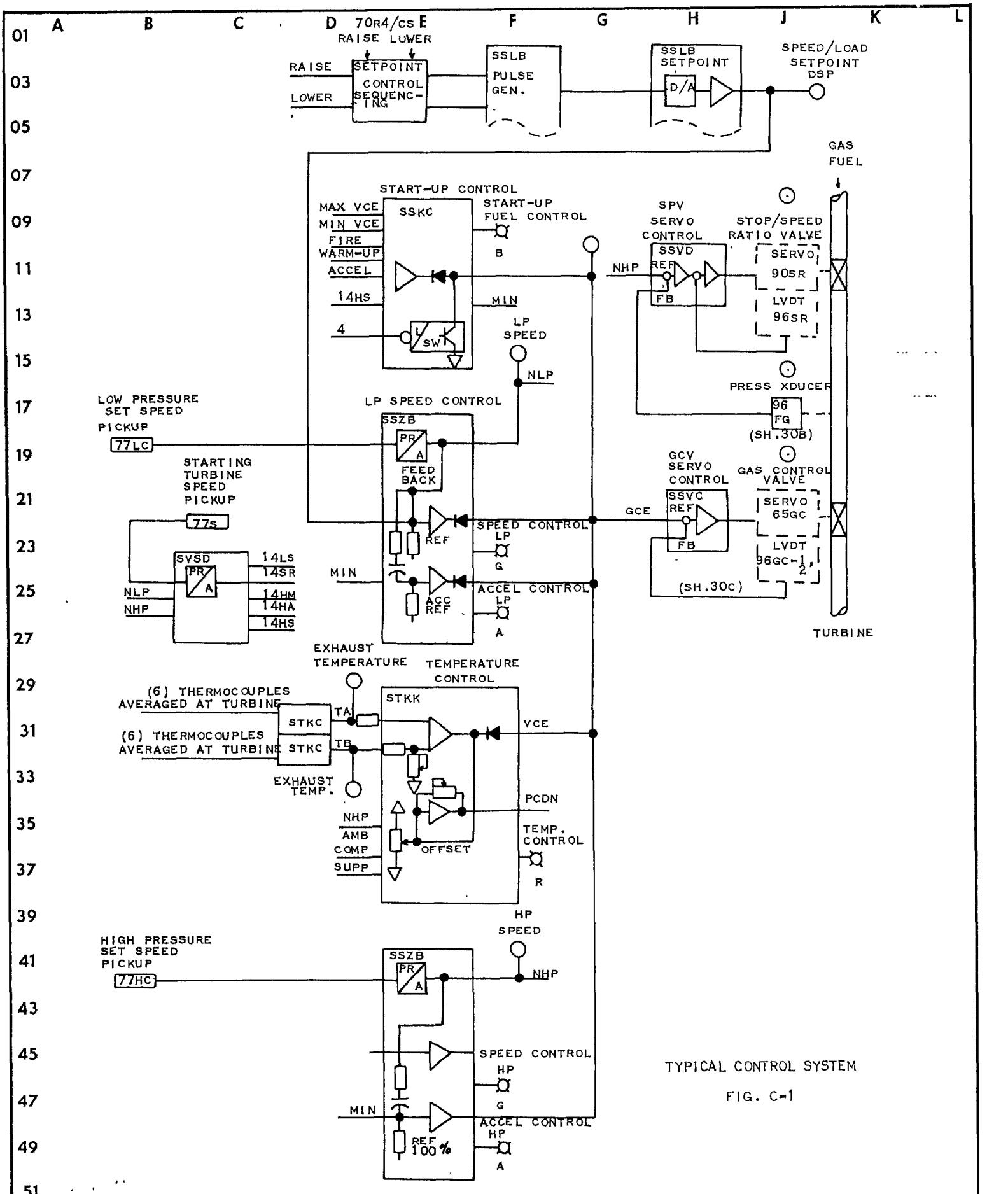
PUSHBUTTONS (MAX, MIN, FIRE, WU, ACC) ARE PROVIDED ON THE CARD FOR TESTING AND ADJUSTING THEIR CORRESPONDING PRESET LEVELS. DEPRESSING EITHER THE MIN, FIRE, OR WU PUSHBUTTONS WILL SET VCE TO ITS CORRESPONDING VOLTAGE LEVEL (SEE RANGES BELOW). DEPRESSING THE MAX PUSHBUTTON WILL SET VCE AT ITS MAXIMUM VALUE, ONLY IF THE "43" SWITCH IS IN THE OFF OR CRANK POSITION (SEE RANGE BELOW). DEPRESSING THE ACC PUSHBUTTON WILL ALLOW VCE TO RISE EXPONENTIALLY TO ITS ACCELERATE LIMIT. DEPRESSING THE TCC AND ACC PUSHBUTTONS SIMULTANEOUSLY, WILL REDUCE THE EXPONENTIAL TIME CONSTANT BY ABOUT A FACTOR OF 10. WITH ANY OF THE PREVIOUS RANGES SELECTED, THE ACTUAL VALUE OF VCE CAN BE ADJUSTED BY THE POTENTIOMETERS LISTED BELOW.

POTENTIOMETER RHI IS PROVIDED FOR MANUAL CONTROL OF VCE. VCE CAN BE ADJUSTED FROM MAX VCE TO ZERO USING RHI IF SOME OTHER CONTROL LOOP IS NOT PULLING VCE DOWN. A LIMIT SWITCH ON THE POTENTIOMETER TIES PINS SW1 AND SW2 TOGETHER WHEN RHI IS NOT FULLY COUNTER-CLOCKWISE.

ADJUSTMENT RANGES

MAX - R106-8.00 TO 10.5 VOLTS
MIN - R104-2.75 TO 4.00 VOLTS
FIRE - R102-3.25 TO 5.00 VOLTS
WU - R101-2.50 TO 4.0 VOLTS
ACC - R100-3.50 TO 6.5 VOLTS

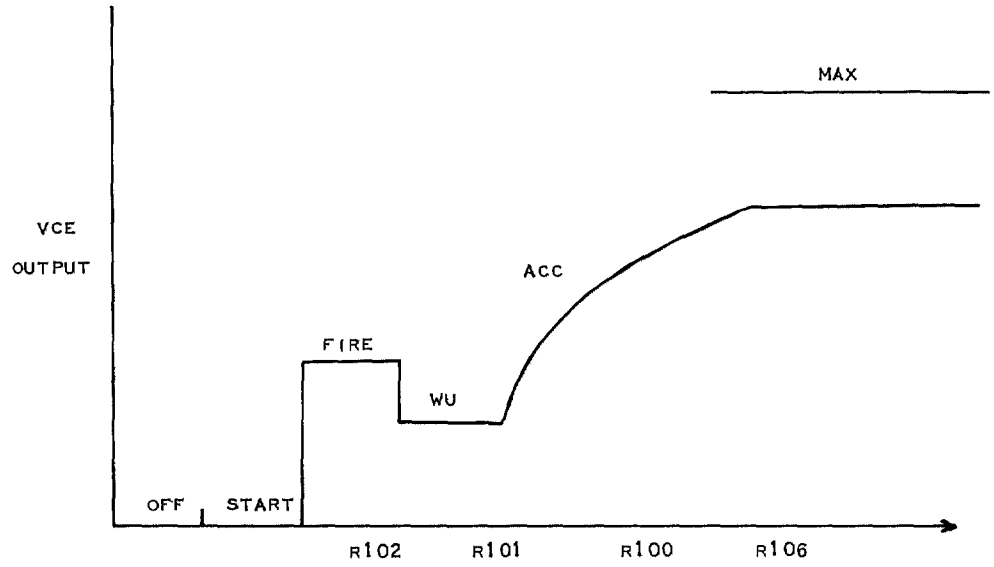
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 42 SH. NO. 41



TYPICAL CONTROL SYSTEM

FIG. C-1

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR	SALEM, VA. U.S.A.	CONT. ON SH 43 SH. NO. 42
REV. 3	MADE BY			I.C. NO.		



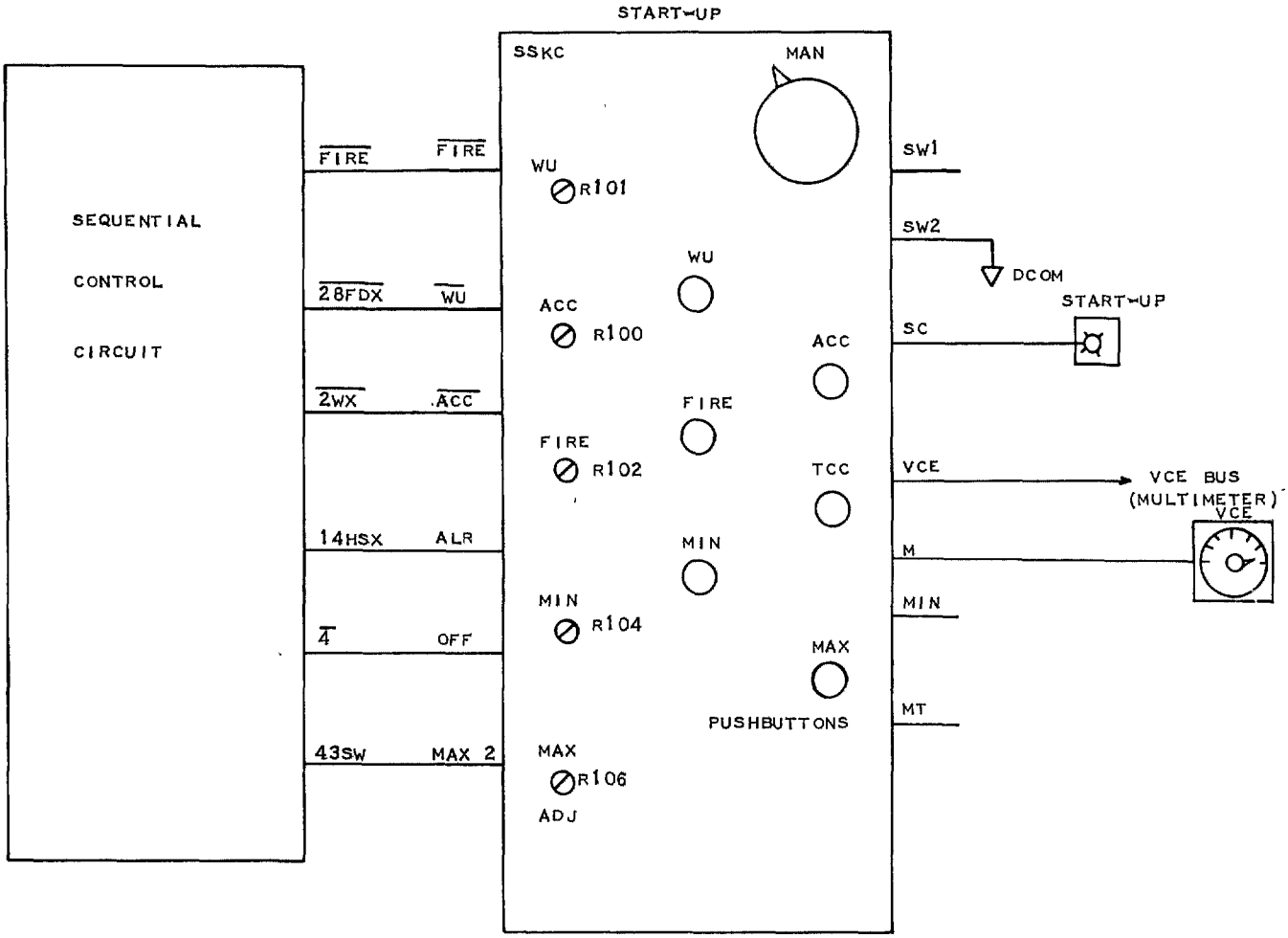
OFF	1	0	0	0	0	0
$\overline{\text{FIRE}}$	ø	1	0	0	0	0
$\overline{\text{WU}}$	ø	1	1	0	0	0
$\overline{\text{ACC}}$	ø	1	1	1	0	0
ALR	ø	0	0	0	0	1

ø = DON'T CARE "0" OR "1"

OPEN LOOP START-UP PROFILE

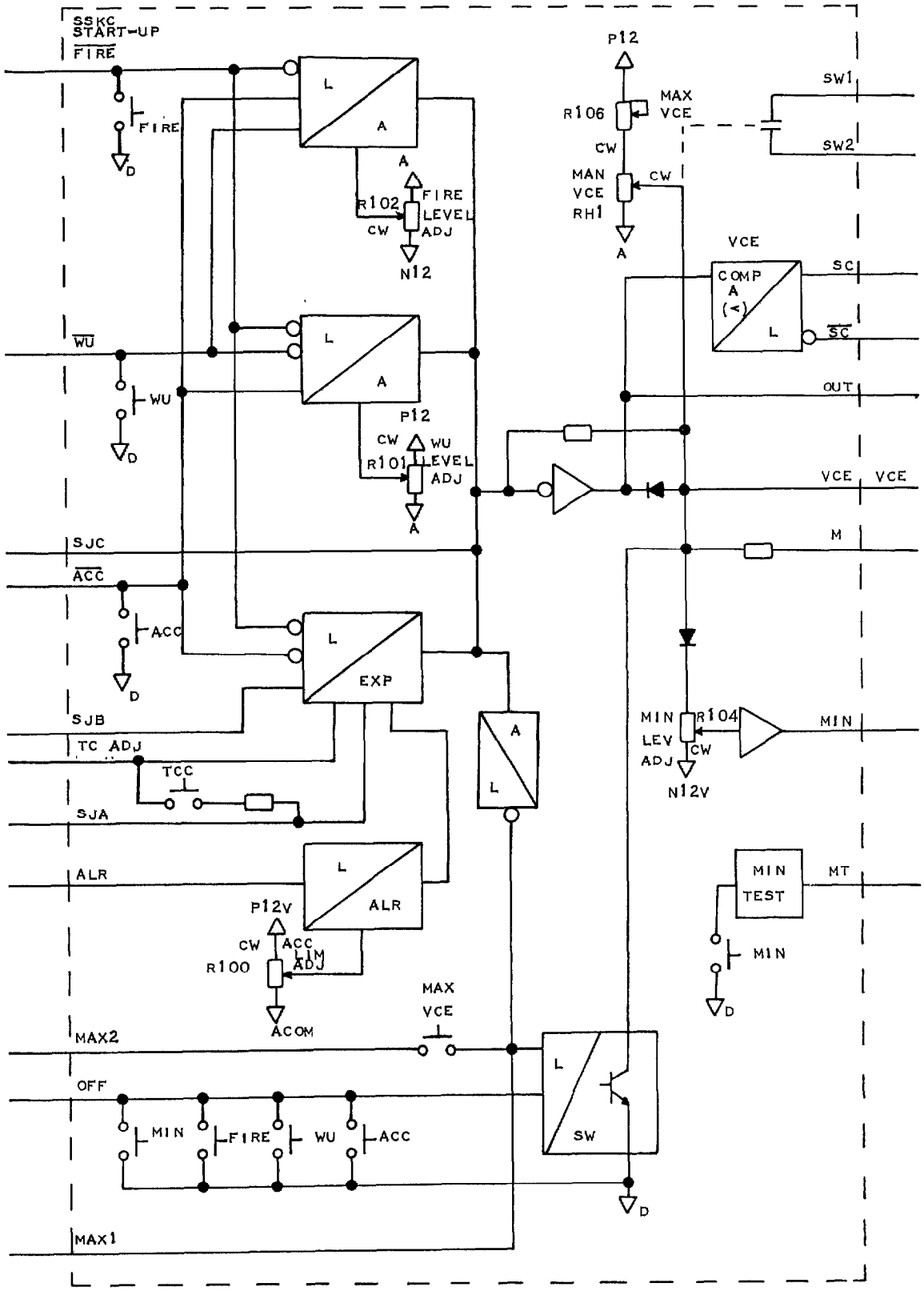
FIG. C-2

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 44 SH. NO. 43



TYPICAL START-UP CONTROL SYSTEM FIGURE C-3

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 45 SH. NO. 44



SSKC CARD

FIG. C-4

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM: CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 46 SH. NO 45

C-3 DIGITAL SETPOINT

THE DIGITAL SETPOINT PROVIDES A PROGRAMMABLE ANALOG SIGNAL USED AS A CALLED FOR SPEED OR LOAD SIGNAL TO THE SPEED CONTROL CARD. THE SYSTEM CONSISTS OF CARDS SSLB, VRCC, AIAD (OPTIONAL) AND ASSOCIATED CONTROL SIGNALS.

VRCC-CARD (SEE FIG. C-5)

THE VRCC CARD CONTAINS A 10 BIT REVERSIBLE COUNTER. THE RATE OF THE COUNTER IS DETERMINED BY THE FREQUENCY OF THE PULSES COMING IN ON PUL. THE DIRECTION OF THE COUNT IS CONTROLLED BY INPUTS ON ST1 AND ST0. A "1" ON ST1 AND A "0" ON ST0 WILL STEER THE COUNTER UP. A "0" ON ST1 AND A "1" ON ST0 WILL STEER THE COUNTER DOWN. IF BOTH ST1 AND ST0 ARE "0" THE COUNTER WILL NOT COUNT. IF BOTH ST1 AND ST0 ARE "1", ALL BITS WILL FLIP ON THE NEXT COUNT PULSE. A "0" ON CL WILL CLEAR ALL BITS.

SSLB-CARD (SEE FIG. C-5)

THE SSLB CARD PROVIDES THREE FUNCTIONS:

(1) STEERING LOGIC AND COUNT PULSES FOR VRCC;
(2) A DIGITAL TO ANALOG CONVERTER
(3) A COMPARATOR CIRCUIT

OUTPUT ST1 IS A "1" IF INPUT UP IS A "1" AND DN IS "0". OUTPUT ST0 IS A "1" IF DN IS "1" AND UP IS "0". COUNT PULSES ARE PRESENT ON OUTPUT SFT IF EITHER UP OR DN IS A "0" AND COUNTER IS NOT FULL OR EMPTY. IF THE COUNTER IS FULL, INPUT MSB AND C WILL BE A "1" AND INHIBIT THE PULSES AND OUTPUT UST WILL BE A "0". IF THE COUNTER IS EMPTY, INPUT MSB AND C WILL BE A "1" AND ALSO INHIBIT THE COUNT PULSE GENERATOR AND OUTPUT DST WILL BE A "0". THE FREQUENCY OF THE GENERATOR MAY BE DECREASED BY CONNECTING A CAPACITOR BETWEEN INPUT CAP AND COMMON.

THE D/A CONVERTER RECEIVES ITS INPUT FROM THE REVERSIBLE COUNTER ON PINS C0-C5 AND $\overline{C6}$-$\overline{C9}$. THE D/A CONVERTER OUTPUT PIN SP VOLTAGE RANGE PROVIDES THE DIGITAL SETPOINT SIGNAL TO THE SPEED CONTROL CARD. OUTPUT SPM PROVIDES A 1MA SIGNAL FOR DSP EQUAL TO 10 VOLTS WHICH CAN BE USED TO DRIVE A SETPOINT INDICATOR METER. ZERO AND SPAN ADJUSTMENT POTENTIOMETERS ARE PROVIDED TO ENABLE ADJUSTMENT OF SP OVER VARIOUS RANGES CORRESPONDING TO COUNTER EMPTY AND COUNTER FULL CONDITIONS.

THE COMPARATOR RECEIVES ONE OF ITS INPUTS FROM THE OUTPUTS OF THE D/A CONVERTER. THIS INPUT IS COMPARED WITH SIGNALS ON INPUTS SJ1, $\overline{REF}$, AND PRST. IF SP IS GREATER THAN THE SIGNAL OR SIGNALS BEING COMPARED TO IT , OUTPUT "$\overline{DN}$" WILL BE A "0". IF SP IS LESS THAN THE SIGNAL OR SIGNALS, OUTPUT UP WILL BE A "1". SERVO ZERO AND SERVO SPAN POTENTIOMETERS ARE PROVIDED TO ENABLE MATCHING THE SP VOLTAGE RANGE TO THE RANGE OF THE VOLTAGE THAT IS BEING COMPARED TO IT. THE OUTPUT SP CAN BE MADE TO SERVO TO A REFERENCE ON INPUTS SJ1 OR REF BY CONNECTING THE $\overline{DN}$ OUTPUT THROUGH AN INVERTER TO INPUT DN AND CONNECTING OUTPUT UP TO INPUT UP.

ADJUSTMENTS

R124 - SETPOINT SPAN
R125 - SETPOINT ZERO
R126 - SERVO ZERO
R127 - SERVO SPAN

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM: CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 47 SH. NO. 46

A1AD-CARD (OPTIONAL) (SEE FIG.C-5)

THE A1AD CARD PROVIDES ISOLATION BETWEEN THE PROCESS SPEED SETPOINT CIRCUIT AND THE SSLB CARD. THE CARD WILL ACCEPT THE FOLLOWING SIGNALS AT THE DESIGNATED INPUTS BY MAKING THE SPECIFIED JUMPERS:

SIGNAL	INPUT CONNECTION	JUMPERS
0 TO 5V	+D TO 0	E TO F
0 TO 10V	+D TO 0	E TO L
0 TO 25V	+M TO 0	E TO F, D TO N
1 TO 5 MA	+A TO 0	E TO F, D TO A
4 TO 20 MA	+B TO 0	E TO F, D TO B
10 TO 50 MA	+C TO 0	E TO F, D TO C

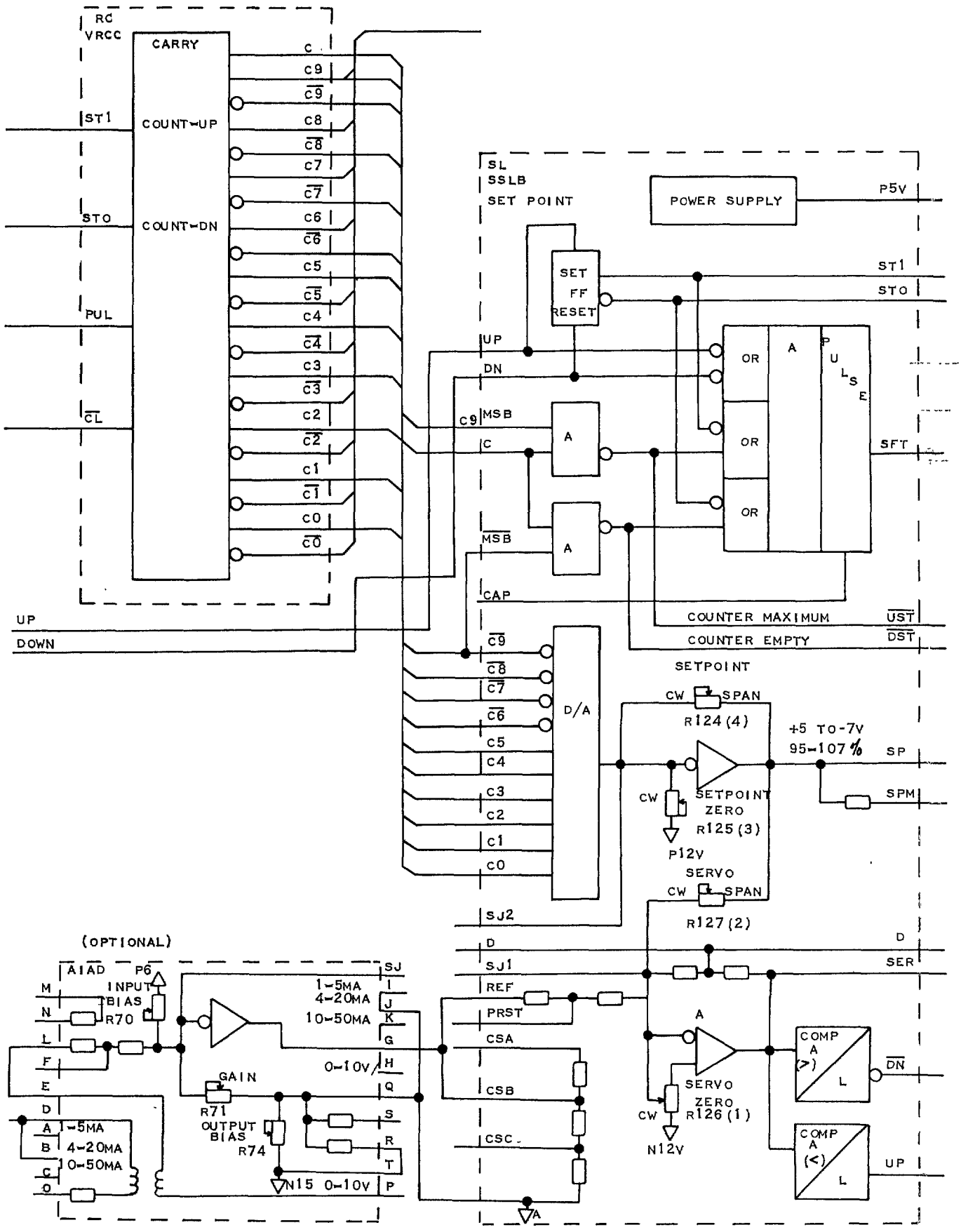
THE FOLLOWING OUTPUTS ARE POSSIBLE:

OUTPUT	CONNECTION	JUMPERS
0 TO 10V	+G TO P	G TO H, Q TO I
1 TO 5 MA	+G TO Q	Q TO I, S TO T
4 TO 20 MA	+G TO Q	Q TO J, R TO T
10 TO 50 MA	+G TO Q	Q TO K, R TO T

ADJUSTMENTS

INPUT BIAS	-R70	0 TO 30% ON ANY SCALE
OUTPUT BIAS	-R74	0 TO 2.5 MA (S CONNECTED TO T) 2.5 TO 15MA (R CONNECTED TO T)
GAIN	-R71	0.5 TO 2 WHEN USED AS CURRENT TO CURRENT CONVERTER +15% MIN ON ANY VOLTAGE RANGE.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 48 SH. NO. 47



DIGITAL SETPOINT CONTROL
FIG. C-5

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH 49 SH. NO. 48

C-4 LP SPEED

THE LP SPEED CONTROL LOOP MODULATES VCE TO CONTROL THE SPEED AND ACCELERATION OF THE LOW PRESSURE SHAFT. THE REFERENCE FOR THE SPEED REGULATOR IS THE SUM OF A 100 % SPEED REFERENCE AND A DIGITAL SETPOINT SIGNAL. A TYPICAL SPEED LOOP IS ILLUSTRATED IN FIG. C-6. THE SPEED LOOP UTILIZES THE SSZB CARD.

SSZB CARD (SEE FIG. C-7 AND C-8)

THE SSZB CARD CONTAINS A PULSE RATE TO ANALOG (PR/A) CONVERTER WHICH RECEIVES AN ALTERNATING SIGNAL FROM THE SPEED SNESOR AT INPUT IN AND PROVIDES A PROPORTIONAL SIGNAL AT OUTPUT N. THIS SPEED FEEDBACK SIGNAL IS SUMMED WITH THE REFERENCE SIGNAL AND OTHER INPUT SIGNALS TO MODULATE NVCE TO OBTAIN THE DESIRED SPEED AND / OR LOAD. JUMPERS ARE MADE ON FA, FC AND FD TO PROGRAM THE PR/A CONVERTER TO GIVE THE APPROPRIATE VOLTAGE FOR VARIOUS MACHINE SPEEDS.

AN OVERSPEED TEST PUSHBUTTON WHICH WILL FEED A NEGATIVE BIAS INTO SJ AND INCREASE NVCE IS PROVIDED FOR OVERSPEED TESTING.

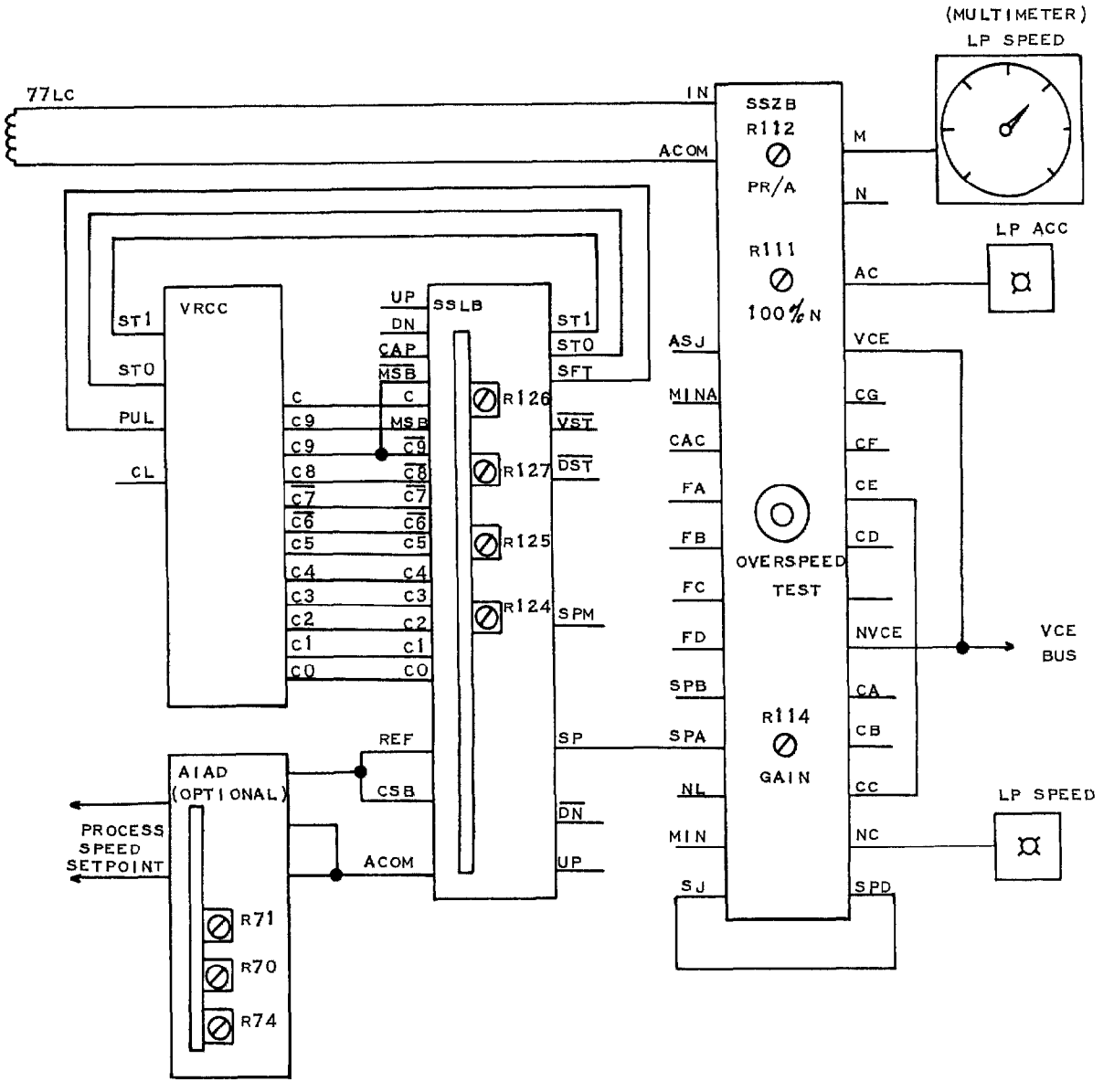
JUMPERS ARE MADE BETWEEN EITHER PIN CA, CB, CC AND EITHER PIN CG, CF, CE OR CD TO SELECT TRANSFER FUNCTION DESIRED. POTENTIOMETER R114 PROVIDES A GAIN ADJUSTMENT. INPUTS SPA AND SPB PROVIDE INPUTS FOR THE DIGITAL SETPOINT SIGNAL. SPA IS USED WHEN 0.2 VOLT DSP IS EQUAL TO 1 % CHANGE IN SPEED, AND SPB IS USED WHEN 1 VOLT DSP IS EQUAL TO 1 % CHANGE IN SPEED.

A DIFFERENTIATOR CIRCUIT LOOKS AT THE SPEED FEEDBACK SIGNAL N AND MODULATES VCE TO LIMIT THE ACCELERATIONS OF THE TURBINE. OUTPUT "AC" WILL BE A "1" WHEN VCE IS UNDER ACCELERATION CONTROL. CONNECTING HR AND ACOM WILL HAVE THE ACCELERATION CONTROL RATE.

ADJUSTMENTS:

GAIN (PR/A) - R112
GAIN (NVCE) - R114
100 %N - R111 - 96 % TO 104 % RATED SPEED

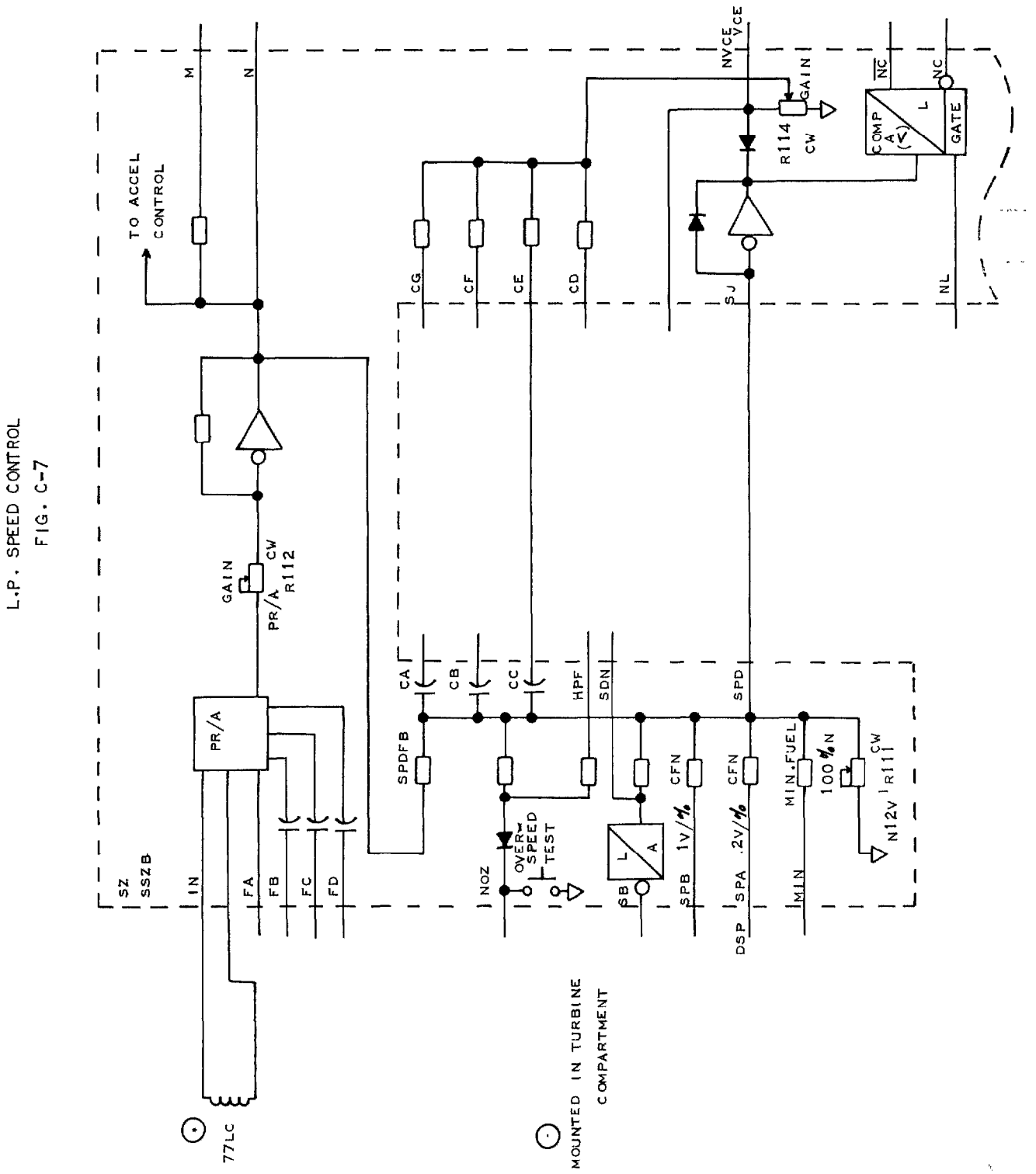
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
REV. 2	ISSUED			FIRST MADE FOR		CONTROL	
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 51	SH. NO. 49



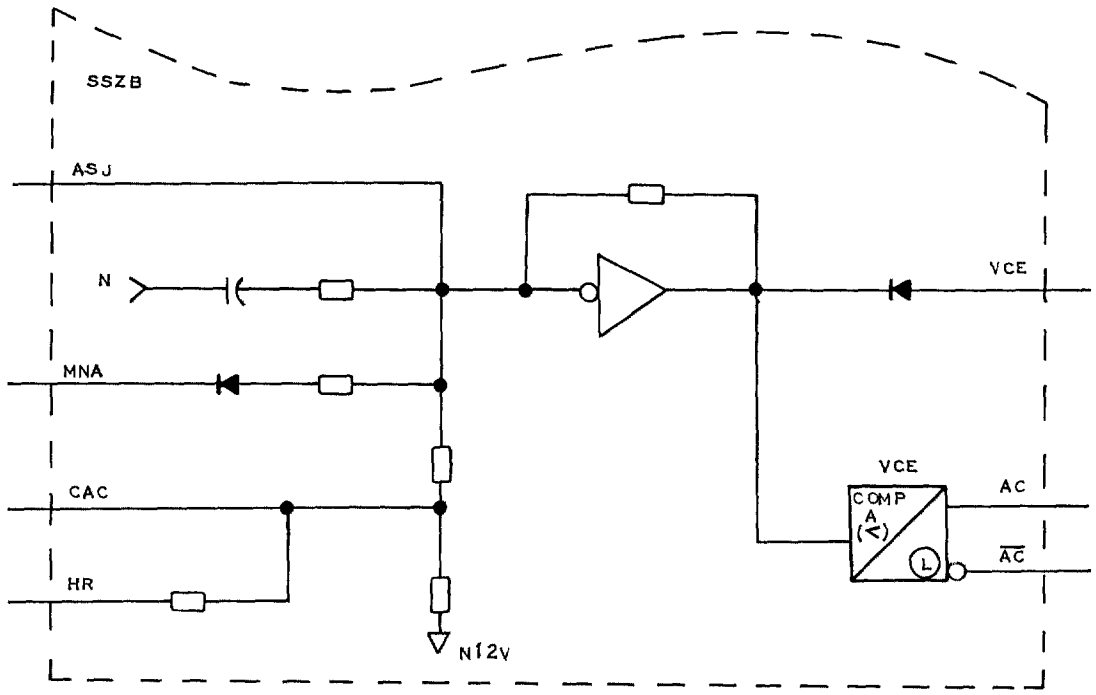
L.P SPEED CONTROL

FIG. C-6

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 52 SH. NO. 51



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C NO	SALEM, VA. U.S.A.	CONT ON SH 53 SH NO. 52



ACCELERATION PORTION OF SSZB

FIG. C-8

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
REV. 2	ISSUED			FIRST MADE FOR		CONTROL	
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 54	SH. NO. 53

C-5 NOZZLE CONTROL (OPTIONAL)

THE NOZZLE CONTROL IS A CLOSED LOOP SYSTEM WHICH REGULATES THE TURBINE'S EXHAUST TEMPERATURE BY MODULATING NCE. NCE MODULATES THE NOZZLE ANGLE WHICH REGULATES THE EXHAUST TEMPERATURE BY VARYING THE HP-SHAFT SPEED. THE NOZZLE REGULATOR CONSISTS OF CARDS SSZB, AND SNCF. (SEE FIG. C-10) THE NOZZLE SERVO VALVE CONTROL IS SIMILAR TO THE GAS CONTROL SERVO VALVE CONTROL AND WILL NOT BE CONVERED SEPARATELY IN THIS PUBLICATION.

SSZB CARD (SEE FIG. C-8, C-9, C-10)

THE SSZB CARD CONTAINS A PULSE RATE TO ANALOG (PR/A) CONVERTER WHICH RECEIVES AN ALTERNATING SIGNAL AT INPUT IN AND PROVIDES A PROPORTIONAL SIGNAL AT OUTPUT N. JUMPERS ARE MADE ON FA, FB, FC AND FD TO PROGRAM THE PR/A CONVERTER TO GIVE THE APPROPRIATE VOLTAGE OUT FOR VARIOUS SPEED MACHINES.

FOR THE STANDARD MANUAL NOZZLE CONTROL R111 IS ADJUSTED SO THAT VCE IS NOT ALLOWED TO INCREASE ABOVE THE VALUE THAT PERMITS 100% SPEED ON THE HP SHAFT.

UNDER THE NOZZLE CONTROL OPTION SJ RECEIVES THE SUM OF NHP AND A TEMPERATURE REFERENCE SIGNAL FROM THE SNCF CARD AND MODULATES OUTPUT NVCE. WHEN SSZB IS CONTROLLING NCE, OUTPUT NC WILL BE A "1" PROVIDED THAT PERMISSIVE TC IS A "1".

A DIFFERENTIATOR CIRCUIT LOOKS AT THE SPEED FEEDBACK SIGNAL N AND MODULATES VCE TO LIMIT THE ACCELERATIONS OF THE TURBINE. OUTPUT AC WILL BE A "1" WHEN VCE IS UNDER ACCELERATION CONTROL. CONNECTING HR TO ACOM WILL HALVE THE ACCELERATION CONTROL RATE.

ADJUSTMENTS:

GAIN (PR/A) - R112
GAIN (NVCE) - R114

SNCF CARD (OPTIONAL) (SEE FIG. C-9, C-10)

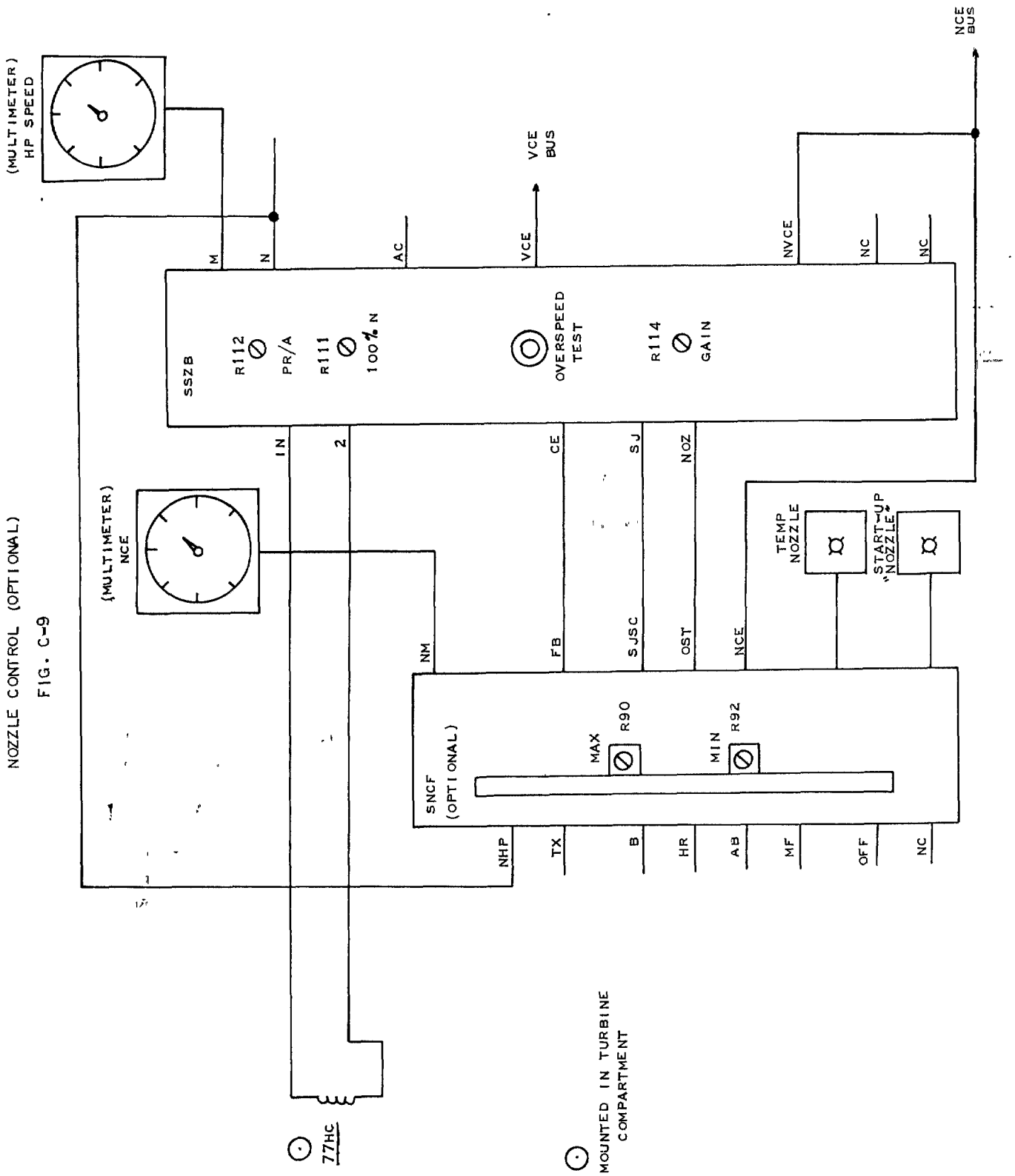
THE SNCF PROVIDES THE DRIVE FOR THE NCE BUS. THE DRIVE IS DEVELOPED SIMILAR TO VCE ON THE SSKC CARD. HOWEVER, THE MAXIMUM LIMIT IS FIXED AND NOT ADJUSTABLE AS VCE IS.

THE SNCF CARD ALSO RECEIVES THE HP SPEED SIGNAL ON INPUT NHP AND SUMS IT WITH A SPEED REFERENCE AND OUTPUTS A SIGNAL ON SJSC TO THE REGULATOR OPERATIONAL AMPLIFIER ON THE SSZB CARD TO MODULATE NCE. THE SPEED REFERENCE SIGNAL (TEMPERATURE ERROR) IS THE SUM OF THE INPUTS ON TX, B AND AB. IF TX PLUS AB IS LESS THAN B, THE REFERENCE CHANGES IN THE POSITIVE DIRECTION TO CLOSE THE NOZZLE AND DECREASE SPEED. IF TX PLUS AB IS GREATER THAN B, THE REFERENCE CHANGES IN THE NEGATIVE DIRECTION TO OPEN THE NOZZLE AND INCREASE HP SPEED. THE TEMPERATURE OPERATING RANGE CAN BE ALTERED VIA HR. THE SPEED REFERENCE SIGNAL IS LIMITED TO A MINIMUM AND MAXIMUM CALLED FOR VALUE.

THE NOZZLE MAY BE FORCED IN THE OPEN POSITION BY INSERTING A NEGATIVE BIAS AT MF. TYPICALLY MF IS CONNECTED TO MIN ON SSKC CARD TO OPEN THE NOZZLE AT MINIMUM VCE. A NEGATIVE BIAS ON ST WILL ALSO FORCE THE NOZZLE OPEN. TYPICALLY, $\overline{ST}$ IS TIED TO THE OVERSPEED TEST PUSH-BUTTON ON THE SSZB CARD TO FACILITATE OVERSPEED TESTING OF THE TURBINE. OUTPUT TC WILL BE A "1" WHEN NOZZLE IS ON TEMPERATURE CONTROL.

ADJUSTMENTS:

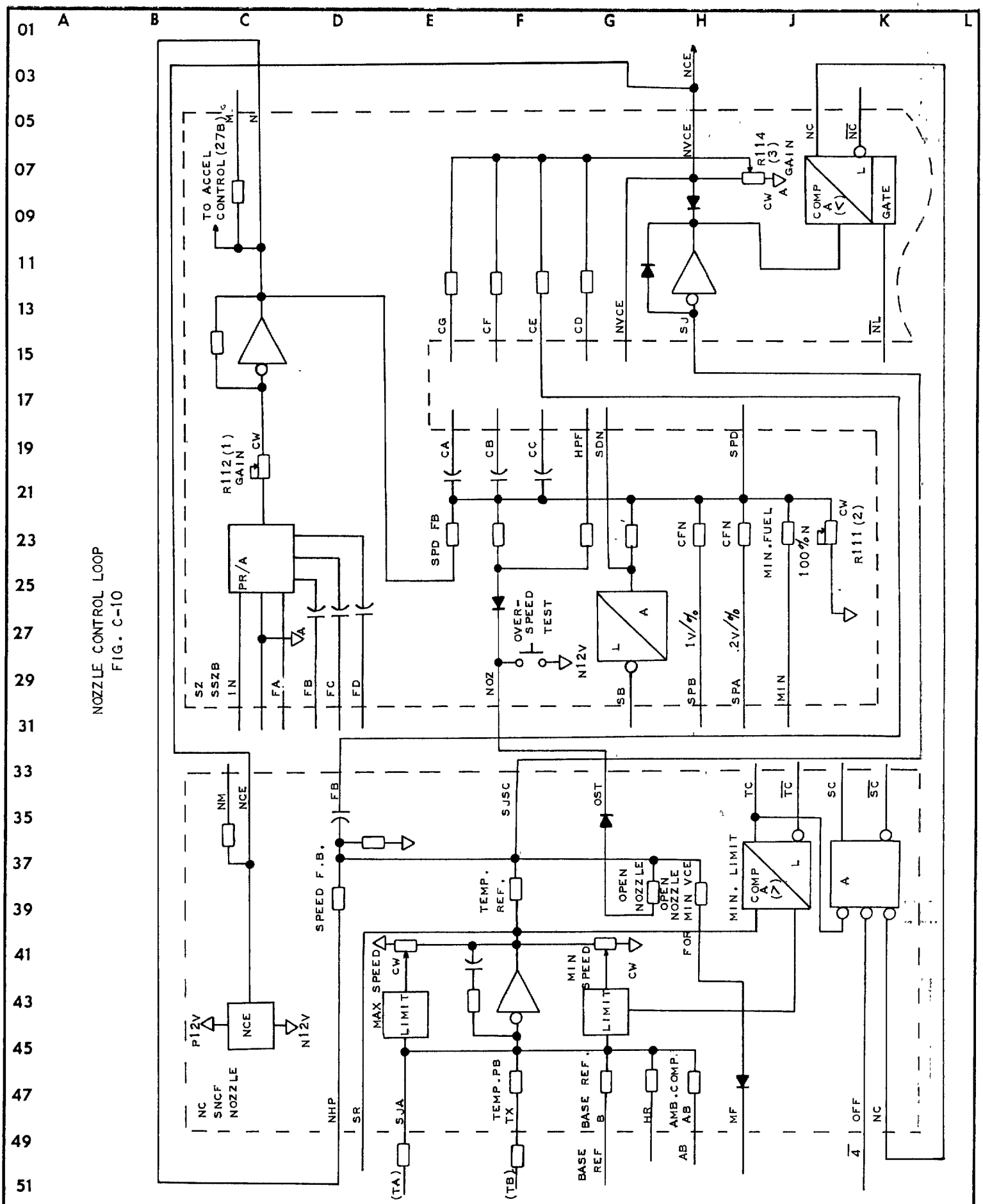
R90-MAX SPEED 80 TO 105%
R92-MIN SPEED 70 TO 100%



NOZZLE CONTROL (OPTIONAL)

FIG. C-9

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH. 57 SH. NO. 56



REV. 1	REV. 4	REV 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO	SALEM, VA. U.S.A.	CONT ON SH 58 SH NO. 57

C-6 - TEMPERATURE

THE TEMPERATURE SIGNALS COME NORMALLY FROM 12 CHROMEL-ALUMEL THERMOCOUPLES PLACED AT SELECTED POINTS OF THE TURBINE EXHAUST DUCT. THE THERMOCOUPLE SIGNALS ARE AVERAGED, COLD JUNCTION COMPENSATED, AND THEN SUMMED WITH REFERENCE SIGNALS TO MODULATE VCE. THE REFERENCE MAY CONSIST OF SEVERAL SIGNALS. NORMALLY THESE SIGNALS ARE BASE REFERENCE AND AMBIENT COMPENSATION. A TYPICAL TEMPERATURE CONTROL LOOP CONSISTS OF CARDS STKJ, STKK, AND A THERMOCOUPLE PROCESSING MODULE. (SEE FIG. C-11, C-12)

THERMOCOUPLE PROCESSING MODULE

THE THERMOCOUPLE PROCESSING MODULE CONSISTS OF TWO STKJ CARDS AND ON SGDE CARD. THE INPUTS TO THE THERMOCOUPLE PROCESSING MODULE ARE TWO SIGNALS, EACH OF WHICH IS AN AVERAGE SIGNAL FROM SIX OF THE TWELVE CHROMEL-ALUMEL THERMOCOUPLES MOUNTED IN THE TURBINE EXHAUST DUCT. THE TWO AVERAGED THERMOCOUPLE SIGNALS ARE COLD JUNCTION COMPENSATED AND AMPLIFIED IN THE STKJ CARDS TO PROVIDE TWO VOLTAGE SIGNALS PROPORTIONAL TO THE EXHAUST DUCT TEMPERATURE. THESE VOLTAGE SIGNALS, TA AND TB, ARE FED TO THE SOTJ CARDS FOR OVERTEMPERATURE PROTECTION AND TO THE STKK CARD FOR TEMPERATURE CONTROL. THE SGDE CARD IN THE THERMOCOUPLE PROCESSING MODULE PROVIDES A FAULT SIGNAL INDICATION IN THE CASE OF THERMOCOUPLE GROUND DETECTION.

STKK CARD (SEE FIG. C-12)

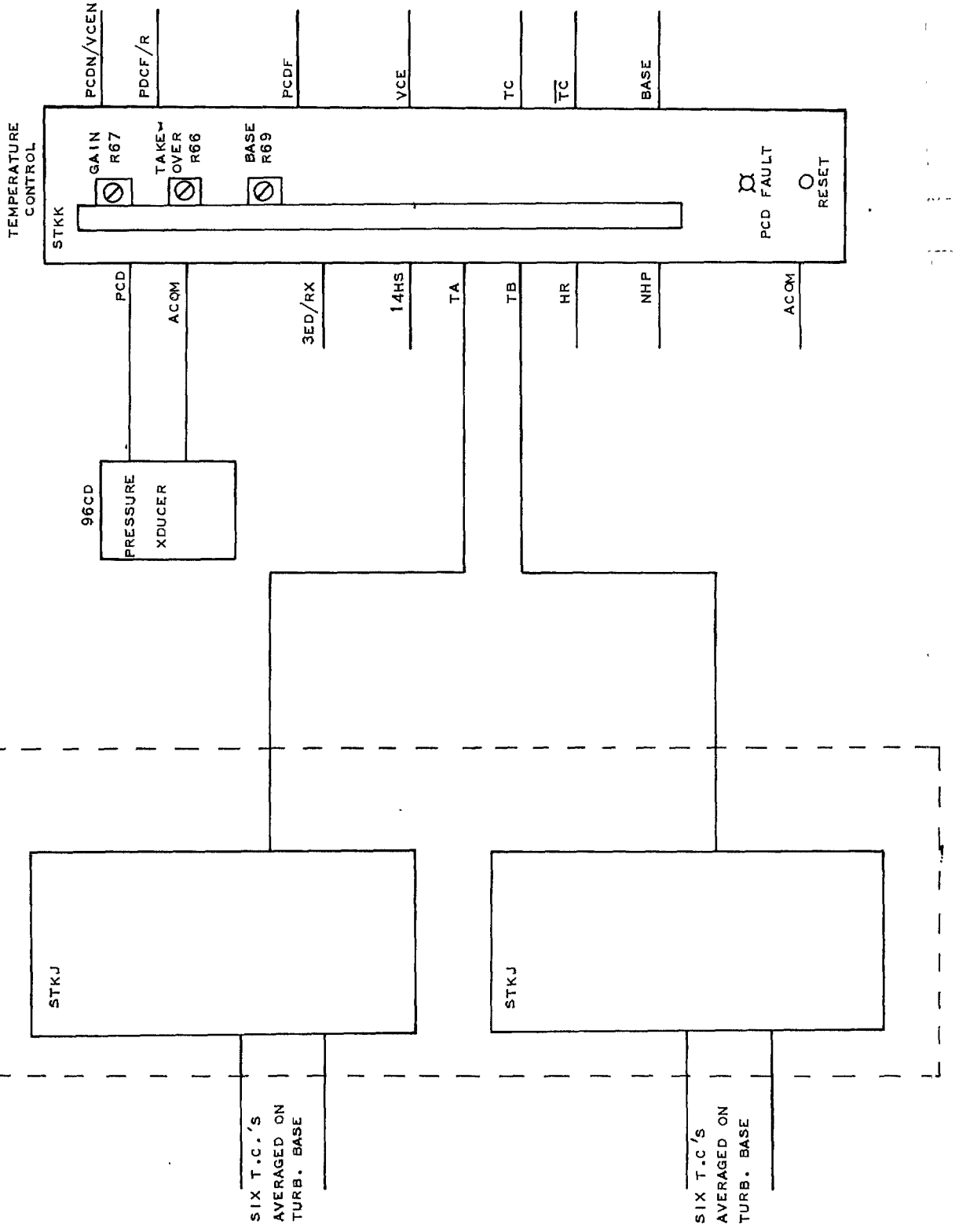
THE STKK CARD RECEIVES THE EXHAUST TEMPERATURE SIGNALS (TA, TB) FROM THE THERMOCOUPLE PROCESSING MODULE. TA AND TB ARE SUMMED WITH THE REFERENCE SIGNALS AND MODULATE OUTPUT VCE TO CONTROL FUEL FLOW. WHEN TA AND TB ARE IN CONTROL OF THE VCE BUS, OUTPUT TC IS A "1" AND $\overline{TC}$ IS "0"/

THE BASE REFERENCE FOR THE CONTROL IS GENERATED ON THE CARD AND IS ADJUSTABLE WITH R69. THE BASE REFERENCE IS ALSO AVAILABLE AT OUTPUT BASE. THE AMBIENT BIAS REFERENCE IS ALSO DEVELOPED ON THE CARD. THE AMBIENT BIAS CIRCUIT CONSISTS OF AN OPERATIONAL AMPLIFIER WITH AN OFFSET AND GAIN ADJUSTMENT. THE INPUT TO THE AMBIENT BIAS CIRCUIT CAN BE A "0" TO 10V SIGNAL ON BVCE OR A "0" TO 5V SIGNAL ON PCD. A 10-50 MA INPUT SIGNAL MAY BE USED BY CONNECTING TO PCD AND JUMPERING PCD TO CSA/B AND CSA. THE OUTPUT OF THE AMBIENT BIAS AMPLIFIER IS AVAILABLE AT OUTPUT PIN PCDN/VCEN. INPUT HR IS AVAILABLE FOR CHANGING THE RANGE OF THE BASE ADJUSTMENT.

ADJUSTMENTS:

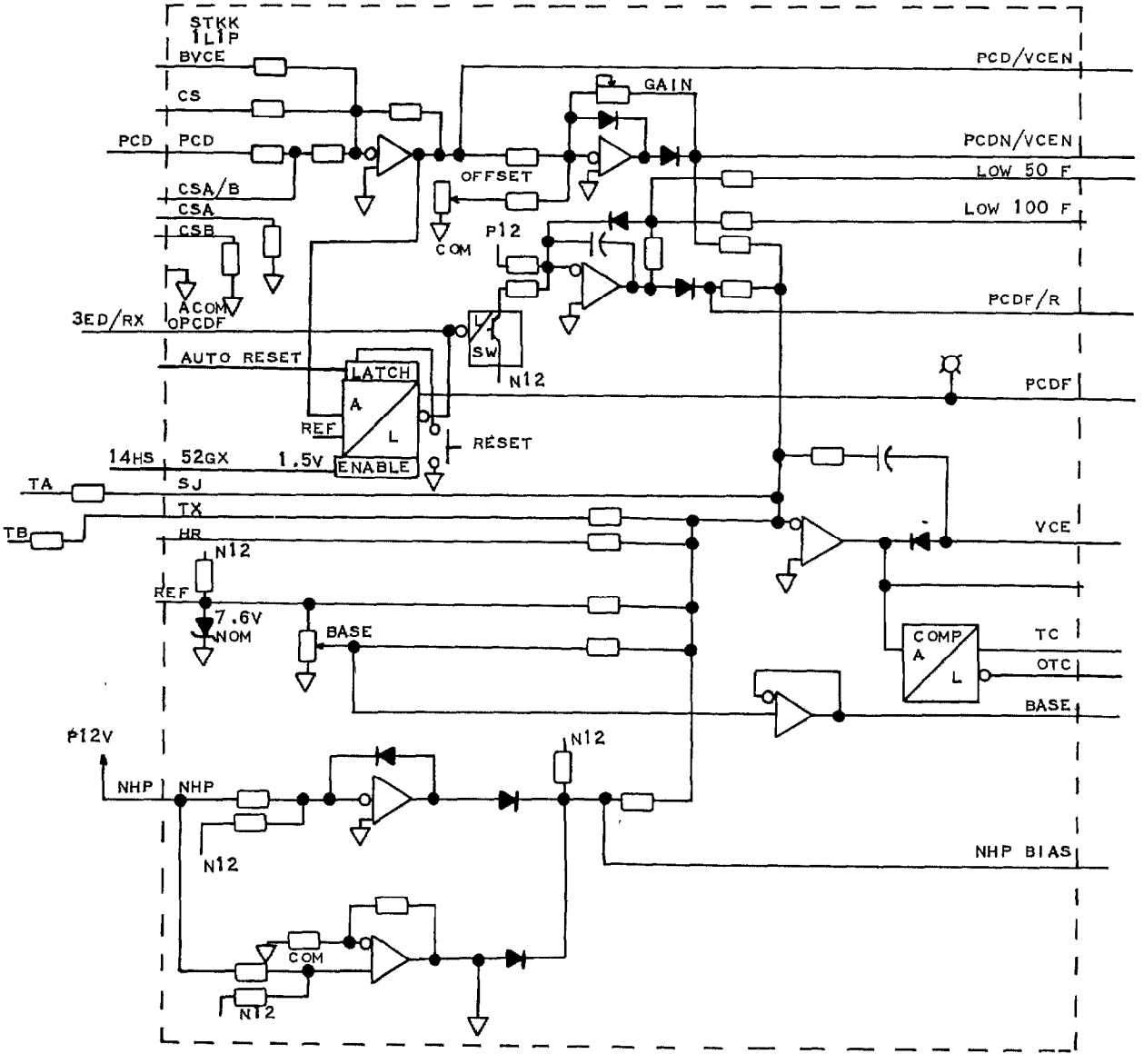
TAKEOVER-R66 - MV/V AMPLIFIER OFFSET ADJUSTMENT
GAIN -R67 - MV/V AMPLIFIER GAIN ADJUSTMENT
BASE -R69 - BASE REFERENCE ADJUSTMENT

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH 59 SH. NO. 58



TEMPERATURE CONTROL
FIG. C-11

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 60 SH NO 59



TEMPERATURE CONTROL LOOP

FIG. C-12

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 62 SH. NO. 60

C-7 FUEL SPLITTER (OPTIONAL)

A FUEL SPLITTER IS PROVIDED FOR SYSTEMS THAT HAVE BOTH GAS AND LIQUID FUEL SYSTEMS. THE TURBINE CONTROL SPECIFICATION AND CONTROL ELEMENTARY DIAGRAM WILL DEFINE THE SPECIFIC TYPE OF FUEL SPLITTER THAT IS USED ON A PARTICULAR SYSTEM.

THE FUEL SPLITTER IS FURNISHED TO PERMIT TRANSFER FROM ON TYPE OF FUEL TO ANOTHER WITHOUT INTERRUPTION OF TURBINE OPERATION. THE FUEL SPLITTER RECEIVES ITS COMMANDS FROM A PANEL MOUNTED SELECTOR SWITCH. A PURGE PERIOD IS AVAILABLE WHEN GOING FROM FULL GAS TO LIQUID OR FROM FULL LIQUID TO GAS. THE FUEL MIXTURE CAN BE STOPPED AT ANY PROPORTION, WHILE IN LOCAL CONTROL. THE FUEL SPLITTER CONSISTS OF CARDS SFKD AND VRCC. THE VRCC CARD WAS DISCUSSED IN THE DIGITAL SETPOINT SECTION. THE SFKD WILL BE COVERED NOW.

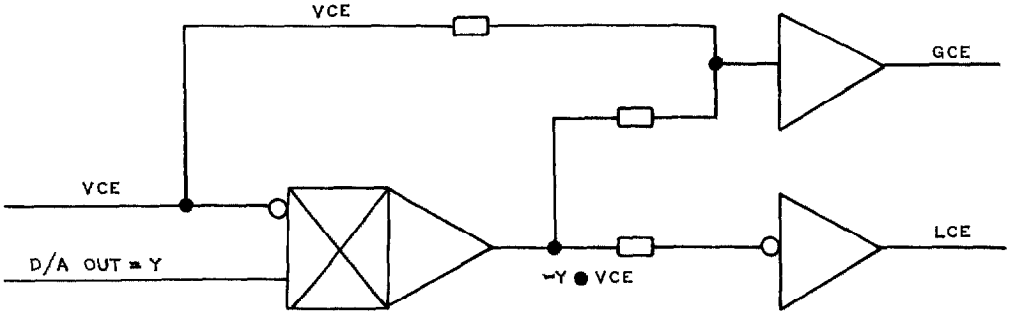
SFKD CARD (SEE FIG. C-13, C-14)

THE SFKD CARD GENERATES THE STEER SIGNALS FOR THE VRCC REVERSIBLE COUNTER ON COMMAND OF THE FUEL SELECTOR SWITCH. A "1" ON GAS WILL STEER THE COUNTER UP (OUTPUT ST1 = "1") AND A "1" ON LIQ WILL STEER THE COUNTER DOWN (OUTPUT ST0 = "1"). OUTPUT PUL PROVIDES COUNT PULSES TO VRCC. PULSES ARE INHIBITED WHEN ANY OF THE FOLLOWING ARE TRUE.

1. LIQ AND GAS ARE BOTH "0" OR "1".
2. INPUT CARRY IS A "1".
3. DGAS IS "0" AND GAS IS "1".
4. DLIQ IS "1", NPG IS "0" AND LIQ IS "1".

INPUTS DLIQ AND DGAS PROVIDES MEANS OF ALLOWING A PURGE TIME AFTER SIGNALS LIQ OR GAS ARE RECEIVED. NPG PROVIDES A WAY TO BYPASS THE PURGE WHEN TRANSFERRING FROM GAS TO LIQUID. INPUT CAP MAY BE USED TO ALTER THE FREQUENCY OF THE COUNT PULSES ON PUL. OUTPUT $\overline{FG}$ IS A "0" WHEN FULLY ON GAS AND $\overline{FL}$ IS A "0" WHEN FULLY ON LIQUID.

THE OUTPUT OF THE VRCC COUNTER IS THE INPUT TO THE D/A CONVERTER ON THE SFKD CARD. THE SIGNAL ON INPUT VCE IS DIVIDED INTO TWO SIGNALS WITH THEIR VALUES DEPENDING ON THE OUTPUT OF THE D/A CONVERTER. A SIMPLIFIED DIAGRAM FOLLOWS. THIS DIAGRAM IS USED ONLY AS AN EXAMPLE AND DOES NOT INCLUDE ALL ASPECTS OF THE ACTUAL CIRCUIT.



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL	
REV. 2	ISSUED			FIRST MADE FOR			
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 64	SH. NO. 63

FOR EXAMPLE, THE SIGNALS VCE, GCE AND LCE ARE EACH BROKEN INTO ACTIVE AND INACTIVE SIGNALS WITH THE INACTIVE SIGNAL BEING THAT PORTION REQUIRED TO CRACK THE VALVE. ALL ABOVE THIS IS CONSIDERED TO BE THE ACTIVE PART. THE ACTUAL CIRCUIT IS DESIGNED SUCH THAT THE ACTIVE PORTION OF LCE AND GCE ARE ALWAYS EQUAL TO THE ACTIVE PORTION OF VCE.

LOOKING AT THE DIAGRAM AND CONSIDERING THE ACTIVE PORTION ONLY, IT IS SEEN THAT

$$LCE = -(Y)\ (VCE).$$

WHERE Y IS PROPORTIONAL TO THE COUNT IN THE COUNTER. ALSO IT IS SEEN THAT

$$GCE = VCE - (Y)\ (VCE).$$

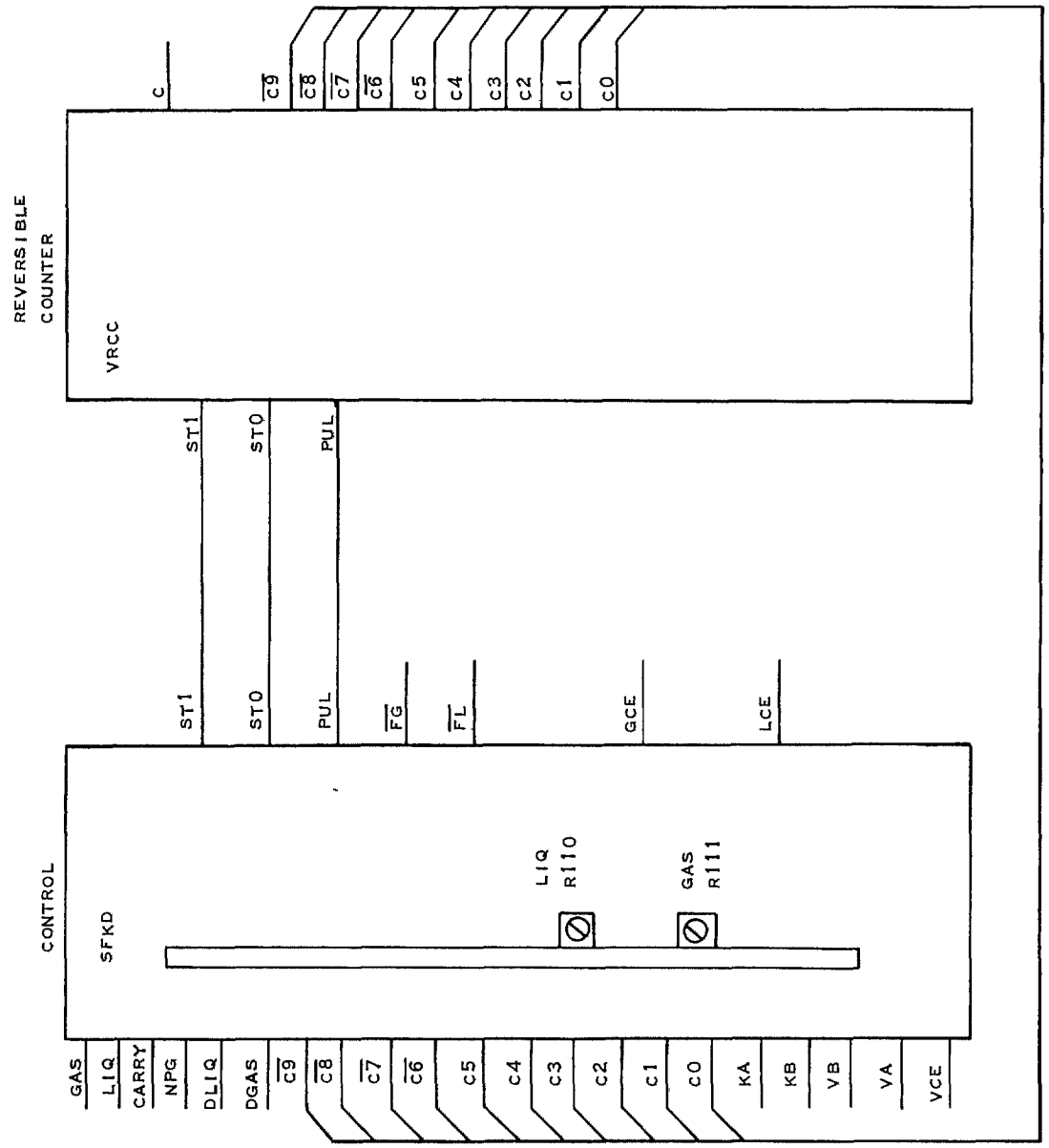
THEREFORE, WHEN Y EQUALS 1, LCE IS EQUAL TO VCE AND GCE EQUALS ZERO. WHEN Y EQUAL ZERO, LCE EQUAL ZERO AND GCE EQUAL VCE. THE COUNTER AND D/A CONVERTER ARE CONNECTED SUCH THAT WHEN THE COUNTER IS FULL, Y IS ZERO, AND WHEN THE COUNTER IS EMPTY, Y EQUAL 1. FOR MIX OPERATION, Y IS SOMEWHERE BETWEEN 0 AND 1. FIG. C-15 THROUGH C-17 (SH. 67) ILLUSTRATE SOME TYPICAL FUEL TRANSFERS.

ADJUSTMENTS:

GAS - R111 - ADJUSTMENT FOR INACTIVE PORTION OF GCE

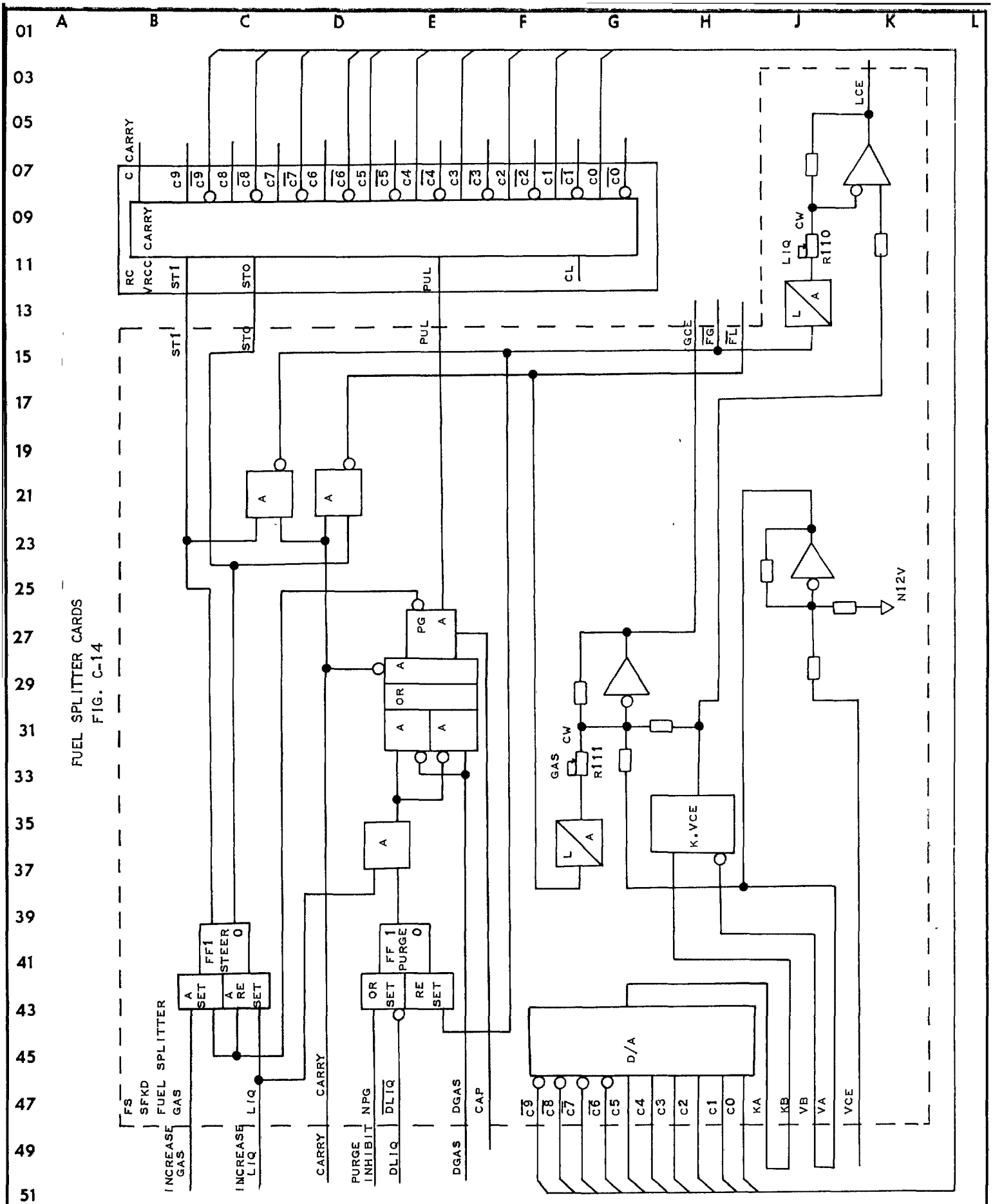
LIQ - R110 - ADJUSTMENT FOR INACTIVE PORTION OF LCE

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 65 SH. NO. 64

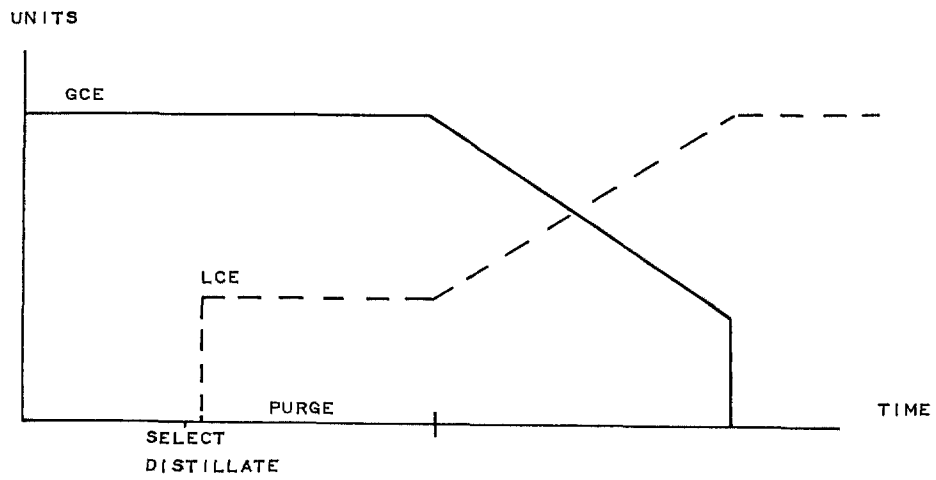


TYPICAL FUEL SPLITTER
FIG. C-13

REV. 1	REV. 4	REV 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 66 SH. NO 65

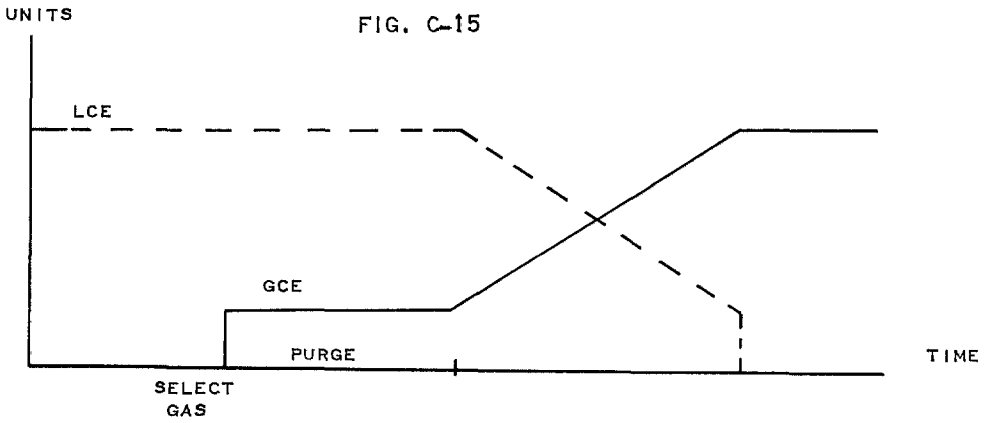


REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 67 SH. NO 66



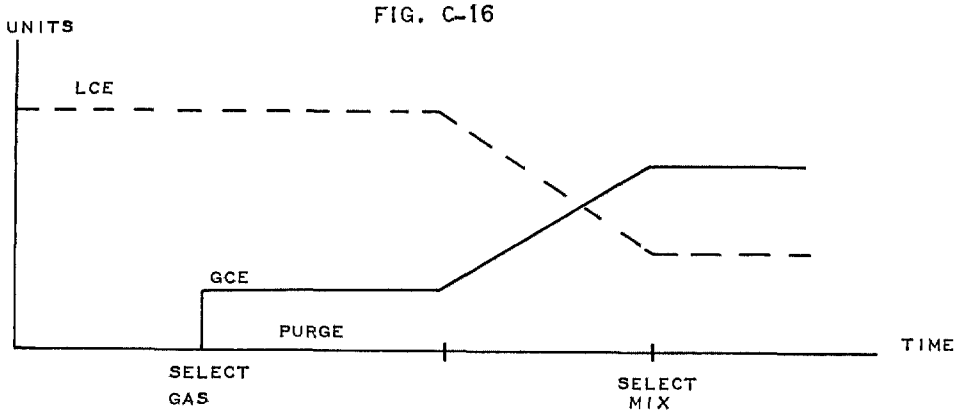
TRANSFER FROM FULL GAS TO FULL DISTILLATE

FIG. C-15



TRANSFER FROM FULL DISTILLATE TO FULL GAS

FIG. C-16



TRANSFER FROM FULL DISTILLATE TO MIXTURE

FIG. C-17

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 68 SH. NO. 67

C-8 LIQUID FUEL CONTROL (OPTIONAL) (SEE FIG. C-18, C-19)

THE LIQUID FUEL CONTROL LOOP CONTAINS THE SSVC CARD WHICH RECEIVES A FEEDBACK SIGNAL AND COMPARES IT TO A FUEL REFERENCE SIGNAL TO MODULATE THE FUEL PUMP SERVO VALVE. THE FEEDBACK IS DEVELOPED USING A LINEAR VARIABLE DISPLACEMENT TRANSFORMER (LVDT) WHOSE COUPLING COEFFICIENCY IS A FUNCTION OF THE FUEL PUMP STROKE. AN OSCILLATOR OF APPROXIMATELY 3KHZ PROVIDES THE INPUT FOR THE LVDT. THE AMPLITUDE OF THE LVDT SECONDARY VARIES PROPORTIONALLY WITH THE FUEL PUMP STROKE.

SSVC CARD (SEE FIG. C-19)

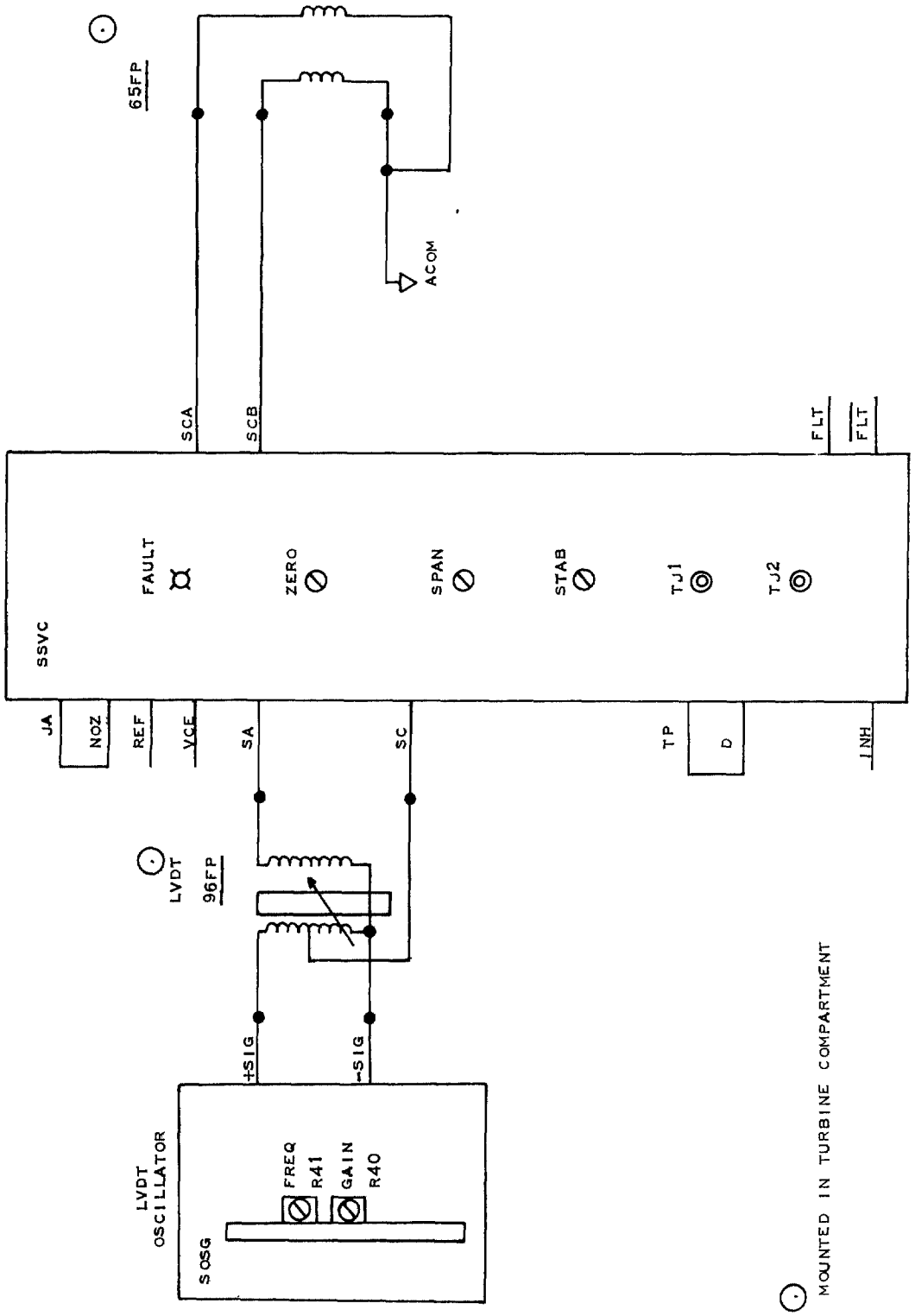
THE SSVC CARD RECEIVES A FEEDBACK SIGNAL FROM THE LVDT ON PINS SA AND SC. A DETECTOR CIRCUIT SENSES THE FEEDBACK VOLTAGE AND HAS GAIN AND OFFSET ADJUSTMENTS TO PROVIDE A FEEDBACK SIGNAL OF -2 VOLTS TO -10 VOLTS FOR ZERO TO FULL PUMP STROKE. A +4 VCE TO +20 VCE (+2 VOLTS TO +10 VOLTS) FUEL REFERENCE SIGNAL (LCE OR VCE) IS APPLIED TO THE REF INPUT. THE REFERENCE AND FEEDBACK SIGNALS ARE SUMMED AND AMPLIFIED TO PROVIDE A SIGNAL TO MODULATE THE FUEL PUMP SERVO VALVE POSITION. A COMPARATOR CIRCUIT CHECKS THE VOLTAGE OUTPUT OF THE FEEDBACK DETECTOR.

A COMPARATOR CIRCUIT CHECKS THE VOLTAGE OUTPUT OF THE FEEDBACK DETECTOR. IF THE DETECTOR OUTPUT EVER DROPS TO BETWEEN 0 AND -1.2 VOLTS, INDICATING A LOSS OF FEEDBACK, THE COMPARATOR WILL ALARM. THE COMPARATOR ALSO ALARMS IF FEEDBACK VOLTAGE DIFFERS BY MORE THAN 0.8 VOLTS FROM THE PROPER -2.0 VOLT VALUE WHENEVER THE TURBINE IS NOT OPERATING ON LIQUID FUEL OR SPEED DROPS BELOW 14HM.

ADJUSTMENTS:

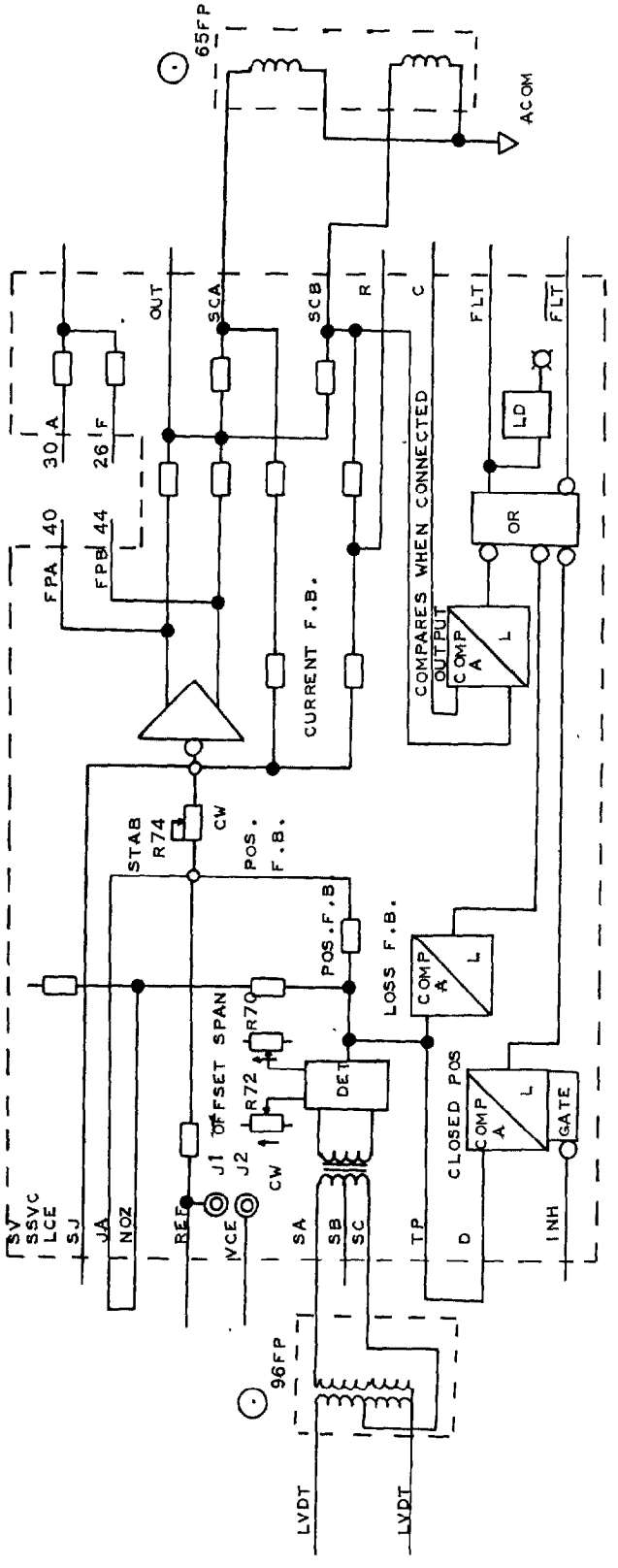
SPAN. - R70
OFFSET - R72
STAB - R74

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
REV. 2	ISSUED			FIRST MADE FOR		CONTROL	
REV. 3	MADE BY			I.C. NO	SALEM, VA. U.S.A.	CONT. ON SH. 69	SH. NO. 68



TYPICAL LIQUID FUEL SERVO
FIG. C-16

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 70 SH NO 69



LIQUID FUEL CONTROL

FIG. C-19

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO	SALEM, VA. U.S.A.	CONT. ON SH 71 SH NO. 70

C-9 GAS CONTROL VALVE (SEE FIG. C-20, C-21)

THE GAS CONTROL LOOP CONTAINS AN SSVC CARD WHICH RECEIVES A FEEDBACK SIGNAL AND COMPARES IT TO A FUEL REFERENCE SIGNAL TO MODULATE THE GAS CONTROL SERVO VALVE. THE FEEDBACK IS DEVELOPED USING A LINEAR VARIABLE DISPLACEMENT TRANSFORMER (LVDT) WHOSE COUPLING COEFFICIENCY IS A FUNCTION OF THE POSITION OF THE GAS CONTROL VALVE. AN OSCILLATOR OF APPROXIMATELY 3 KHZ PROVIDES THE INPUT FOR THE LVDT. THE AMPLITUDE OF THE LVDT SECONDARY VARIES DIRECTLY WITH THE POSITION OF THE GAS CONTROL VALVE.

SSVC CARD (SEE FIG. C-21)

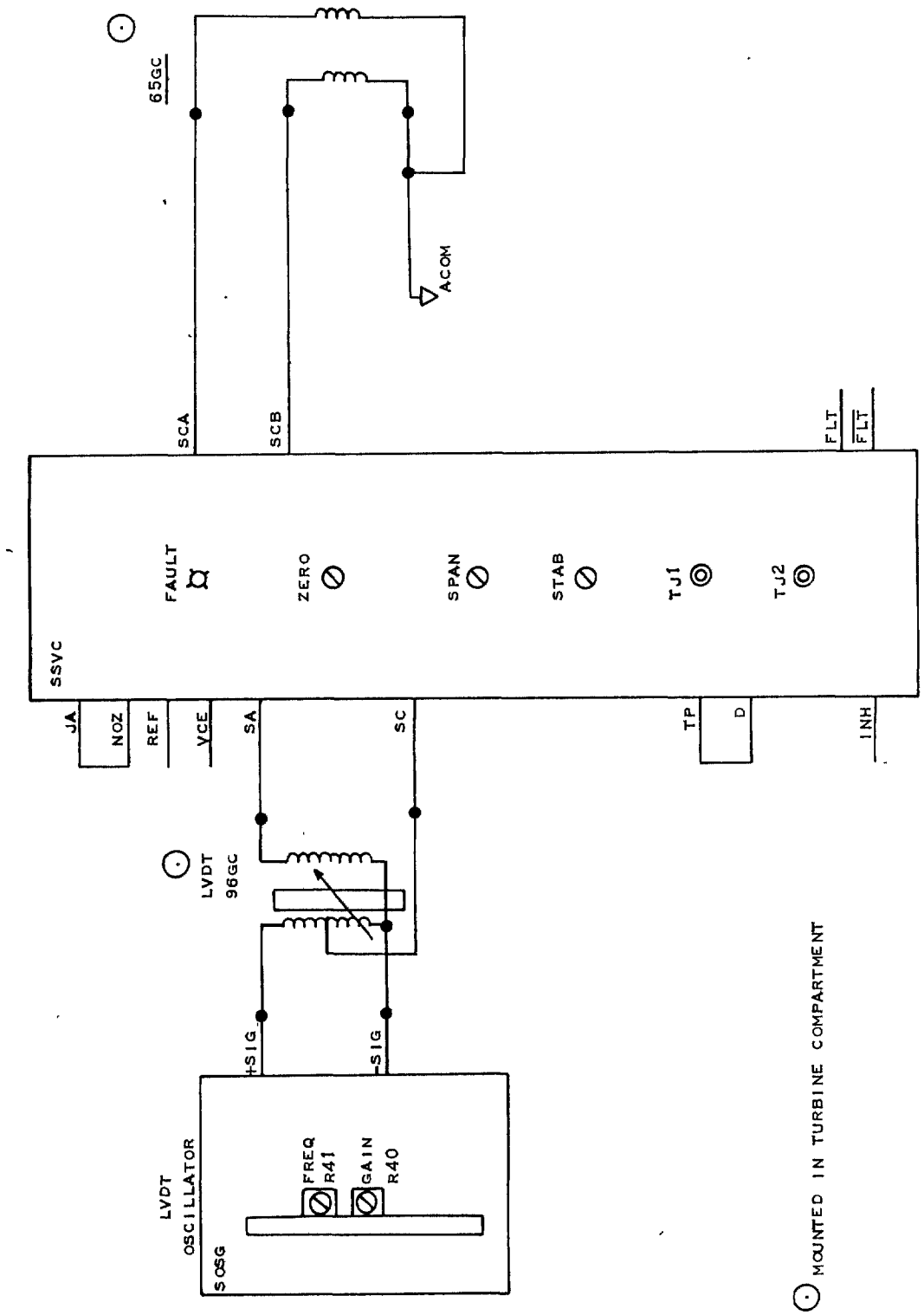
THE SSVC CARD RECEIVES A FEEDBACK SIGNAL FROM THE LVDT ON PINS SA AND SC. A DETECTOR CIRCUIT SENSES THE FEEDBACK VOLTAGE AND HAS GAIN AND OFFSET ADJUSTMENTS TO PROVIDE A FEEDBACK SIGNAL OF -2 VOLTS -10 VOLTS FOR COMPLETELY CLOSED TO COMPLETELY OPEN. (GCE OR VCE) IS APPLIED TO THE REF INPUT. THE REFERENCE AND FEEDBACK SIGNALS ARE SUMMED AND AMPLIFIED TO PROVIDE A SIGNAL TO MODULATE THE GAS CONTROL SERVO VALVE.

A COMPARATOR CIRCUIT CHECKS THE OUTPUT OF THE FEEDBACK DETECTOR. IF THE DETECTOR OUTPUT EVER DROPS TO BETWEEN 0 AND -1.2 VOLTS, INDICATING A LOSS OF FEEDBACK, THE COMPARATOR WILL ALARM. THE COMPARATOR ALSO ALARMS IF THE DETECTOR OUTPUT DIFFERS BY MORE THAN 0.8 VOLTS FROM THE PROPER -2.0 VOLT VALUE WHENEVER THE TURBINE IS BELOW 14HM OR IS NOT OPERATING ON GAS FUEL.

ADJUSTMENTS:

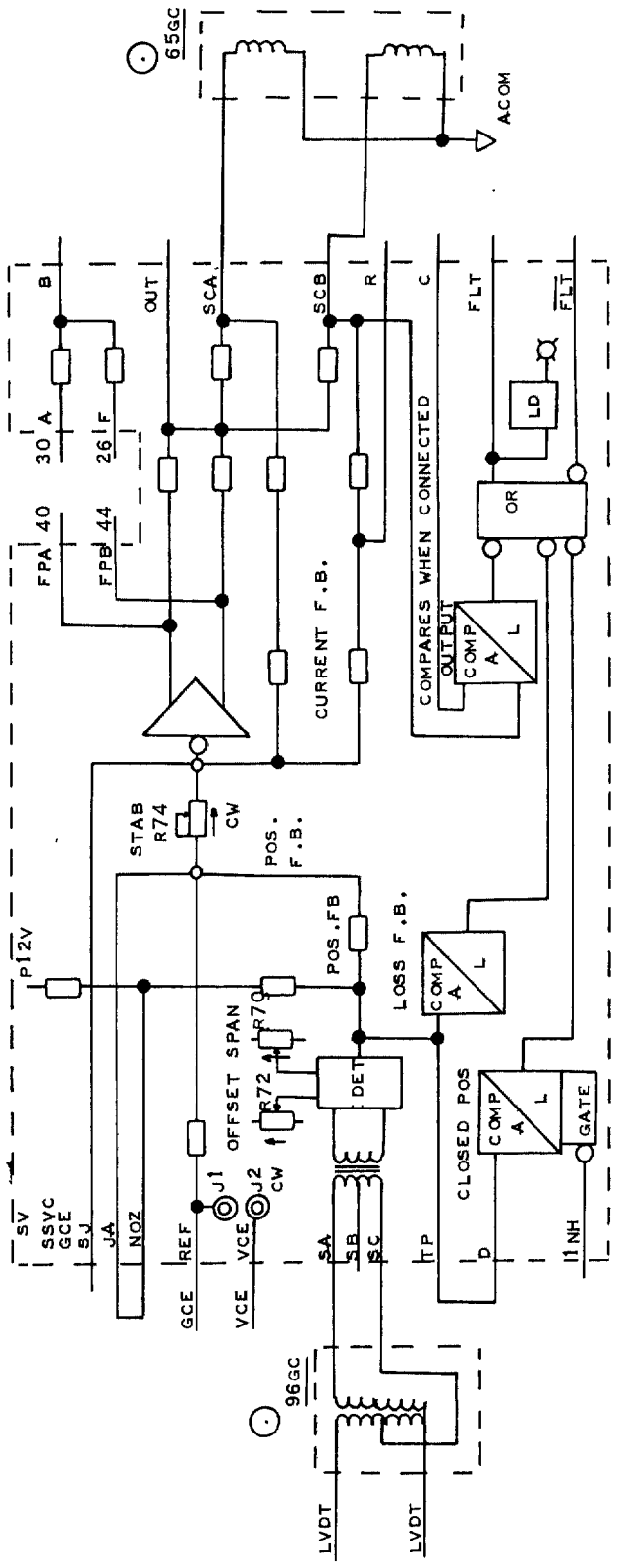
SPAN - R70
OFFSET - R72
STAB - R74

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM	
REV. 2	ISSUED			FIRST MADE FOR		CONTROL	
REV. 3	MADE BY			I C NO.	SALEM, VA. U.S.A.	CONT. ON SH 72	SH. NO. 71



TYPICAL GAS CONTROL VALVE
FIG. C-20

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I C NO.	SALEM, VA. U.S.A.	CONT. ON SH. 73 SH. NO. 72



GAS FUEL CONTROL

FIG. C-21

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 74 SH. NO 73

C-10 STOP/SPEED RATIO VALVE

WHEN A GAS FUEL SYSTEM IS USED, A STOP/SPEED RATIO VALVE IS REQUIRED TO MAINTAIN THE PRESSURE UP STREAM OF THE GAS CONTROL VALVE AT A VALUE THAT IS A FUNCTION OF TURBINE SPEED (SEE FIG. C-22, C-23)

SSVD CARD (SEE FIG. C-23)

THE SSVD RECEIVES A SIGNAL ON P1 THAT IS PROPORTIONAL TO PRESSURE. THIS SIGNAL IS COMPARED WITH THE SPEED SIGNAL ON NHP AND PROVIDES AN OUTPUT ON TRD. AN OFFSET AND GAIN IS PROVIDED ON THE NHP SIGNAL. THE SIGNAL ON TRD IS THE REFERENCE FOR THE SERVO VALVE REGULATOR ON THE CARD. A "1" ON INPUT SR WILL FORCE TRD TO ZERO VOLTS. THE FEEDBACK SIGNAL FOR THE REGULATOR IS DERIVED USING A LVDT AS WAS DONE IN THE GAS CONTROL VALVE. ONLY ONE LVDT IS USED IN THIS CASE. ALARM OUTPUT FLT WILL BE A "1" AND FAULT LIGHT WILL BE ON FOR THE FOLLOWING:

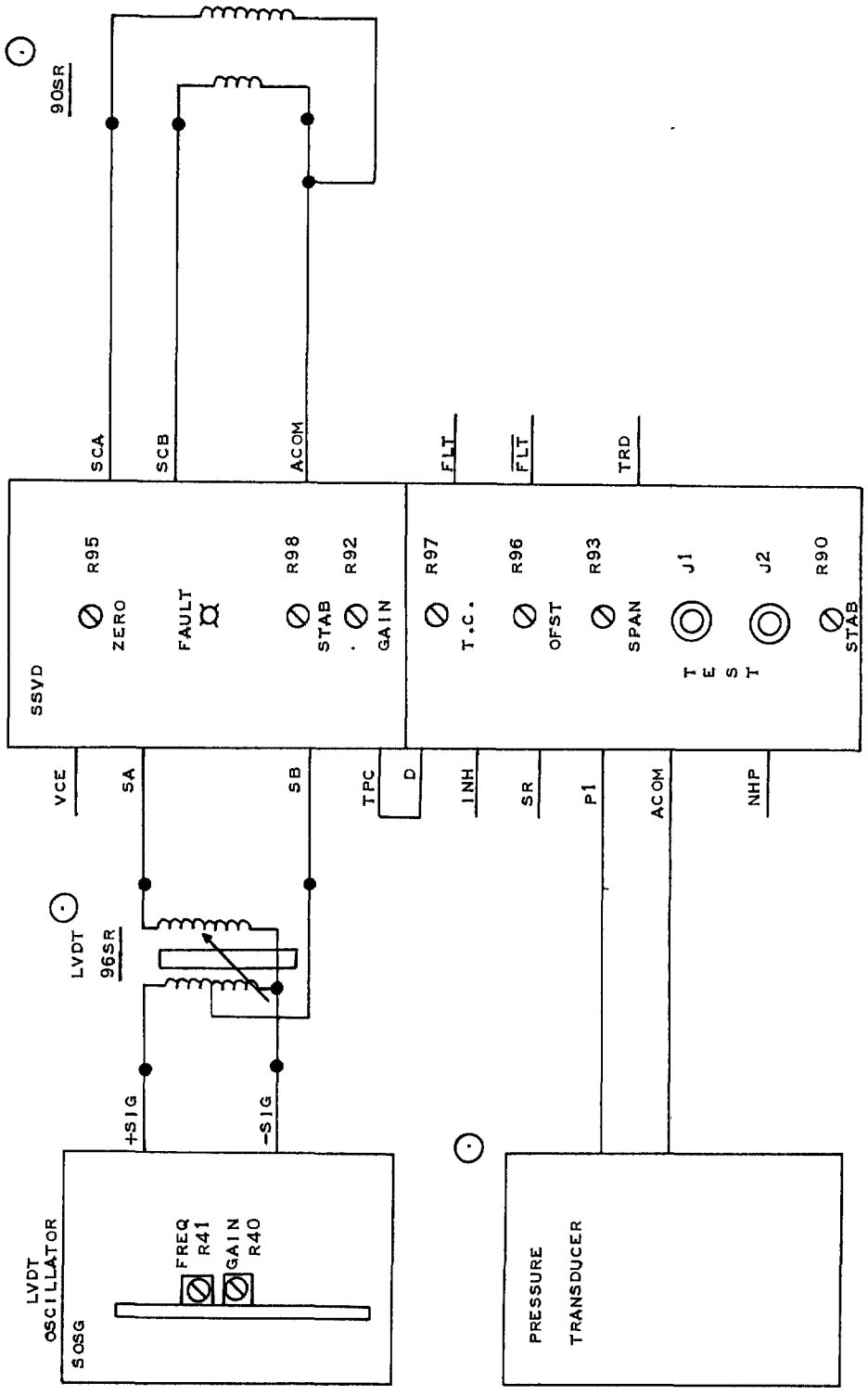
(1) IF TPC GOES MORE POSITIVE THAN -1.2 VOLTS INDICATING LOSS OF FEEDBACK.

(2) IF INPUT D IS NOT BETWEEN -1.2 AND -2.8V AND INPUT INH IS A "0".

ADJUSTMENTS:

ZERO	-R95
GAIN	-R92
PRESS SPAN	-R93
NHP OFFSET	-R96
STAB	-R90

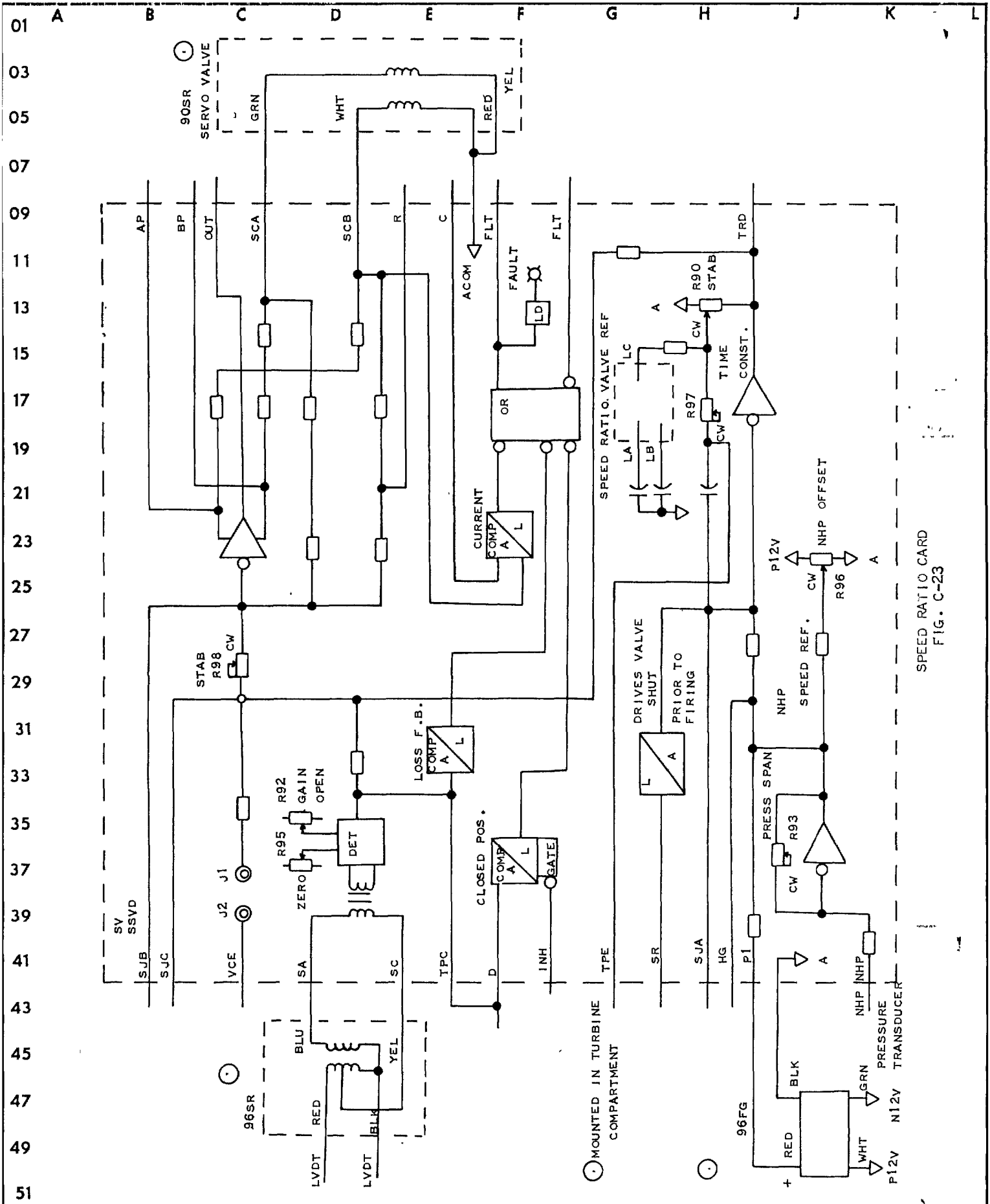
REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH. 75 SH. NO. 74



MOUNTED IN TURBINE COMPARTMENT

TYPICAL SPEED RATIO VALVE SERVO
FIG. C-22

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		CONTROL
REV. 3	MADE BY			I C NO.	SALEM, VA. U.S.A.	CONT. ON SH. 76 SH. NO. 75



SPEED RATIO CARD
FIG. C-23

REV. 1	REV 4	REV 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM CONTROL
REV. 2	ISSUED			FIRST MADE FOR		
REV 3	MADE BY			I C NO.	SALEM, VA. U.S.A.	CONT ON SH 77, SH NO. 76

SECTION D

PROTECTION

D-1 "4" CIRCUIT

THE FOCAL POINT OF THE PROTECTION SYSTEM IS THE VMPA CARD WHICH GENERATES THE "4" SIGNAL AND THE SHUTDOWN BIAS TO THE SERVO-VALVES. (DUE TO THE MAGNITUDE OF ITS IMPORTANCE, A DETAILED DESCRIPTION WILL BE PRESENTED.) THE SYMBOL FOR THE VMPA CARD IS SHOWN IN FIGURE D-1. TO HELP ELIMINATE SOME CONFUSION, A ("*") IS PLACED NEAR THOSE PINS NOT NORMALLY USED IN INDUSTRIAL APPLICATIONS. A TYPICAL PROTECTION SYSTEM ILLUSTRATING THE "4" BUS IS SHOWN IN FIGURE D-2.

THE VMPA SERVES THE DUAL FUNCTION OF MONITORING INPUT SIGNALS AS PERMISSIVES TO START THE TURBINE AND AS PROTECTIVE SIGNALS TO TRIP THE TURBINE. THE "4" SIGNAL OUTPUTS ARE A FUNCTION OF A FLIP-FLOP WHICH IS SET UPON RECEIPT OF A START COMMAND WHEN ALL PERMISSIVES HAVE BEEN SATISFIED. WITH THE FLIP-FLOP SET, THE "4" SIGNAL OUTPUTS PERMIT THE TURBINE TO BE STARTED AND OPERATED. THE FLIP-FLOP IS RESET WHEN AN ABNORMAL OPERATING CONDITION IS DETECTED BY PROTECTIVE MONITORING DEVICES. RESETTING THE FLIP-FLOP TRIPS THE TURBINE. THE VMPA FUNCTIONS, INPUTS AND OUTPUTS, ARE DISCUSSED BELOW. THE DISCUSSION IS REFERENCED TO FIGURE D-1 BY NUMBERS SHOWN INSIDE A CIRCLE.

SUM "1" GATE

THE SUM "1" GATE (REFER TO (1) ON SYMBOL) GENERATES THE "START-CHECK" PERMISSIVE LEG TO SET THE FLIP-FLOP. THE PERMISSIVE IS SATISFIED WHEN ALL INPUTS TO THE SUM "1" GATE ARE "1"S. ONCE THE FLIP-FLOP IS SET, THIS PERMISSIVE IS NO LONGER REQUIRED. AN "AND EXPANDER" POINT, SAE, IS PROVIDED TO EXPAND THE SUM "1" GATE BY THE USE OF LGXA CARDS.

INPUTS: 14X, $\overline{14HMX}$, 12HT-1, 12HT-4, 12LT-1, 12LT-4, 14HSX (SIGNAL INVERTED) SAE.

OUTPUT: $\overline{STCK}$ - GOES TO LOGIC "0" WHEN ALL INPUTS TO SUM "1" GATE ARE "1"S.

MAJORITY LOGIC

THERE ARE TWO MAJORITY LOGIC ELEMENTS IN THE VMPA- ONE FOR HP OVERSPEED (2), AND ANOTHER FOR LP OVERSPEED (3). FOR THE MS1002 ONLY TWO OF THE THREE INPUTS TO EACH OF THE TWO LOGIC ELEMENTS ARE USED. THEREFORE, THE OUTPUT OF THE HP LOGIC WILL BE A "0" IF EITHER 12HT-1 OR 12HT-4 IS A "1". THE LP LOGIC IS "OR"'ED WITH A SPEED SIGNAL "AND" CIRCUIT WHICH BYPASSES THE MAJORITY LOGIC ELEMENT OUTPUT DURING START-UP. WHEN IN THE RUN MODE, A "1" ON EITHER 12LT-1 OR 12LT-4 WILL GIVE A "1" FOR A TRIP SIGNAL ON THE LP LOGIC OUTPUT.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		PROTECTION
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH. 78 SH. NO 77

MAJORITY LOGIC (CONTINUED)

HP LOGIC INPUTS: 12HT-1, 12HT-4

HP LOGIC OUTPUT: INTERNAL OUTPUT TO THE START/RUN GATE.

LP LOGIC INPUTS: 12LT-1, 12LT-4, 14LS, 14HSX

LP LOGIC OUTPUT: 12LT (CONNECTED TO LT INPUT OF THE START/RUN GATE)

SUM "0" GATE

THE SUM "0" GATE (4) MONITORS TRIP PROTECTIVE DEVICES. THE GATE OUTPUTS A "1" WHEN ALL INPUTS ARE "0"S. THIS "1" OUTPUT IS A PERMISSIVE TO OPERATE CONDITION. IF ANY INPUT GOES TO A "1" STATE, THE OUTPUT WILL BECOME A "0" AND TRIP THE TURBINE. THEREFORE, ANY UNUSED INPUT MUST BE TIED TO COMMON TO ALLOW THE TURBINE TO RUN. THE OUTPUT OF THE SUM "0" GATE IS CONNECTED TO THE OUTPUT OF THE START/RUN GATE LOGIC TO FORM A HARDWIRED "OR". APPROXIMATELY 0.7 MSEC AFTER THE TRIP INPUT SIGNAL IS INITIATED, THE TRIP BUS WILL BECOME A "0" TO TRIP THE TURBINE.

INPUTS: T01, T02, T03, T04, T05, T06, T07, T08, T09, T10, T11, T12, T13.

OUTPUTS: INTERNALLY TIED TO OUTPUT OF START/RUN GATE LOGIC. COMBINED OUTPUT OPERATES A TIME DELAY CIRCUIT, WHICH IS CONNECTED TO THE TRIP BUS. THIS OUTPUT ALSO DRIVES THE $\overline{PS}$ OUTPUT, WHICH IS A LOGIC "0" WHEN THE SUM "0" GATE AND THE START/RUN GATE ARE SATISFIED.

START/RUN GATE

THE START/RUN GATE CONSISTS OF A GATE FOR START-UP (5) AND A GATE FOR RUNNING (6). THE OUTPUTS OF THESE TWO GATES ARE "OR" ED TO SUPPLY AN OUTPUT WHICH IS HARDWIRE "OR"ED WITH THE OUTPUT OF THE SUM "0" GATE PREVIOUSLY DESCRIBED.

THE START-UP PORTION IS AN "AND" GATE WITH "OR" EXPANSION CAPABILITIES. THIS GATE MUST BE SATISFIED PRIOR TO AND DURING THE START-UP SEQUENCE. THE "OR" EXPANDER PORTION IS USED TO MAINTAIN THE OUTPUT DURING THE FIRING PERIOD. THIS GATE IS DE-ENERGIZED WHEN ITS INPUTS CHANGE STATE, NORMALLY UPON DETECTION OF FLAME (28FDX). THE OUTPUT HAS A 0.5 SECOND TIME DELAY ON TRANSITION FROM A "1" TO A "0" TO ALLOW THE RUN GATE TO BE ENERGIZED AND MAINTAIN TURBINE OPERATION. THE OUTPUT ALSO HAS A 40 MILLISECOND TIME DELAY ON TRANSITION FROM A "0" TO A "1" TO INSURE TRIPPING THE TURBINE ON LOSS OF FLAME.

INPUTS: $\overline{14HSX}$ FROM INTERNAL INVERTER, $\overline{28FDX}$, ST, OR EXP.

OUTPUTS: INTERNAL CONNECTION TO "OR" GATE WITH RUN PORTION AND INVERTED OUTPUT "$\overline{SM}$" FOR INDICATION OF START-UP MODE SATISFIED.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROTECTION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH. 79 SH. NO. 78

START/RUN GATE (CONTINUED)

THE RUN MODE PORTION OF START/RUN GATE PROVIDES MONITORING OF PROTECTIVE TRIP SIGNALS WHICH MUST BE BYPASSED DURING START-UP. THIS RUN GATE IS AN "AND" GATE USED FOR MONITORING HP OVERSPEEDS, LP OVERSPEEDS, AND HYDRAULIC DUMP PRESSURE. THESE SIGNALS ARE MONITORED FOR PERMISSABLE STATE AFTER FLAME HAS BEEN ESTABLISHED. A TRIP STATE ON ANY ONE OF THE INPUTS, INCLUDING LOSS OF FLAME, WILL TRIP THE TURBINE.

INPUTS: INTERNAL HP OVERSPEED SIGNAL FROM MAJORITY LOGIC, "$\overline{\text{LT}}$" NORMALLY CONNECTED TO OUTPUT "12LT" FROM LP MAJORITY LOGIC, "$\overline{\text{63HD}}$", "$\overline{\text{28FDX}}$".

OUTPUTS: INTERNAL CONNECTION TO ("OR") GATE WITH START-UP PORTION, AND INVERTED OUTPUT "RM" FOR INDICATION OF RUN MODE SATISFIED.

MASTER FLIP-FLOP

THE STATE OF THE MASTER FLIP-FLOP (7) DETERMINES WHETHER THE UNIT MAY RUN OR IS TRIPPED. SETTING THE FF PERMITS THE UNIT TO OPERATE. RESETTING THE FF TRIPS THE TURBINE.

THE FF IS SET BY A FOUR INPUT "AND" GATE. THIS GATE SETS THE FF ONLY WHEN THE START-UP CHECKS ARE SATISFIED, LUBE OIL PRESSURE IS ESTABLISHED, THE "TRIP BUS" IS SATISFIED, AND A START SIGNAL HAS BEEN RECEIVED.

THE FF IS RESET WHEN THE "TRIP BUS" BECOMES DISSATISFIED (LOGIC "0" STATE). THE "TRIP BUS" CAN BE DRIVEN TO THE TRIP STATE BY SEVERAL METHODS. DETECTION OF A TRIP CONDITION BY THE SUM "0" GATE WILL DRIVE THE "TRIP BUS" TO THE TRIP STATE. DROPPING OUT THE START-UP/RUN GATE WILL ALSO TRIP THE TURBINE. LOSS OF LUBE OIL PRESSURE, 63QT, OR LOSS OF POWER AS DETECTED BY INTERNAL SENSORS WILL ALSO CAUSE A TRIP. THE "TRIP BUS" TRIPS THE TURBINE ON NORMAL SHUTDOWNS WHEN INPUT TG BECOMES A LOGIC "1" AS A RESULT OF EXTERNAL LOGIC SEQUENCING. OPERATING THE EMERGENCY STOP PUSH-BUTTON TRIPS THE UNIT BY DISCONNECTING 28V POWER FROM INPUT "ESTOP" AND TRIGGERING A VOLTAGE SENSOR CIRCUIT CONNECTED TO THE "TRIP BUS".

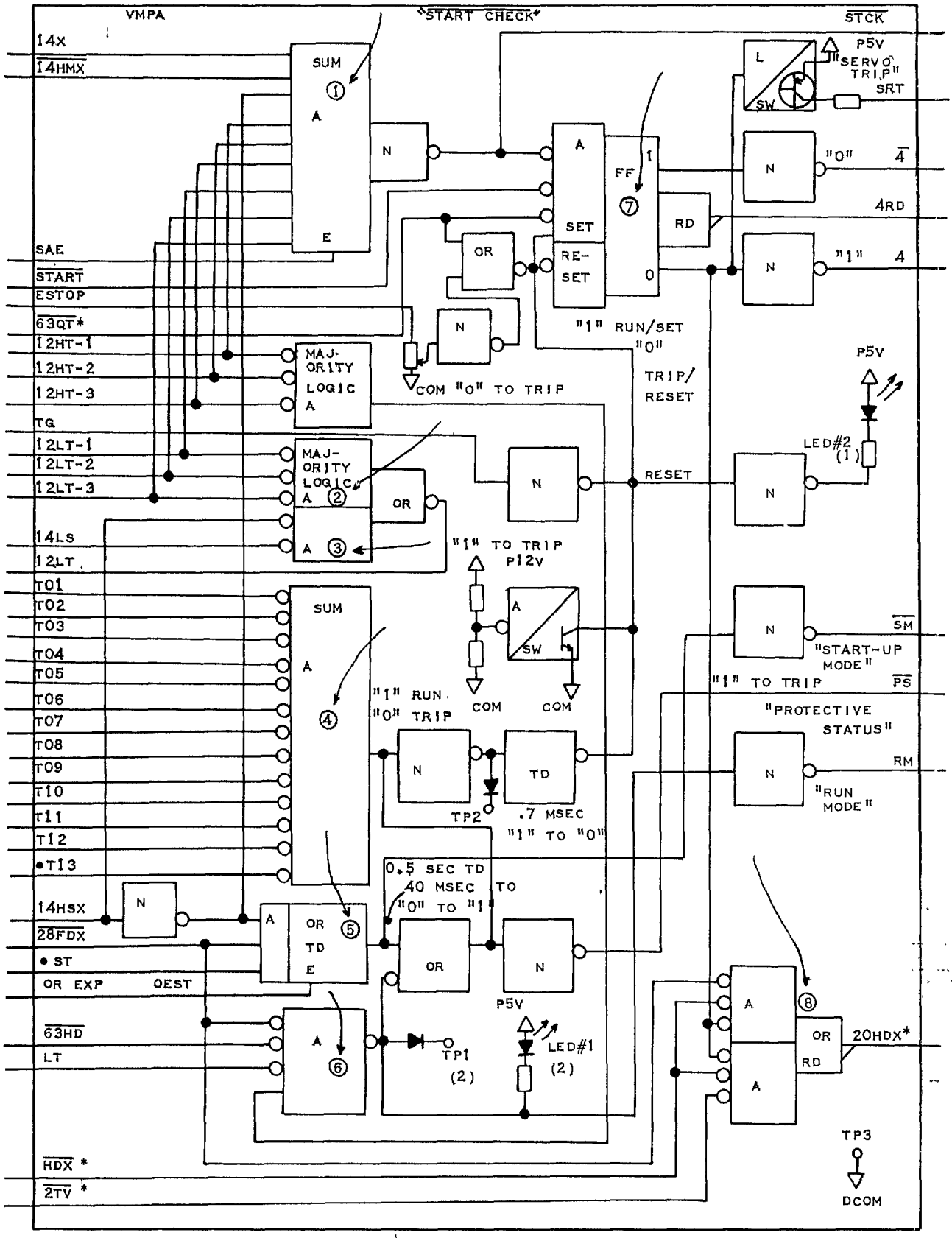
INPUTS: $\overline{\text{START}}$, ESTOP, $\overline{\text{63QT}}$, TG, AND INTERNAL OUTPUTS FROM THE SUM "1" GATE, SUM "0" GATE, START-UP/RUN GATE, AND POWER SUPPLY VOLTAGE SENSING CIRCUITS.

OUTPUTS: 4, $\overline{4}$-LOGIC OUTPUTS FOR SEQUENCING.
4RD -RELAY DRIVER OUTPUT TO ENERGIZE A RELAY WHEN THE FF IS SET.
SRT -SERVO TRIP BIAS TO DRIVE CONTROL VALVES TO THEIR PROPER SHUTDOWN STATE WHEN THE TURBINE IS TRIPPED.

HYDRAULIC PUMP CIRCUIT

THE HYDRAULIC SOLENOID MAY BE CONTROLLED BY THIS DRIVER OUTPUT THROUGH A RELAY, (8). TWO "AND" FUNCTIONS ARE "OR"ED. THE LOWER "AND" PROVIDES AN OUTPUT PRIOR TO FIRING, AND THE TOP "AND" PROVIDES AN OUTPUT DURING THE RUN MODE. ALL INPUTS ON THE "AND"S MUST BE "0"S TO ENABLE. BOTH "AND" CIRCUITS ARE CONTROLLED BY THE MASTER FLIP-FLOP OUTPUT AND THE $\overline{\text{HDX}}$ INPUT.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROTECTION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 80 SH. NO. 79



VMPA CARD
FIG. D-1

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		PROTECTION
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH. 81 SH NO 80

D-2 OVERSPEED

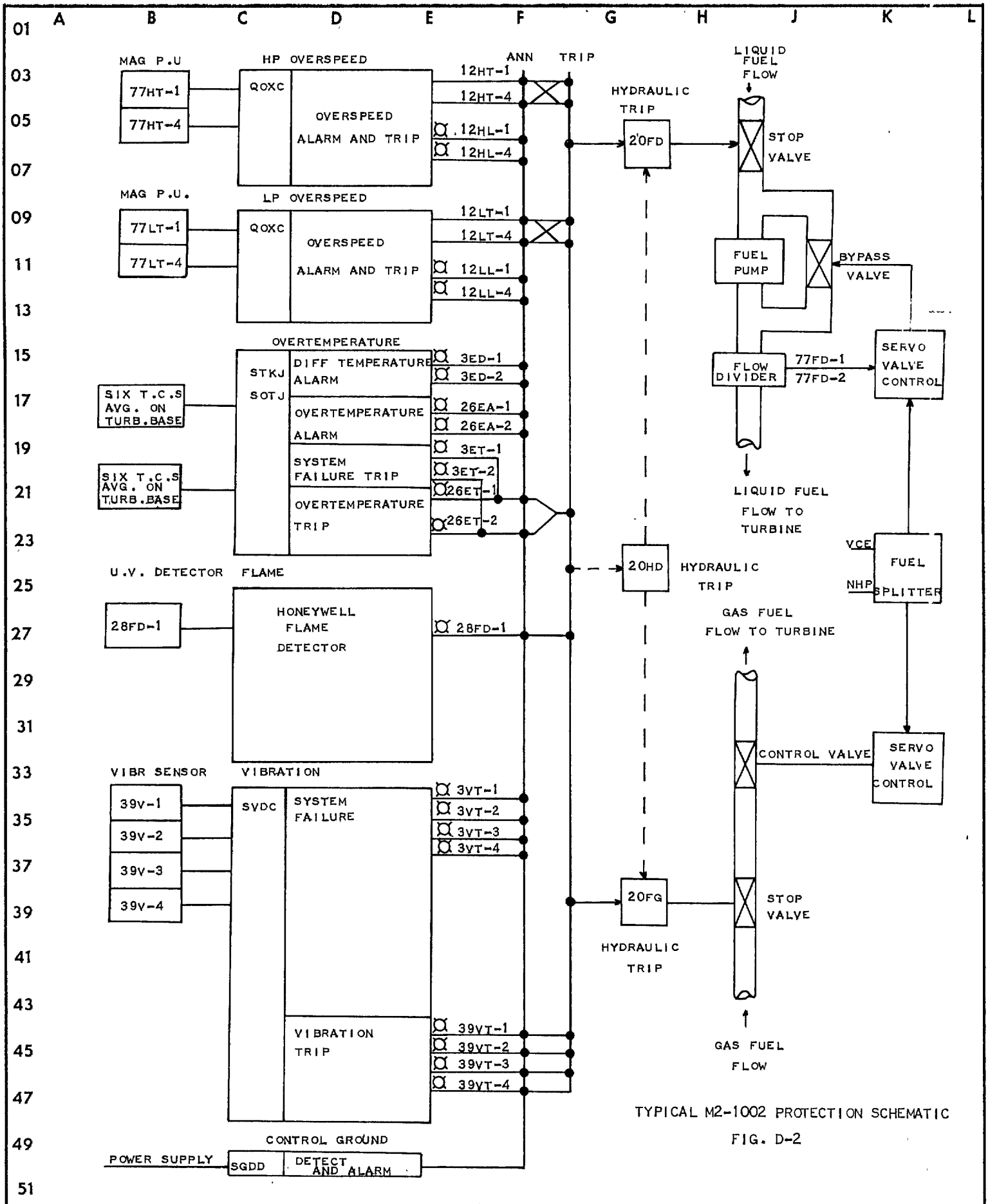
OVERSPEED PROTECTION IS PROVIDED FOR BOTH THE LP AND HP SHAFT. THE LP AND HP PROTECTION SYSTEMS ARE VERY SIMILAR, AND EACH CONSISTS OF TWO SEPARATE SENSING CIRCUITS TO PROVIDE INCREASED RELIABILITY. A DIAGRAM OF A TYPICAL HP SHAFT PROTECTION IS SHOWN IN FIG. D-4

QOXC CARD (SEE FIG. D-3, D-4)

THE QOXC CARD RECEIVES A FREQUENCY PROPORTIONAL TO THE SPEED OF THE TURBINE BETWEEN PINS IN AND ACOM. OUTPUTS TRP1 AND TRP2 WILL BE A "1" FOR THE FOLLOWING CONDITIONS:

(1) INPUT SPEED IS 10% ABOVE RATED SPEED.
(2) INPUT SPEED IS BELOW 9% OF RATED SPEED WHEN TURBINE IS ACCELERATING.
(3) INPUT SPEED IS BELOW 4% OF RATED SPEED ON DECELERATION OF TURBINE.

OUTPUT $\overline{\text{CHKOR}}$ IS A "0" IF EITHER TRP1 OR TRP2 IS A "1". OUTPUT $\overline{\text{CHKAND}}$ IS A "0" IF BOTH TRP1 AND TRP2 ARE "1". HOWEVER THE PRIMARY PURPOSE OF THE CIRCUIT IS TO GENERATE A "1" ON OUTPUTS TRP1 AND TRP2 WHEN THE TURBINE IS OPERATING OVER A PREDETERMINED SPEED. A LATCHING CIRCUIT IS PROVIDED TO HOLD THE TRIP INDICATION SIGNAL ON TPL. OUTPUT TPL WILL LATCH IN THE "1" STATE IF EITHER TRP1 OR TRP2 IS A "1" AND INPUTS 14HM AND 4 ARE "1". THE PUSHBUTTON ON THE CARD FRONT MUST BE DEPRESSED TO CLEAR THE LATCH.



TYPICAL M2-1002 PROTECTION SCHEMATIC

FIG. D-2

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS
REV. 2	ISSUED			FIRST MADE FOR
REV. 3	MADE BY			I.C. NO.

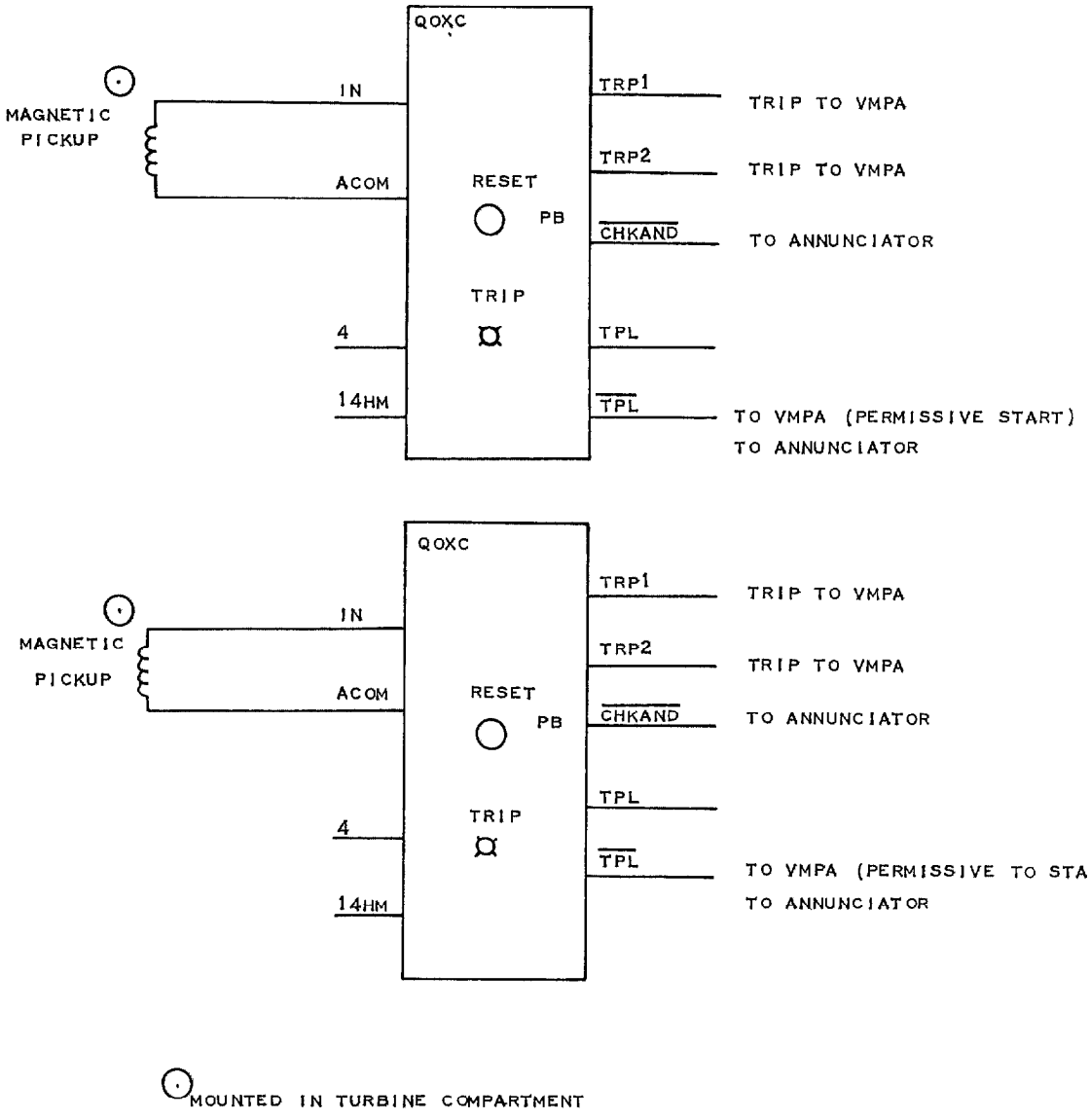
GENERAL ELECTRIC

SALEM, VA. U.S.A.

ELEMENTARY DIAGRAM

PROTECTION

CONT ON SH 83 SH NO. 82

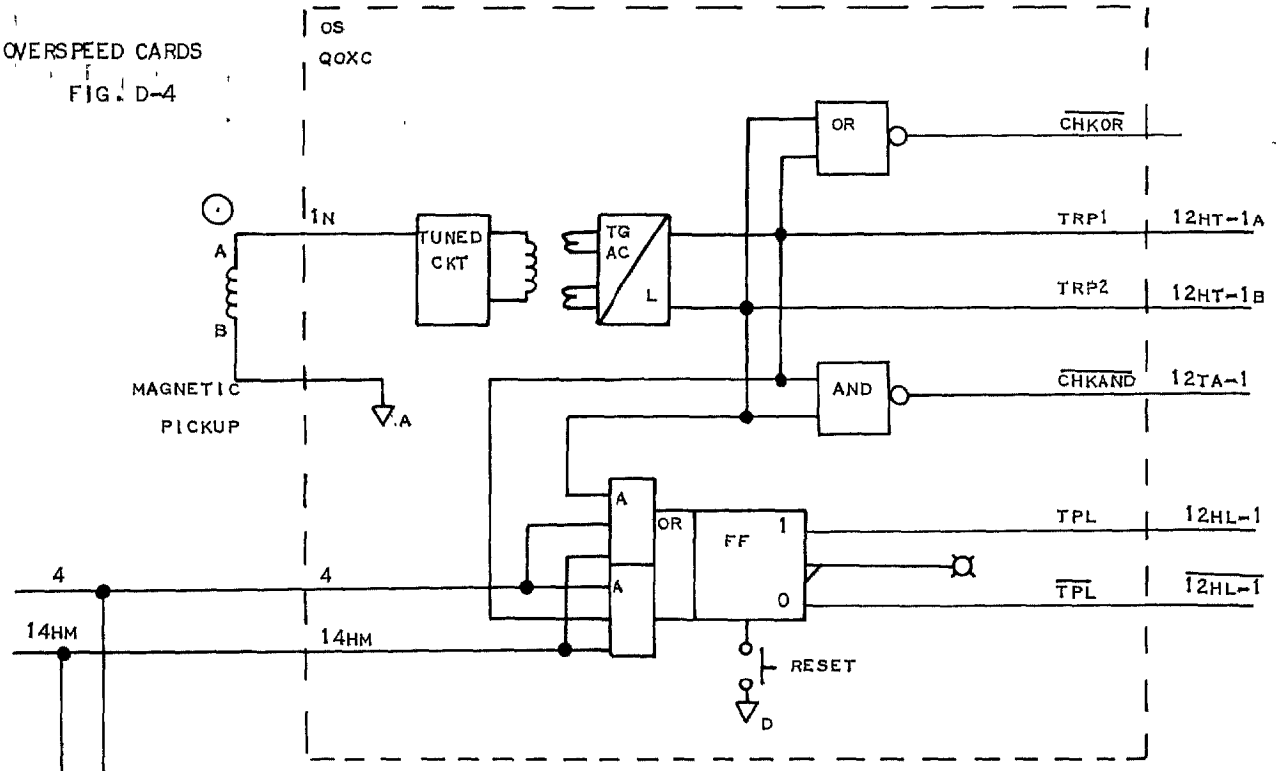


TYPICAL OVERSPEED TRIP

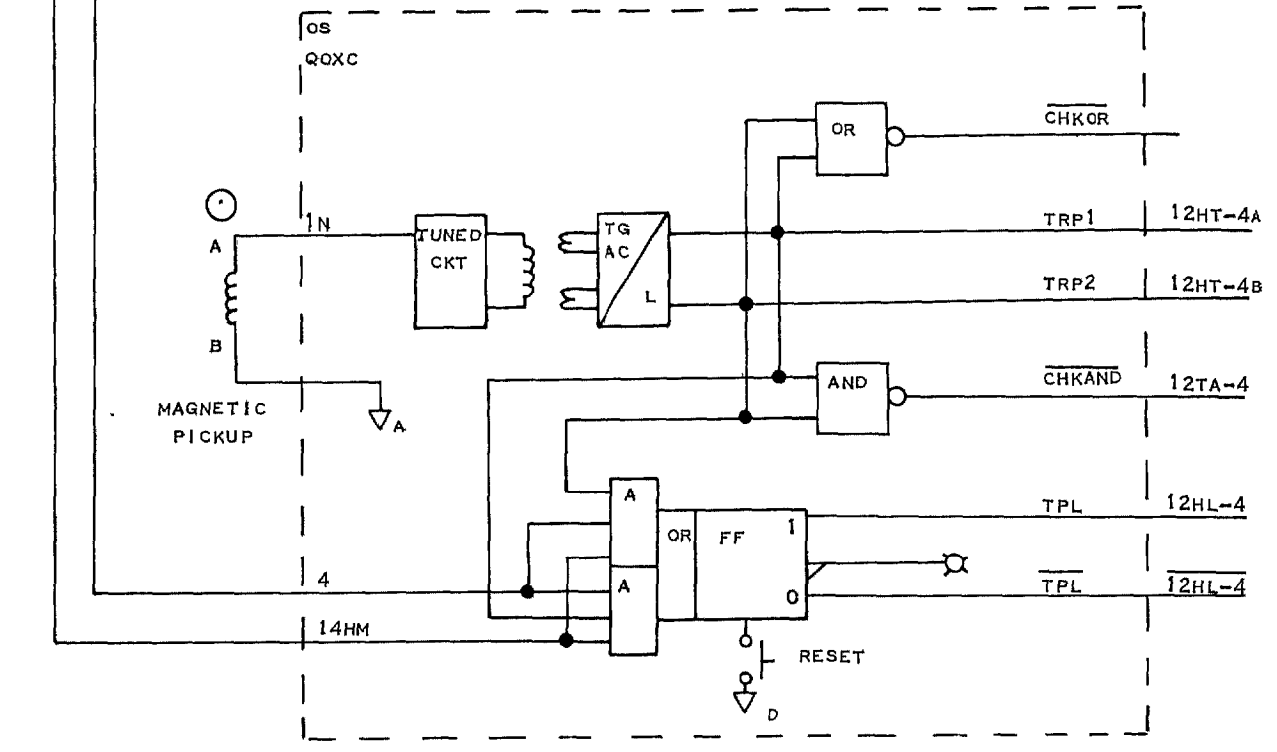
FIG. D-3

REV. 1	REV. 4	REV 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROTECTION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH 84 SH. NO. 83

OVERSPEED CARDS
FIG. D-4



(·) MOUNTED IN TURBINE COMPARTMENT



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROTECTION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 85 SH. NO. 84

D-3 OVERTEMPERATURE TRIP AND ALARM

THE OVERTEMPERATURE PROTECTION SYSTEM CONSISTS OF TWO ALARM CHANNELS AND TWO TRIP CHANNELS WITH ONE ALARM CHANNEL AND ONE TRIP CHANNEL ON EACH OF THE TWO SOTJ CARDS. EACH CHANNEL LOOKS AT THE AVERAGED OUTPUT OF THERMOCOUPLES PLACED IN THE TURBINE EXHAUST DUCT. THE OVERTEMPERATURE SYSTEM INCLUDES THE TWO SOTJ CARDS AND THE TWO STKJ CARDS. (SEE FIG. D-5 FOR DIAGRAM OF THE OVERTEMPERATURE PROTECTION SYSTEM).

SOTJ CARD:

EACH OF THE SOTJ CARDS RECEIVES THE AVERAGED AND AMPLIFIED THERMOCOUPLE SIGNALS TA AND TB FROM THE STKJ CARDS. THESE SIGNALS ARE APPLIED TO PINS OT AND TX. THE SIGNAL ON OT IS SUMMED WITH THE SIGNALS APPLIED TO PINS PCDF/R, PCDN/VCEN AND AN INTERNALLY GENERATED NHP BIAS. IF OT, PCDF/R, PCDN/VCEN AND THE NHP BIAS EXCEED THE ALARM SETPOINT REFERENCE, A "1" IS GENERATED ON AL AND THE ALARM LAMP COMES ON. IF THESE SIGNALS EXCEED THE TRIP SETPOINT REFERENCE A "1" IS GENERATED ON TP AND THE TRIP LAMP COMES ON. THE TRIP CIRCUIT IS LATCHING AND MAY BE RESET BY DEPRESSING THE RESET PUSHBUTTON. HR MAY BE USED TO ALTER THE RANGE OF THE ALARM AND TRIP ADJUSTMENTS.

THE SOTJ CARD ALSO PERFORMS TWO CIRCUIT PERFORMANCE CHECKS: (1) IT COMPARES THE TEMPERATURE DIFFERENTIAL BETWEEN TA AND TB AND GENERATES AN ALARM WHEN THE DIFFERENTIAL EXCEEDS A PREDETERMINED VALUE. (2) IT GENERATES A SYSTEM FAILURE TRIP SIGNAL IF OT DROPS BELOW 200° F WHEN SPEED IS ABOVE 14HS.

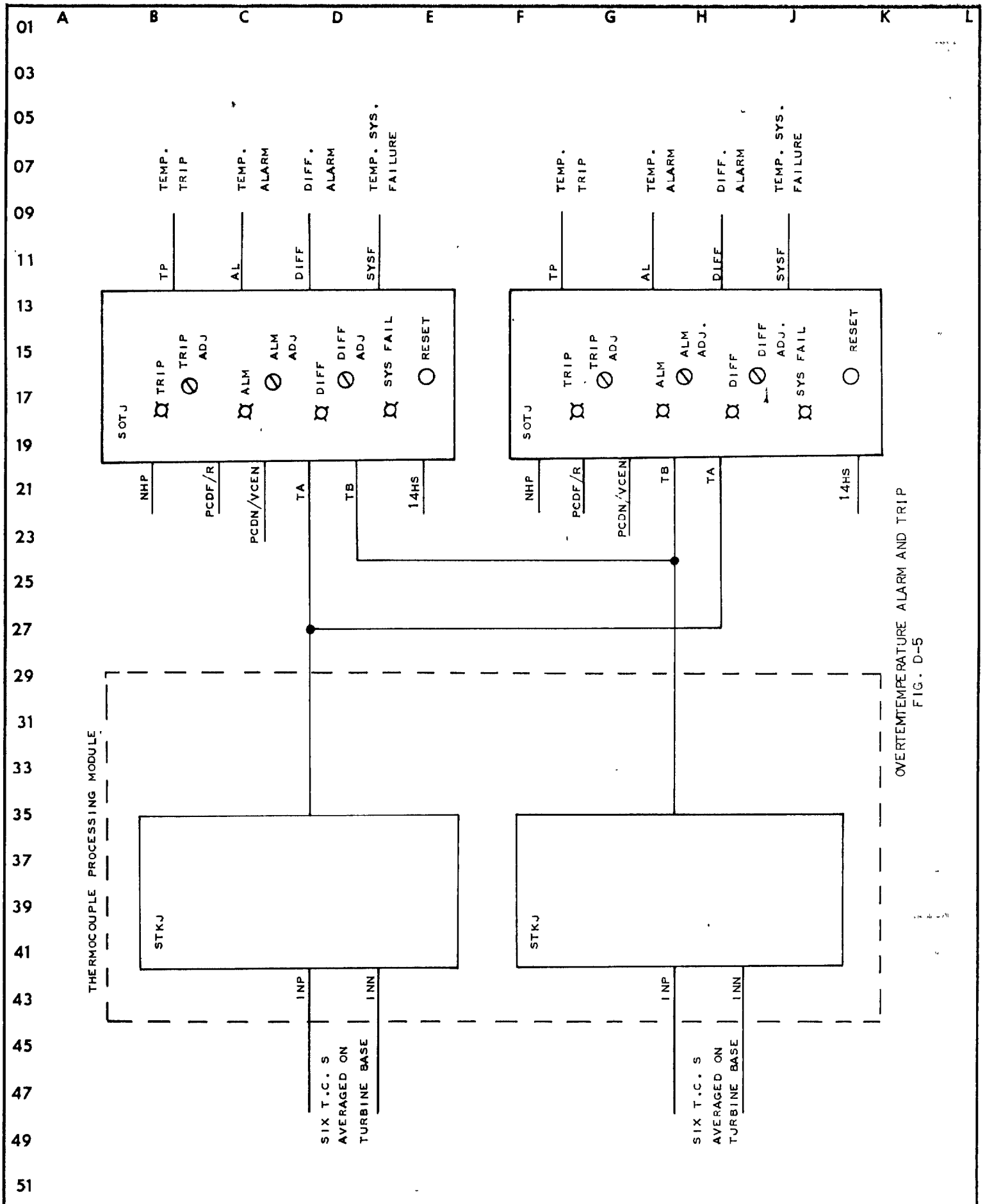
IF DIFFERENTIAL IS DETECTED, THE TEMP. CONTROL AND OVERTEMP. ALARM AND TRIP POINTS ARE AUTOMATICALLY LOWERED 35°F. THE TEMPERATURE CONTROL MUST BE RESET WITH A CARD MOUNTED PUSHBUTTON, 3ED/R, TO REGAIN THE ORIGINAL TEMPERATURE CONTROL POINTS.

IF "PCD" GOES BELOW 2.0 VOLTS, THE TEMPERATURE CONTROL ALARM AND TRIP POINTS WILL ALSO BE ALTERED AS THEY WILL FOR A TEMPERATURE DIFFERENTIAL.

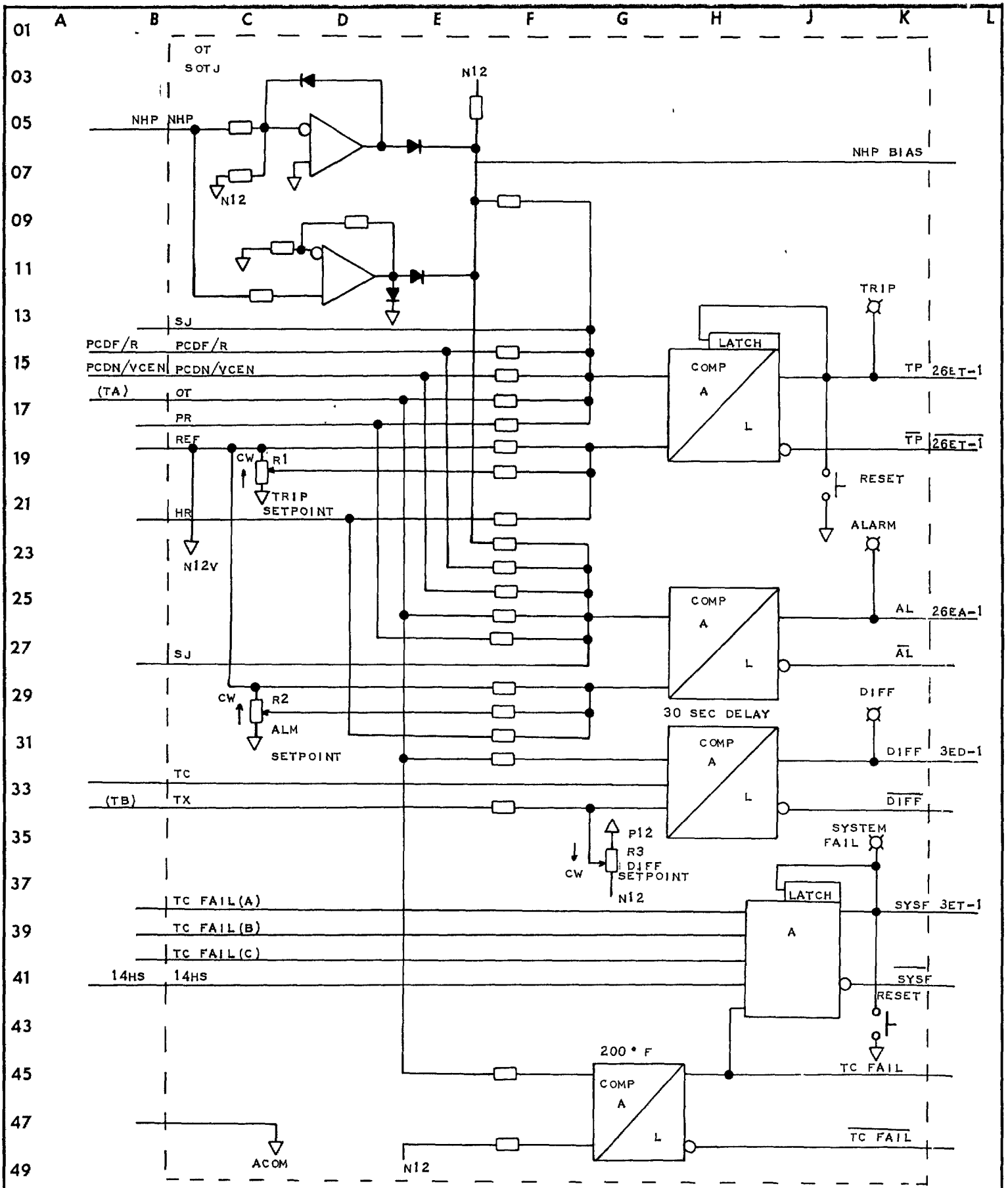
ADJUSTMENTS:

TRIP	-R1
ALARM	-R2
TEMP DIFF	-R3

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROTECTION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH 86 SH NO. 85



REV 1	REV 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		PROTECTION
REV. 3	MADE BY			I C NO	SALEM, VA. U.S.A.	CONT ON SH 87 SH. NO 86



OVERTEMPERATURE ALARM AND TRIP CARD

FIG. D-6

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROTECTION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 88 SH. NO 87

D-4 VIBRATION PROTECTION

A STANDARD VIBRATION PROTECTION SYSTEM CONSISTS OF TWO TURBINE TRIP CHANNELS. TWO ADDITIONAL VIBRATION TRIP CHANNELS FOR THE COMPRESSOR ARE OPTIONAL. A TRIP ON ANY CHANNEL WILL TRIP THE TURBINE. A TRIP IS LATCHING AND MUST BE RESET BY PUSHBUTTON OR REMOTE SIGNAL. VELOCITY VIBRATION SENSORS PROVIDE THE INPUT SIGNALS. EACH CHANNEL HAS CIRCUITRY TO TEST FOR AN OPEN OR SHORTED SENSOR. A TYPICAL DIAGRAM IS SHOWN IN FIG. D-7.

SVDC CARD (SEE FIG. D-8)

THE SVDC CARD RECEIVES THE SIGNAL FROM THE SENSOR BETWEEN PINS V1B AND ACOM. THE PEAK VALUE OF THIS SIGNAL IS DETECTED AND COMPARED TO A REFERENCE SET BY R1. IF THE PEAK VALUE OF THE SIGNAL EXCEEDS THE REFERENCE A "1" IS LATCHED UP ON OUTPUT T AND A LIGHT IS ILLUMINATED. THE LATCHING CIRCUIT MAY BE RESET VIA THE LOCAL RESET PUSHBUTTON OR A "1" ON INPUT RST. THE SETPOINT REFERENCE MAY BE MONITORED BETWEEN TP3 AND TP4. THE PEAK VALUE (VIB LEVEL) OF THE INPUT SIGNAL MAY BE MONITORED BETWEEN TP1 AND TP2. IF THE SENSITIVITY OF THE SENSOR USED IS 0.2 VOLTS PEAK FOR "1"/SEC. PIN GAIN IS CONNECTED TO ACOM TO DOUBLE THE GAIN OF THE DETECTOR.

A SENSOR FAULT DETECTOR IS PROVIDED TO DETECT A SHORTED OR OPEN SENSOR. THIS IS A LATCHING CIRCUIT AND MUST BE RESET BY LOCAL PUSHBUTTON. THE TROUBLE LIGHT IS ILLUMINATED AND OUTPUT "A" BECOMES A "1" FOR A FAULTY SENSOR. THE FOLLOWING TABLE LISTS THE IMPEDANCE REQUIRED FOR NO FAULT CONDITIONS:

JUMPERS	NO FAULT IMPEDANCE
NONE	2.9K TO 50K OHMS
PIN 9 - P12V	80 TO 800 OHMS
PIN 6 - P12V	160 TO 1.5K OHMS

ADJUSTMENTS:

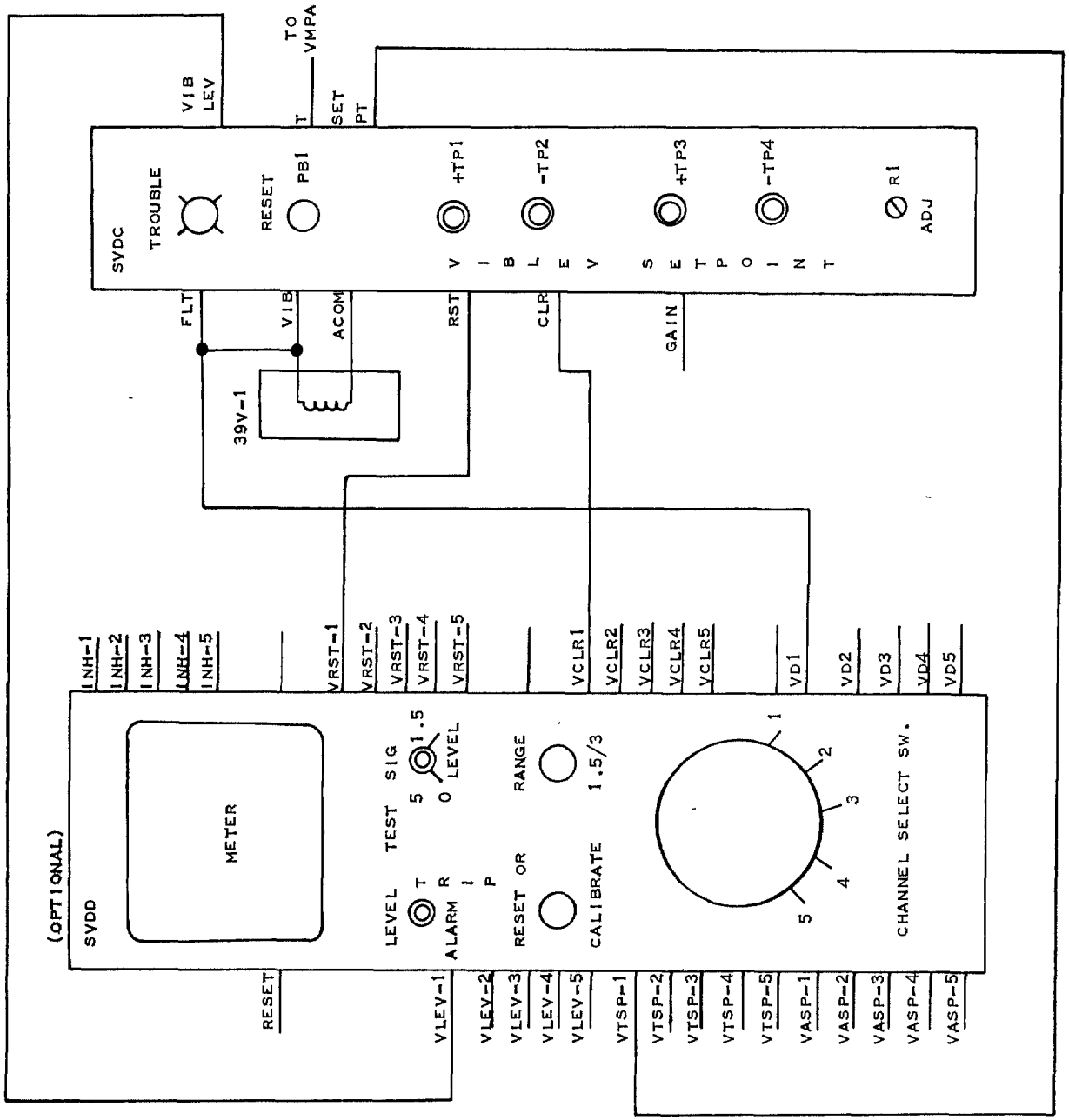
SET PT - R1 0.25 TO 1.6 IN/SEC.

SVDD CARD (OPTIONAL) - SEE FIG. D-9

THE SVDD CARD PROVIDES A MEANS OF CHECKING AND ADJUSTING THE SVDC CARD. THE "CHANNEL SELECTOR" SWITCH ON SVDD SELECTS THE SVDC CARD TO BE CHECKED. THE SVDD CALIBRATOR CARD HAS FOUR OUTPUT SIGNALS. (1) AN INHIBIT SIGNAL WHICH MAY BE USED TO PREVENT TRIPPING THE SYSTEM WHEN THE CALIBRATE PUSHBUTTON IS DEPRESSED. (2) A CLEAR SIGNAL TO CLEAR TRIP AT COMPLETION OF TEST. (3) A RESET SIGNAL TO RESET LATCHED TRIPS, (4) A 100 HERTZ SIGNAL WITH ADJUSTABLE MAGNITUDE TO SIMULATE THE SENSOR SIGNAL. THE INPUTS TO THE CALIBRATOR ARE THE VIBRATION LEVEL AND SETPOINT LEVEL FROM THE SVDC CARD. THE "METER SELECTOR SWITCH" DETERMINES WHICH OF THESE WILL BE DISPLAYED ON THE SVDD METER. THE "CALIBRATE" PUSHBUTTON MUST BE DEPRESSED TO CONNECT THE SVDD INPUTS AND OUTPUTS TO THE SVDC CARD. ANY TRIPS IN A SVDC CARD WILL BE RESET UPON DEPRESSING OR RELEASING THE CALIBRATE PUSHBUTTON.

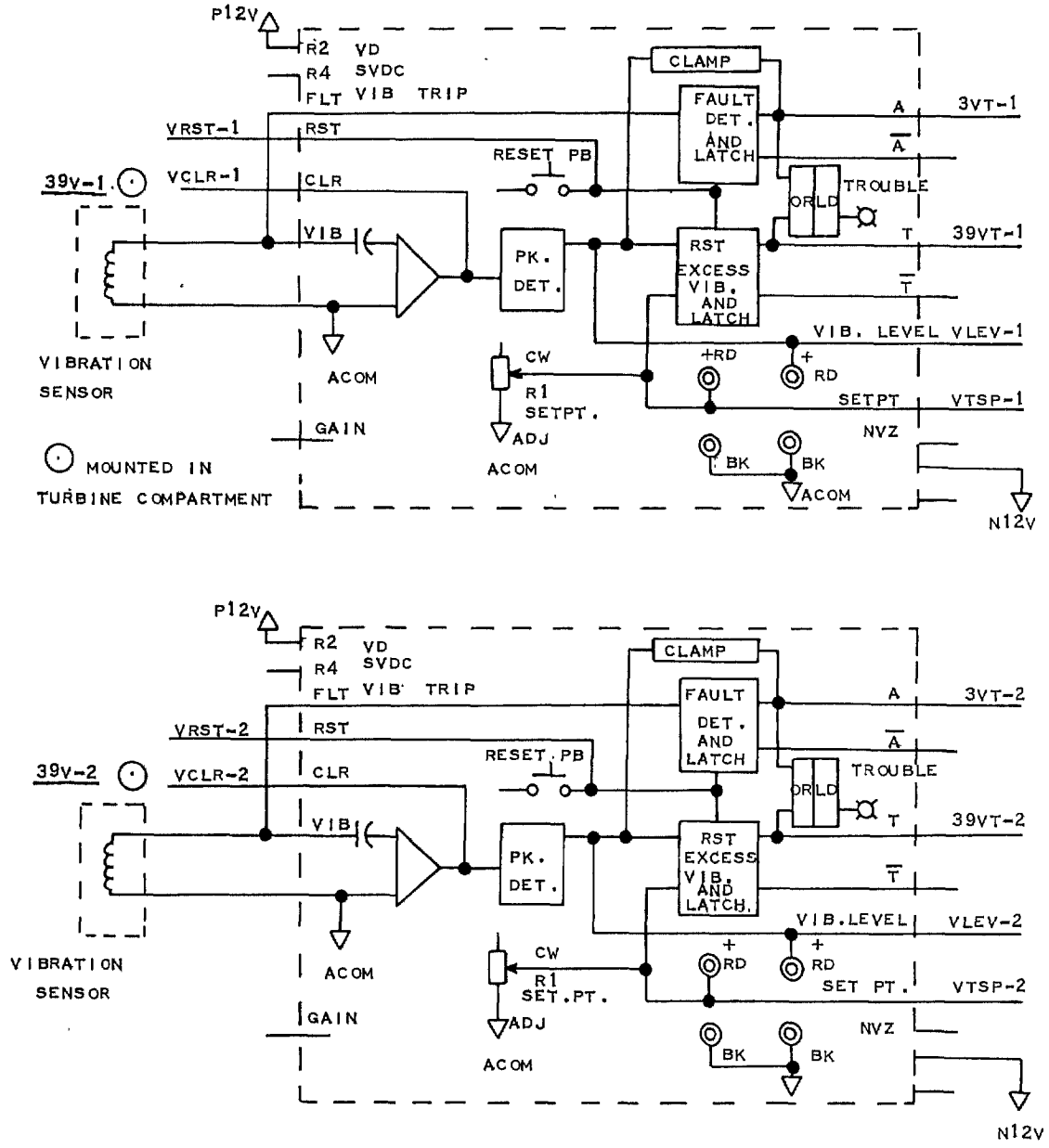
ADJUSTMENTS:

LEVEL - R56 - 0 TO 0.7 VOLTS RMS



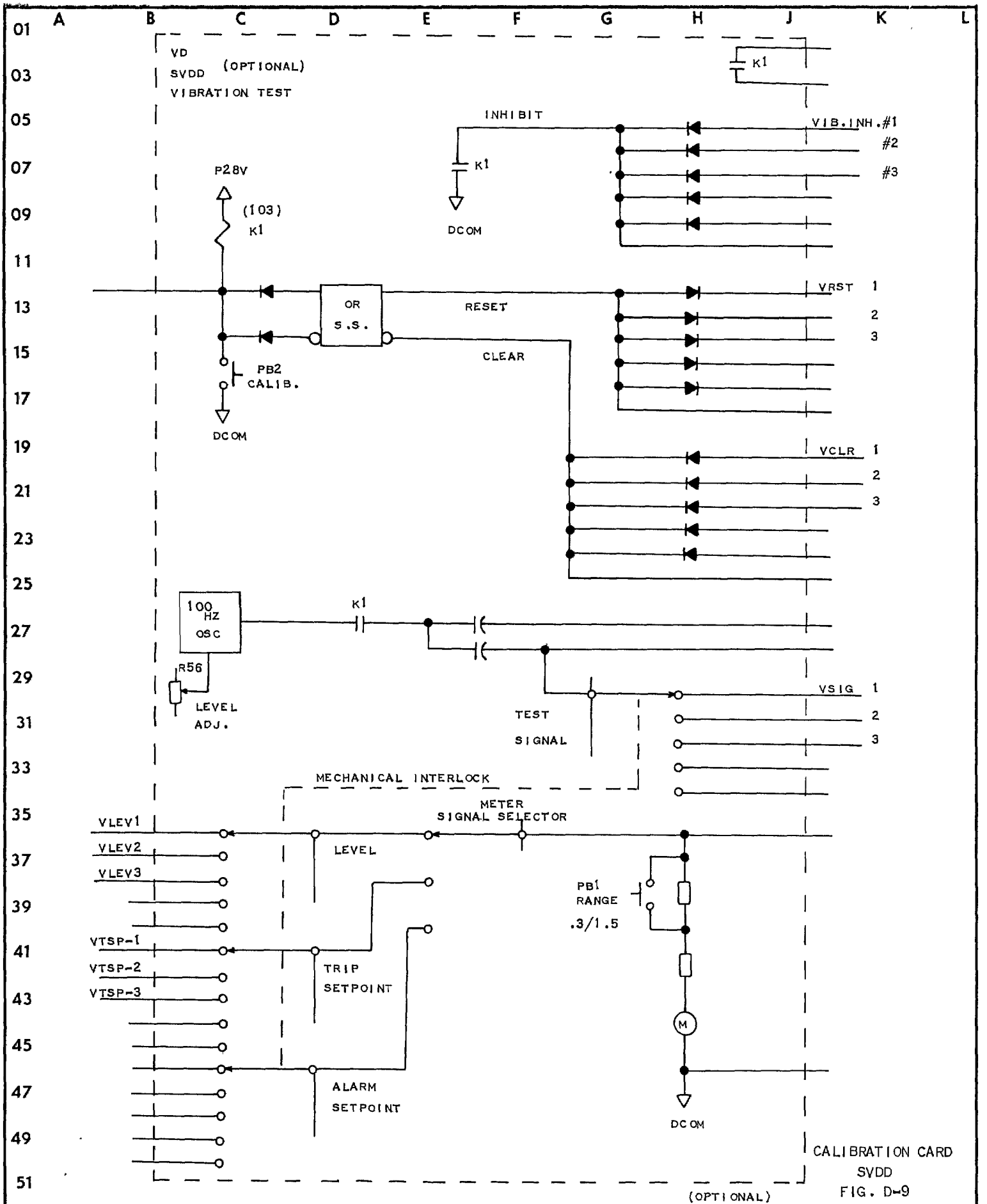
VIBRATION TRIP
D-7

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		PROTECTION
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 91 SH. NO. 90



VIBRATION PROTECTION CARDS
FIG. D-8

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROTECTION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH. 92 SH. NO. 91



REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		PROTECTION
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 93 SH. NO. 92

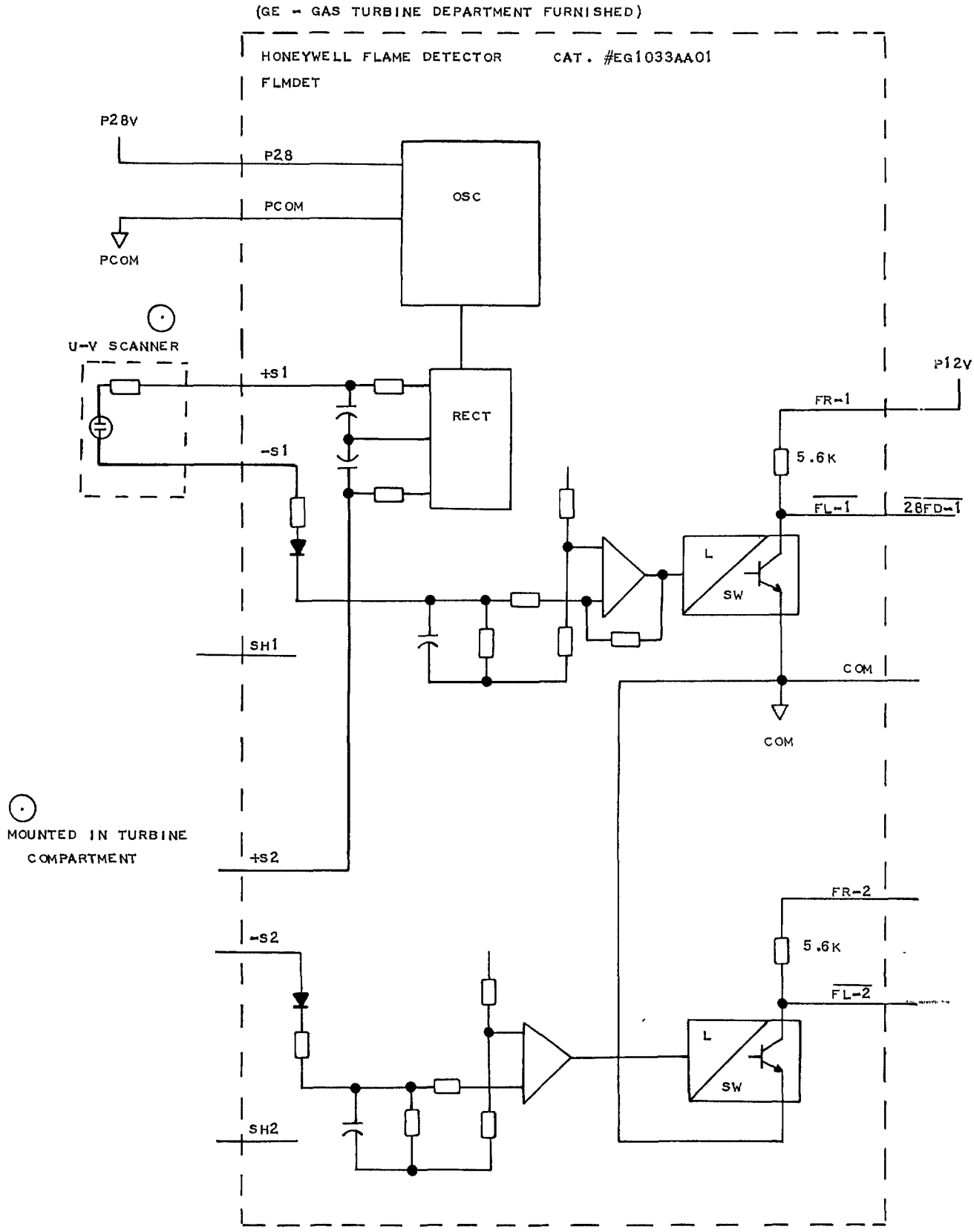
D-5 FLAME DETECTOR (SEE FIG. D-10)

THE HONEYWELL FLAME DETECTION SYSTEM IS SHOWN IN FIG. D-10. THE FLAME DETECTION SYSTEM RECEIVES AN INPUT FROM AN ULTRAVIOLET SCANNER. THE HONEYWELL SCANNER AND DETECTION SYSTEM WHICH IS USED IS EXPLAINED IN THE HONEYWELL PUBLICATION IN VOLUME II OF THE INSTRUCTION BOOK.

D-6 GROUND DETECTORS (SEE FIG. D-11)

GROUND DETECTOR CIRCUITS ARE PROVIDED FOR INDICATION OF A GROUND ON THE P28, P12, P5.3, N12, ACOM, P125 AND XCOM BUSSES. A GROUND ON ANY OF THESE BUSSES WILL BE ANNUNCIATED. THE FOLLOWING TABLE GIVES THE IMPEDANCE TO GROUND REQUIRED FOR DETECTION.

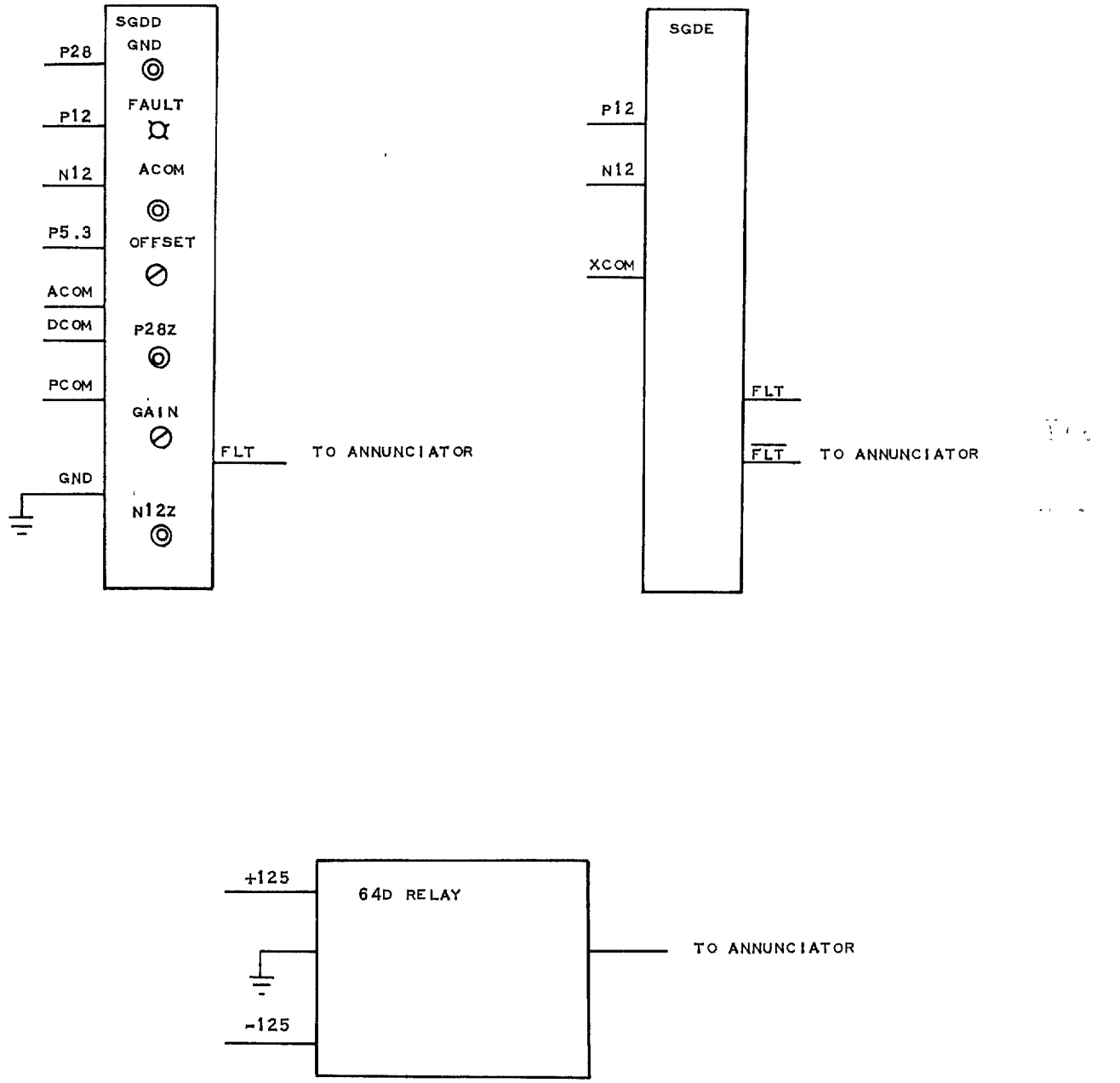
BUS	IMPEDANCE FOR FAULT
P28	31-51K
P12	13.1K-21.7K
P5.3	3.5K-7.5K
N12	170K-270K
ACOM	6K-10K
P125	NOMINAL - 5K OHMS
XCOM	24.2K $\pm$ 50%



FLAME DETECTOR MODULE

FIG. D-10

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM PROTECTION
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 95 SH. NO 94



GROUND DETECTORS

FIG, D-11

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		PROTECTION
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 96 SH. NO. 95

SECTION E

POWER SUPPLY SYSTEM

A BLOCK DIAGRAM OF THE POWER SUPPLY SYSTEM IS GIVEN IN FIGURE E-1. THE SYSTEM IS BROKEN DOWN INTO THE FOLLOWING TWO FUNCTIONS (PARTS):

(A) DC BULK SUPPLY
(B) FINAL REGULATOR

E-1 GENERAL DESCRIPTION

(A) DC BULK SUPPLY

BASICALLY THE DC BULK SUPPLY PERFORMS THE FOLLOWING:

(- I) ACCEPTS A NOMINAL 125V DC VOLTAGE
(II) CONVERTS TO AC (350 TO 450 HERTZ)
(III) TRANSFORMS VOLTAGE TO APPROPRIATE LEVELS TO FEED THE P28V, P12V, N12V, AND P5V POWER SUPPLY CIRCUITS
(IV) PROVIDES ISOLATION FROM THE PRIMARY (I.E. THE 125 DC CIRCUITRY)
(V) RECTIFIES AND FILTERS THE ABOVE VOLTAGES
(VI) FEEDS THE P28V BUS DIRECTLY FROM THE DC BULK SUPPLY
(VII) FEEDS THE P12V, N12V, P5V BUSES VIA THE FINAL REGULATOR AS ILLUSTRATED IN FIGURE E-3.

THE MAIN FEATURES OF THE DC BULK SUPPLY ARE ILLUSTRATED IN FIGURE E-3. WORKING FROM LEFT TO RIGHT IN FIGURE E-2.

(I) ON-OFF SWITCH
(II) FUSING, 20 AMP
(III) THYRECTOR, FOR SPIKE SUPPRESSION
(IV) INDUCTOR, TO ISOLATE SWITCHING SPIKES (GENERATED IN BULK SUPPLY) FROM GETTING BACK ONTO THE SOURCE BUS
(V) 11,800 MICROFARADS OF CAPACITANCE TO ALLOW THE POWER SUPPLIES TO RIDE THROUGH A 65 MSEC SOURCE VOLTAGE DIP
(VI) ISOLATING DIODE, TO PREVENT A VOLTAGE DIP ON THE SOURCE FROM DISCHARGING THE CAPACITANCE OF STEP (V).
(VII) 100 MICROFARAD CAPACITOR, TO PROVIDE AC COUPLING TO GROUND
(VIII) FUSING, 15 AMPS, CURRENT LIMIT (FAST) TYPE
(IX) TRANSFORMER, THE 125V DC(NOMINAL) IS APPLIED TO THE CENTER OF THE PRIMARY, WITH EACH END ALTERNATELY SWITCHED TO THE RETURN 125 VDC, VIA SWITCHING TRANSISTORS. THE SECONDARIES ARE RECTIFIED AND FILTERED TO PROVIDE DC VOLTAGES TO THE FINAL REGULATOR AND THE P28V BUS (REFER TO FIGURE E-4)
(X) THE EPSU CARD PROVIDES CONTROL OF THE SWITCHING TRANSISTORS OF STEP (IX), AND PROVIDES CONTROL POWER TO EPSV. THE EPSU IS IN TURN CONTROLLED VIA PULSES FROM THE EPSV CARD.
(XI) THE EPSV CARD IS A COARSE P28V VOLTAGE REGULATOR, IT SENSES THE OUTPUT

VOLTAGE ON THE P28V SECONDARY WINDING, COMPARES IT TO A REFERENCE AND SENDS OUT CONTROL PULSES TO EPSU (STEP (X)). EPSV ALSO PROVIDES A SHUNT LOAD CONTROL (POWER TRANSISTOR) TO ENSURE CONTINUOUS CURRENT IN THE FILTER INDUCTOR (MINIMUM OF THREE AMPERES).

(B) FINAL REGULATOR

THE FUNCTIONS OF THE FINAL REGULATOR ARE ILLUSTRATED IN FIGURE E-3, AND ARE AS FOLLOWS:

(I) MODULE IC3601A250B (INCLUDING EPSW):
- -STEPS DOWN A COARSELY REGULATED VOLTAGE, P12X, TO A FINELY REGULATED VOLTAGE, P12V;
- -PROVIDES CURRENT LIMIT;
- -PROVIDES CURRENT AND VOLTAGE JACKS FOR MEASURING PURPOSES;
- -PROVIDES OVERVOLTAGE PROTECTION BY MEANS OF A CROWBAR CIRCUIT (I.E. BY FIRING A SCR).

(II) MODULE IC3601A250A: SAME AS STEP (I) EXCEPT P5V BUS IS CONTROLLED INSTEAD OF THE P12V BUS.

(III) CARD EPSX: SAME AS STEP (I), EXCEPT N12V BUS IS CONTROLLED INSTEAD OF THE P12V BUS AND OVERVOLTAGE PROTECTION IS VIA A DIODE INSTEAD OF AN SCR (PROVIDES PROTECTION AGAINST BEING PULLED POSITIVE).

(IV) CARD EPSY:
- -PROVIDES AN OUTSIDE, FINE VOLTAGE, CONTROL LOOP ON THE P28V BUS; I.E. IT TRIMS THE COARSE VOLTAGE REGULATOR EPSV.
- -PROVIDES CURRENT LIMIT CONTROL VIA A "STOP" SIGNAL TO EPSV.
- -PROVIDES CURRENT AND VOLTAGE JACKS FOR MEASURING PURPOSES.

E-2 APPLICATION INFORMATION

(A) BULK SUPPLY

(I) INPUT VOLTAGE RANGE IS 95 TO 140V DC

(II) INPUT CURRENT IS 2.1 TO 7.3 AMPS AT 95V DC AND 1.75 TO 6.1 AMPS AT 140V DC.

(III) OUTPUTS ARE AS FOLLOWS:

TERMINAL PT.	VOLTAGE (8%-100% LOAD)	NO LOAD VOLTAGE	RATED CURRENT
P30	29.5-31	31	15A
P12X	18-20	34.8	6A
N12X	18-21	32.7	1A
P5X	10-13	20.8	6A
P11	11-11.5		30A

(IV) THE CURRENT SIGNAL ACROSS P30V TO P28X (0.1 OHM SHUNT), IS 0.3V DC TO 1.5V DC FOR A ZERO TO 15 AMP LOAD ON THE P28V BUS. A SHUNT REGULATOR MAINTAINS A MINIMUM TO GIVE CONTINUOUS CURRENT EVEN WHEN THE LOAD ON THE P28V IS ZERO.

(V) THE SUPPLIES WILL NOT DIP DURING A DIESEL START (OR EQUIVALENT), DEFINED AS FOLLOWS:

DIP TO 60V, BACK TO 80V IN 65 MSEC.

(VI) EPSU CARD (PRIMARY SUPPLY CONTROL)

FUNCTIONS: -PRIMARY TRANSISTOR BASE DRIVE,
-PRIMARY CURRENT LIMIT,
-CONTROL POWER SUPPLY,
-PROTECTION-IF BASE AND COLLECTOR HIGH, BASE DRIVE REMOVED AND LED INDICATION.

INPUTS: -COLLECTOR VOLTS (CA, CB)
-EMITTER CURRENTS. EA1-3, EB1-3, SCA, SCB.
-PULSES (CONTROL) FROM COARSE REG. CARD EPSV. (TIME RATIO CONTROL)
-+51V ZENER INPUT FOR CONTROL POWER (FEEDS INVERTER).

OUTPUTS: -BASE DRIVE
-CONTROL POWER SUPPLY TO EPSV, 13V, 5KHz
-LED INDICATION FOR FAILURE

(VII) EPSV CARD (P28V COARSE REGULATOR)

FUNCTIONS: -COARSE VOLTAGE REGULATOR ON 28V SUPPLY OUTPUT VOLTAGE AND PROVIDING CONTROL PULSES TO EPSU (ABOVE)
-CONTROL TO SHUNT CURRENT REG TO ENSURE +12, -12, +5.3 STAY UP.
-PROVIDES CONTROL POWER TO OTHER CARDS VIA P11 (11V DC) WHICH DOES NOT DISAPPEAR WHEN INVERTER SHUTS DOWN.

INPUTS: -CONTROL POWER ACC1, ACC2, FROM EPSU
-VOLTAGE FEEDBACK, AC1, AC2, COM.
-STOP-LOGIC, "1" SHUTDOWN INVERTER "0" STARTS INVERTER
-REG-INPUT FROM FINE VOLTAGE REGULATOR EPSY.
-SIGNAL BIAS OPERATION OF COARSE REGULATOR
-CURRENT BIAS OPERATION OF COARSE REGULATOR
-CURRENT SIGNAL ACROSS SHUNT P30, P28X

OUTPUTS: -PULSES TO EPSU (TIME RATIO CONTROL)
-CONTROL SUPPLY P11, 11V
-S COM - RETURN FOR P11
-SH - BASE DRIVE TO SHUNT CURRENT REG.

(B) FINAL REGULATOR

(I) THE FINAL REGULATOR HAS THE FOLLOWING OUTPUT CHARACTERISTICS:

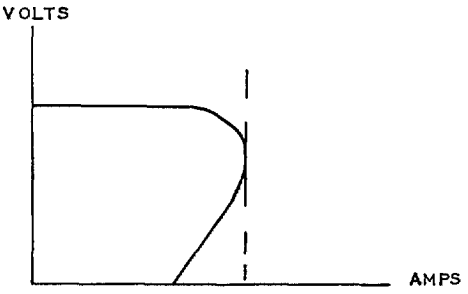
VOLTS	BUS	RETURN	RATED CURRENT	CURRENT LIMIT	RIPPLE UP-P	REG (LD)	DRIFT (TEMP)	100 CURRENT SHUNT-SIGNAL
+28V*	P28V	PCOM	15A	17A	250 MV	300 MV	——	1.5V
+12V	P12V	ACOM	6A	7A	10 MV	30 MV	60 MV	1 V
-12V	N12V	ACOM	1A	1.2A	10 MV	30 MV	60 MV	2 V
+5.3V	P5V	DCOM	6A	7A	100 MV	150 MV	40 MV	1 V

*FOR INPUT OF 105 TO 140V DC; FOR INPUT DOWN TO 95V, REDUCE 28V LOAD TO 7A FOR ACCEPT P28V DROP TO 26V.

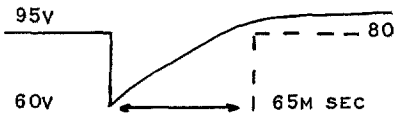
(II) AN SCR IS USED ON EACH OF THE P12V AND P5V BUSES AND IS CALLED A "CROWBAR" CIRCUIT. ITS PURPOSE IS TO MONITOR THE BUSES AND PROTECT THE LOAD EQUIPMENT FROM OVERVOLTAGE. THE P12V BUS CROWBAR OPERATES AT 15.5 TO 19.5V; THE P5V BUS CROWBAR OPERATES AT 8-10.5V; THE N12V BUS HAS A DIODE TO PROTECT AGAINST POSITIVE GOING VOLTAGES; THE P28V HAS NO CROWBAR, SINCE ITS LOAD IS NOT AS VOLTAGE SENSITIVE AS THE OTHER BUSES.

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		POWER SUPPLIES
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH 99 SH NO. 98

(III) THE SUPPLIES HAVE A CURRENT LIMIT FOLDBACK AS FOLLOWS:



(IV) THE SUPPLIES WILL NOT DIP DURING A DIESEL START (OR EQUIVALENT), DEFINED AS FOLLOWS:



(V) AC VOLTAGE, 350-450 HZ, 60V RMS IS AVAILABLE FROM THE BULK SUPPLY; FOR AC LOADING, REDUCE P28V VOLT-AMPS BY AN EQUIVALENT AMOUNT; NON-STANDARD.

(VI) JACKS ON CARD FRONTS, FOR BOTH CURRENT AND VOLTAGE SIGNAL.

(VII) MONITORS OVER/UNDER VOLTAGE ON ALL SUPPLIES, AND +28V SUPPLY OVERCURRENT, MONITORING WITH LED AND LOGIC ANNUNCIATION.

(VIII) EPSY (P28 FINE REGULATOR)

FUNCTIONS: -FINE VOLTAGE REGULATION
-CURRENT LIMIT

INPUTS: -P11-CONTROL SUPPLY
-P28-28V BUS, FOR RING BUS MONITOR
-P30-P28X- FOR CURRENT LIMIT SIGNAL
-P30-PCOM-RAW DC SUPPLY
-SENSE-REMOTE VOLTAGE SENSING

OUTPUTS: -REG.-FINE VOLTAGE REG. BIAS TO EPSV
-$\overline{CL}$-CURRENT LIMIT LOGIC SIGNAL "0" = CL; TO BE USED WITH MONITOR
-STOP-TO SHUT OFF INVERTER ON CL "1" = INVERTER OFF

(IX) EPSW (P12V, P5V REGULATOR)

FUNCTIONS -REGULATE +5.3V OR +12 BUS
-CURRENT LIMIT ON +5.3V OR +12 BUS
-CROWBAR +5.3V OR +12 BUS

INPUTS: -P28X-CONTROL POWER
-P12X OR P5X-POWER
-V SENSING-REMOTE SENSING

OUTPUTS: -GATE TO CROWBAR SCR
-VM-TO MONITOR
-BASE DRIVE TO SERIES TRANSISTOR REGULATOR

(X) EPSX (12V REGULATOR)

FUNCTIONS: -REGULATE -12V
-CURRENT LIMIT

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM: POWER SUPPLIES
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT ON SH. 100 SH NO. 99

INPUTS: -N12X - POWER
-VM(+12V SUPPLY) CONTROL POWER
-VS - REMOTE SENSING

OUTPUTS: -VM - MONITOR VOLTAGE
-N12V - OUTPUT POWER

E-3 TROUBLE SHOOTING GUIDE

THESE INSTRUCTIONS ARE WRITTEN TO AID THE TROUBLESHOOTER IN ISOLATING POWER SUPPLY TROUBLES. THIS IS A GUIDE ONLY, AND CAN NOT BE APPLIED BLINDLY WITHOUT SOME KNOWLEDGE OF TROUBLESHOOTING PROCEDURES AND OF THE POWER SUPPLIES, THEMSELVES. IT IS RECOMMENDED THAT THE TROUBLESHOOTER, IF NOT ALREADY FAMILIAR, BRIEFLY REVIEW THE GENERAL DESCRIPTION OF THE POWER SUPPLY SYSTEM BEFORE PROCEEDING WITH THESE INSTRUCTIONS.

(A) SYMPTOM: P5V OR P12V BUS OUT OF LIMITS, BUT ALL OTHER BUSSES CORRECT.

ACTION:

(I) IT'S POSSIBLE THAT THE "CROWBAR" PROTECTION CIRCUIT HAS BEEN ENERGIZED DUE TO EXTERNAL VOLTAGE SPIKING. VERIFY THAT EXTERNAL INSTRUMENTATION, ETC., IS NOT DISTURBING THE BUS. REMOVE POWER FOR SEVERAL SECONDS (TO ALLOW SCR TO RESET) AND REAPPLY POWER.
(II) IF FAULT STILL PRESENT, REMOVE POWER, REPLACE EPSW, REAPPLY POWER
(III) IF FAULT STILL PRESENT, VERIFY THAT INPUT TO FINAL REGULATOR MODULE (P12X OR P5X) IS AS DEFINED IN APPLICATION SECTION E-2B, (III); IF NOT, THE BULK SUPPLY IS SUSPECT. (I.E. THE DIODE-FILTER SECTION OF THE BULK SUPPLY)
(IV) IF STEP (III) WAS CORRECT, REMOVE POWER, ISOLATE ALL LOADING ON THE POWER SUPPLY IN TROUBLE, REAPPLY POWER. IF THE POWER SUPPLY IS CORRECT NOW, THE TROUBLE PROBABLY EXTERNAL TO SUPPLY, EITHER EXCESSIVE LOAD OR OVERVOLTAGE FROM OTHER SOURCE.
(V) IF POWER SUPPLY IS STILL NOT CORRECT, MODULE IC3601A250A (P5V) OR IC3601A250A (P12V) IS SUSPECT.

(B) SYMPTOM: N12V BUS IS OUT OF LIMITS, BUT OTHER BUSSES ARE CORRECT.

ACTION:

(I) SUSPECT EPSX; REMOVE POWER, REPLACE EPSX.
(II) IF BUS STILL NOT CORRECT, ISOLATE LOAD AND CHECK N12V BUS. IF THE BUS RETURNS TO NORMAL, THE TROUBLE IS PROBABLY EXTERNAL TO THE POWER SUPPLY SYSTEM.
(III) IF BUS STILL NOT CORRECT, MEASURE VOLTAGE ON N12X (INPUT TO EPSX AND VERIFY THAT IT IS AS DEFINED IN APPLICATION SECTION E-2B, (III). IF NOT, THE BULK SUPPLY IS SUSPECT (I.E. DIODE-FILTER SECTION).

REV. 1	REV. 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		POWER SUPPLIES
REV. 3	MADE BY			I.C. NO.	SALEM, VA. U.S.A.	CONT. ON SH 101 SH. NO. 100

(C) SYMPTOM: P28V BUS OUT OF LIMITS, BUT THE OTHER BUSSES CORRECT.

ACTION:

(I) EPSY IS SUSPECT; REMOVE POWER, REPLACE EPSY.
(II) IF BUS IS STILL NOT CORRECT, ISOLATE THE LOAD AND CHECK P28V BUS. IF THE BUS RETURNS TO NORMAL, THE TROUBLE IS PROBABLY EXTERNAL TO THE POWER SUPPLY.
(III) IF BUS STILL NOT CORRECT, MEASURE THE VOLTAGE ON P28X, AND VERIFY THAT IT IS AS DEFINED IN APPLICATION SECTION E-2B, (III); IF NOT, THE BULK SUPPLY IS SUSPECT; I.E. DIODE-RECTIFIER OUTPUT IS SUSPECT.

(D) SYMPTOM: ALL POWER SUPPLY BUSSES ARE LOW, BUT EPSU LIGHT EMITTING DIODE (LED) IS OFF, AND BULK SUPPLY MAKING A NOISE (HUM).

ACTION:

(I) CHECK THAT BULK SUPPLY INPUT IS 95V DC OR GREATER
(II) CHECK THAT THE "STOP" SIGNAL (WITH RESPECT TO P-COM), LABELED INTERCONNECTIONS BETWEEN BULK SUPPLY AND REGULATOR, IS A LOGIC "ZERO" AND P28V SUPPLY IS RUNNING INTO CURRENT LIMIT, IT WILL BOUNCE BETWEEN "ZERO- AND APPROXIMATELY 2V DC. ALSO CHECK THE VOLTAGE SIGNAL ACROSS THE CURRENT JACKS OF THE EPSY CARD AND COMPARE TO THE NOMINAL CURRENT LIMIT SIGNAL OF 1.7V DC.
(III) IF THE TROUBLE IS STILL NOT DETERMINED, REMOVE POWER, ISOLATE THE P28V LOAD, REAPPLY POWER AND CHECK FOR CORRECT BUS VOLTAGES. IF THE BUSES ARE STILL LOW, JUMPER "STOP" TO "PCOM". IF THE BUS VOLTAGES ARE NOW CORRECT, EPSY IS SUSPECT. IF THE BUS VOLTAGES ARE STILL LOW, BULK SUPPLY CARDS EPSU, EPSV ARE SUSPECT.
(IV) IF THE BUSES BECAME CORRECT WHEN THE P28V LOAD IS ISOLATED, SUSPECT P28V SUPPLY CURRENT LIMIT TROUBLE OR PRIMARY CURRENT LIMIT TROUBLE. BUT P28V CURRENT LIMIT WAS CHECKED IN STEP (II). THEREFORE, PRIMARY CURRENT THAT IS EPSU, IS SUSPECT (CHANGE OUT EPSU).

(E) SYMPTOM: EPSU LIGHT EMITTING DIODE "LED" IS ON; BULK SUPPLY WILL NORMALLY BE SHUT DOWN, I.E. WILL NOT BE MAKING A NOISE (HUM).

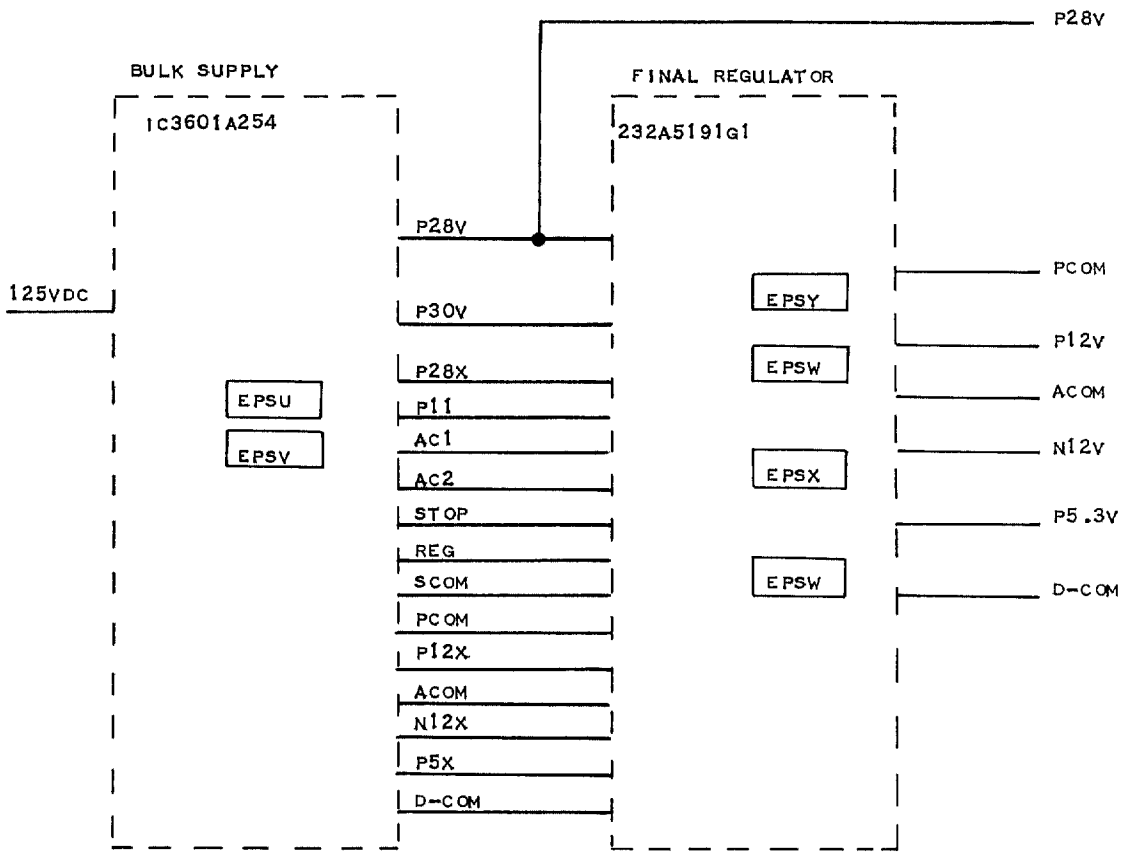
ACTION:

(I) REMOVE POWER AND CHECK ALL POWER TRANSISTORS ON BULK SUPPLY HEAT SINK HS1, HS2.
(II) IF TRANSISTORS ARE CORRECT SUSPECT EPSU AND REPLACE.

(F) SYMPTOM: BULK SUPPLY NOT HUMMING, EPSU "LED" IS OFF, BUT INDICATING LIGHT, DS1 IS ON.

ACTION:

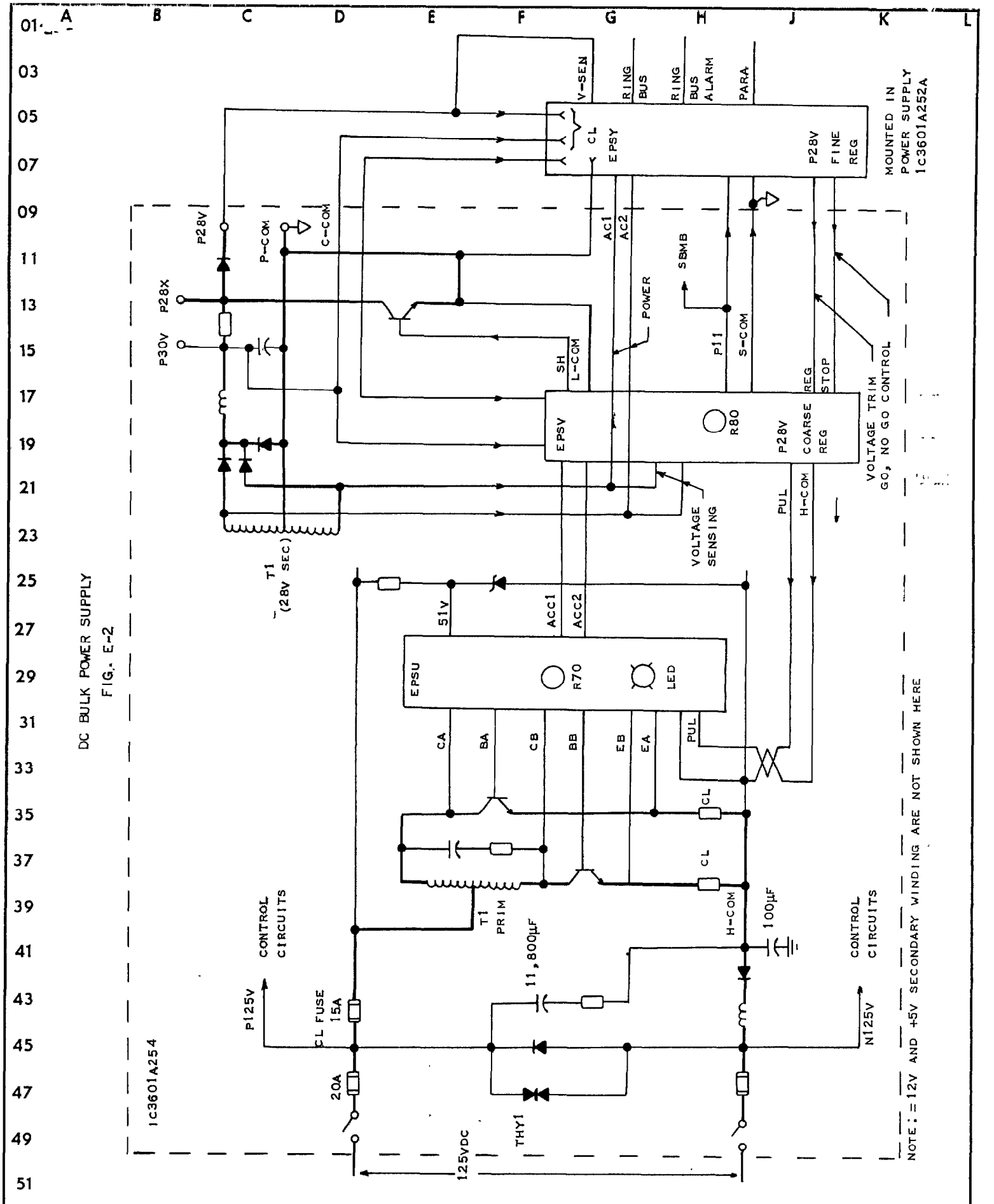
(I) REMOVE POWER AND CHECK FUSES; REPLACE ANY BAD FUSES.
(II) REMOVE EPSU, EPSV AND CHECK POWER TRANSISTORS ON HS1 AND HS2; REPLACE ANY BAD TRANSISTORS.
(III) APPLY POWER AND CHECK FOR BATTERY VOLTAGE FROM CA TO H-COM AND FROM CB TO H-COM (LOCATED ON TB2) BATTERY VOLTAGE MUST BE BETWEEN 95-145V DC. IF THIS VOLTAGE IS NOT PRESENT, SUSPECT BAD POWER TRANSISTOR OR SOME OTHER LOW IMPEDANCE PATH ON THE FRONT END. REMOVE POWER, ISOLATE AND REMOVE THIS LOW IMPEDANCE PATH.
(IV) IF BATTERY VOLTAGE IS CORRECT ON CA, CB, CHECK FOR 51+VDC BETWEEN PINS 11, 51 IN THE EPSU SLOT.
(V) IF THE PROBLEM HAS NOT BEEN IDENTIFIED TO THIS POINT, REMOVE POWER, PLUG IN NEW EPSU AND OLD EPSV CARDS AND CHECK SYSTEM. IF OPERATION IS STILL NOT CORRECT, REMOVE POWER REPLACE EPSV, AND TRY AGAIN.



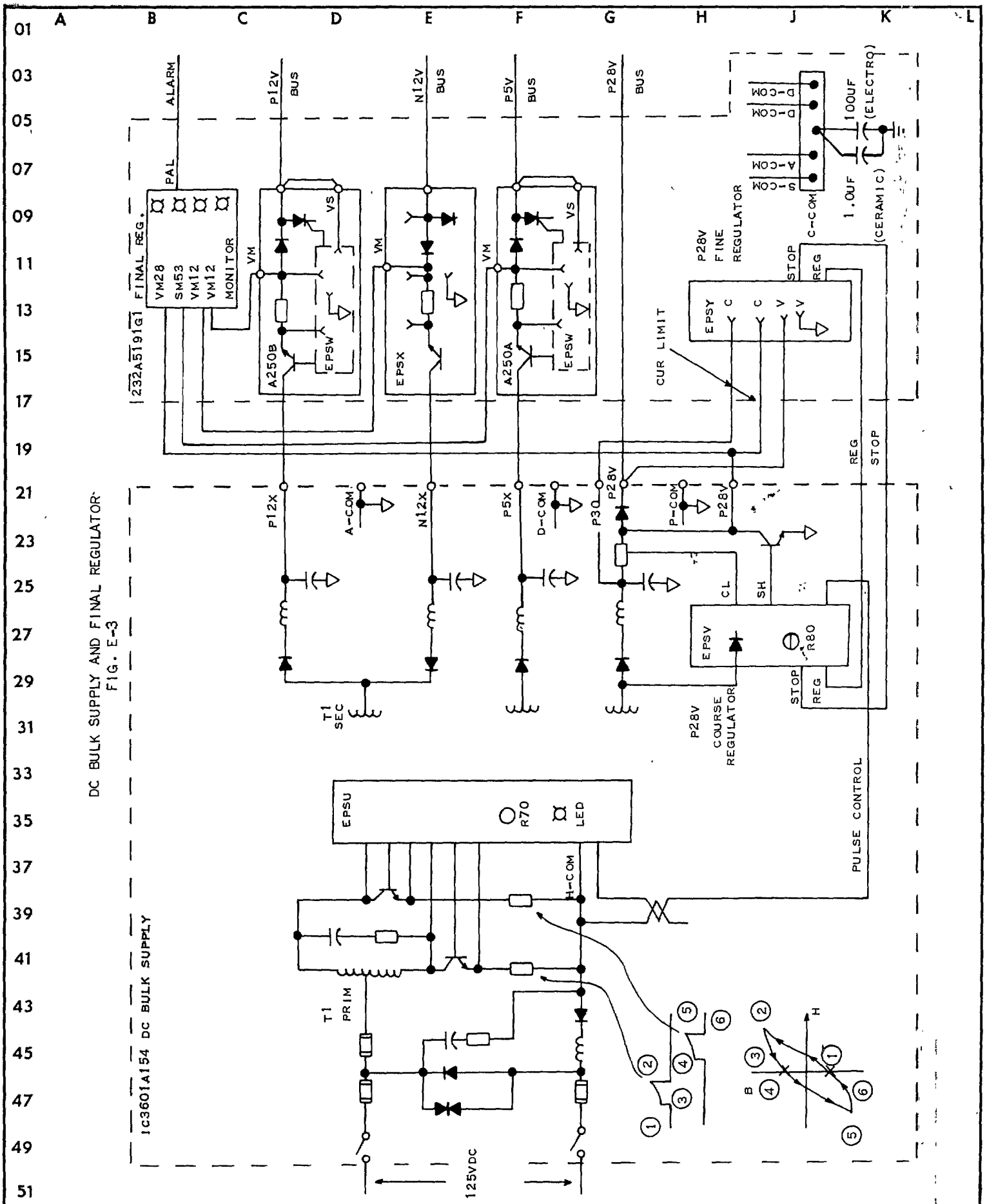
MARK II POWER SUPPLY CONFIGURATION

FIG. E-1

REV. 1	REV. 4	REV 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM POWER SUPPLIES
REV. 2	ISSUED			FIRST MADE FOR		
REV. 3	MADE BY			I.C. NO	SALEM, VA. U.S.A.	CONT. ON SH. 103 SH. NO 102



REV. 1	REV 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		POWER SUPPLIES
REV. 3	MADE BY			I.C NO.	SALEM, VA. U.S.A.	CONT. ON SH 104 SH. NO 103



REV. 1	REV 4	REV. 5	PRINTS TO	APPROVALS	GENERAL ELECTRIC	ELEMENTARY DIAGRAM
REV. 2	ISSUED			FIRST MADE FOR		POWER SUPPLIES
REV. 3	MADE BY			I.C. NO	SALEM, VA. U.S.A.	CONT ON SH FL.. SH NO 104